GRE Prep

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1 Calculus

1.1 Convergence Tests

Theorem 1.1 (Divergence Test)**.** *The series* $\sum_{n\geq 0} a_n$ *diverges if* $\lim_{n\to\infty} a_n \neq 0$.

Theorem 1.2 (p-series Test)**.** *The series* $\sum_{n\geq 0} \frac{1}{n^p}$*, for some* $p \in \mathbb{R}$ *converges if and only if* $p > 1$.

Theorem 1.3. *Suppose that* $\sum a_n, \sum b_n$ *are nonnegative series. Consider the quantity*

$$\lim_{n\to\infty} \frac{a_n}{b_n} = L$$

where $L \in [0, \infty]$*. Then we have the following three conclusions:*

- *If $L = 0$, the series $\sum a_n$ converges if $\sum b_n$ converges.*
- *If $L = \infty$, the series $\sum b_n$ converges if $\sum a_n$ converges.*
- *If $L \in (0, \infty)$, the series $\sum a_n$ converges if and only if $\sum b_n$ converges.*

Theorem 1.4. *Suppose that $\{a_n\}$ is a positive series such that there exists a continuous, decreasing function $f(x)$ satisfying $f(n) = a_n$. Then the series $\{a_n\}$ converges if and only if the improper integral*

$$\int_0^\infty f(x)dx$$

exists (converges).

Theorem 1.5 (Alternating Series Test)**.** *Suppose that $\{a_n\}$ is a positive, (eventually) decreasing series. If $\lim_{n\to\infty} a_n = 0$, the series $\sum_{n\geq 0}(-1)^n a_n$ converges.*

Example 1.6. The series $\sum \frac{1}{n}$ doesn't converge, but the alternating series $\sum \frac{(-1)^n}{n}$ does converge.

1.2 Integration, FTC, and Variants

Recall u-substitution. Suppose we have the integral

$$\int_a^b f(g(x))g'(x)dx.$$

Then, if we set $u = g(x)$, then this integral becomes

$$\int_{g(a)}^{g(b)} f(u)du.$$

Here are some useful specific cases:

$$\int_0^a f\left(\frac{x}{a}\right)dx = a\int_0^1 f(u)du$$
$$\int_a^b f(x-c)dx = \int_{a-c}^{b-c} f(u)du$$
$$\int_a^b f(a+b-x)dx = \int_a^b f(u)du$$

Theorem 1.7 (Leibniz's Rule)**.** *Suppose that $a(x)$ and $b(x)$ are differentiable, and that $f(x,t)$ and $\frac{\partial f}{\partial x}(x,t)$ are continuous. Then*

$$\frac{d}{dx}\int_{a(x)}^{b(x)} f(x,t)dt = f(x,b(x))b'(x) - f(x,a(x))a'(x) + \int_{a(x)}^{b(x)} \frac{\partial f}{\partial x}(x,t)dt.$$

1.3 Fubini

The assumptions for Fubini are quite technical, so I leave it up to the wikipedia page. This is a result from measure theory which essentially states when we can swap integrals. For absolutely integrable functions (or Lebesgue measurable), we can do this.

Exercise 1.8. Assume $b > a > 0$. Calculate the integral:

$$\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x}\, dx.$$

Hint: Write the integrand as an integral from a to b.

(A) $\ln a - \ln b^2$

(B) $\ln b - \ln a^2$

(C) $\ln a^2 - \ln b^2$

(D) $\ln a - \ln b$

(E) $\ln b - \ln a$

Solution: (B). The integral is

$$\int_0^\infty \int_a^b e^{-xy} dy dx.$$

We use Fubini to flip the order of integration. □

1.4 Things to watch out for

We include some common misconseptions.

Remark 1.9. Let $a \in \mathbb{R}$, and suppose we have some $f(x)$ defined on the reals such that $\int_a^\infty f(x)dx$ converges. This does not imply that $f(x) \to 0$ as $x \to \infty$. We can consider some 'spike function', whose spikes become narrow and narrow as x becomes large. For instance, let $f(x)$ be zero everywhere, except a spike of hegiht n and width $\frac{2}{n^3}$, around every positive integer. Then, the integral is $\frac{\pi^2}{6}$, which converges, but clearly the limit doesn't exist. If we want something like this, then it must be like the following statements:

- If the improper integral exists and $\lim_{x\to\infty} f(x) = L$, where L is some finite real, then $L = 0$.
- If the improper integral exists and $f(x)$ is uniformly continuous, then the limit exist (Barbalat's lemma). By the above, it must be zero.
- If f is nonnegative and eventually nonincreasing, then $\lim_{x\to\infty} f(x) = 0$.

Remark 1.10. We note that $|f(x)| \to \infty$ does not imply that $f(x) \to \infty$ or $f(x) \to -\infty$.

1.5 Standard Calculus Facts to Quicken Things

We start withs ome basic limits.

- $\lim_{x\to 0} \frac{\sin x}{x} = 1$
- $\lim_{x\to 0} \frac{\tan x}{x} = 1$
- $\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$

- $\lim_{x\to 0} \frac{\arctan x}{x} = 1$

Useful Taylor expansions:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$
$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$
$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots.$$

2 Multivariable Calculus and Geometry

Here, we include no proofs.

2.1 2D-Geometry

Proposition 2.1. *Given a circle of radisu r and angle θ (in radians). Then, the area is $\frac{1}{2}r^2\theta$ and the arc length $r\theta$.*

Proposition 2.2 (Hyperbola)**.** *Consider the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Then, the following hold:*

- *The foci are $(\pm c, 0)$, where $c^2 = a^2 + b^2$.*
- *Its asymptotes are $y = \pm\frac{b}{a}x$, and the angle it constructs with the x-axis is* $\arctan\left(\frac{b}{a}\right)$.

Proposition 2.3 (Ellipses)**.** *Consider the ellipses $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The focal distance c satisfies $c^2 = a^2 - b^2$.*

Proposition 2.4. *A tetrahedron with three mutually perpendicular edges of lengths a, b, c has volume $\frac{abc}{6}$.*

2.2 Trigonometric Identities

Addition formulae:

- $\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$
- $\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$
- $\tan(a \pm b) = \frac{\tan a \pm \tan b}{1 \mp \tan a \tan b}$

Double and Triple Angles

- $\sin 2x = 2 \sin x \cos x$
- $\cos 2x = 1 - 2\sin^2 x = 2\cos^2 x - 1$
- $\tan 2x = \frac{2\tan x}{1 - \tan^2 x}$
- $\sin 3x = 3\sin x - 4\sin^3 x$
- $\cos 3x = 4\cos^3 x - 3\cos x$
- $\tan 3x = \frac{3\tan x - \tan^3 x}{1 - 3\tan^2 x}$

2.3 3D Geometry and Basic Limits

Exercise 2.5. Find the equation of the plane in $\mathbb{R}^3$ passing through $P = (0, 0, 1), Q = (1, 0, 0), R = (1, 1, 1)$.

Solution: We use a reference point and use the vector cross product:

$$v := \overrightarrow{PQ} = \langle 1, 0, -1 \rangle, \qquad w := \overrightarrow{PR} = \langle 1, 1, 0 \rangle$$

Now, the normal vector to P is $v \times w$:

$$v \times w = \langle 1, -1, 1 \rangle$$

The equation of the plane is given by

$$\langle 1, -1, 1 \rangle \cdot \langle x - 0, y - 0, z - 1 \rangle,$$

which is simply $x - y + z = 1$. □

Theorem 2.6 (Point-to-Plane Distance)**.** *Let $P = (x_0, y_0, z_0)$ be a point and let $ax + by + cz + d = 0$ be a plane, where $(a, b, c) \neq (0, 0, 0)$. Then the distance from P to the plane is*

$$\frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Exercise 2.7. Compute the arc length of the helix $\gamma(t) = (\sin(t), -\cos(t), t)$ from $t = 0$ to $t = 2\pi$.

Solution: We must use the arc-length formula $\int_t \|\gamma'(t)\| dt$.

$$\int_{t=0}^{2\pi} \sqrt{\cos^2(t) + \sin^2(t) + 1^2} dt = \int_{t=0}^{2\pi} \sqrt{2} dt = 2\pi\sqrt{2}.$$

□

The following is a nice way to determine if a limit doesn't exist:

Theorem 2.8. $\lim_{(x,y)\to(a,b)} f(x, y) = L$ *if and only if for all* $\gamma \colon [-1, 1] \to \mathbb{R}^2$ *such that* $\gamma(0) = (a, b)$*, then* $\lim_{t\to 0} f(\gamma(t)) = L$.

Exercise 2.9. Prove that $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$ does not exist.

Solution: We consider the paths $\gamma(t) = (t, t)$ and $\delta(t) = (t, -t)$, which both tend to $(0, 0)$ as $t \to 0$. Using γ, we have as follows:

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2 + y^2} = \lim_{t\to 0} \frac{t^2}{t^2 + t^2} = \frac{1}{2}$$

On the other hand, δ gives:

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2 + y^2} = \lim_{t\to 0} \frac{-t^2}{t^2 + t^2} = -\frac{1}{2}$$

Therefore, the limit doesn't exist. □

Exercise 2.10. Find $\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2+y^2}$.

Solution: We use polar coordinates and change the limit to $r \to 0$.

$$\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2+y^2} = \lim_{r\to 0} \frac{r^3 \cos\theta \sin^2\theta}{r^2} = \lim_{r\to 0} r\cos\theta\sin^2\theta$$

Now,

$$0 \le |\lim_{r\to 0} r\cos\theta\sin^2\theta| \le |\lim_{r\to 0} r| = 0,$$

so the limit is 0 via squeeze. □

Exercise 2.11. Let

$$z = x^2 + y^2, \qquad x = t^2, \qquad y = e^t.$$

Find $\dfrac{dz}{dt}$.

(A) $2t^2 + 2e^t$

(B) $4t + 2e^{2t}$

(C) $4t^3 + 2e^{2t+1}$

(D) $4t^3 + 2e^{2t}$

(E) $2t^2 + e^t$

Solution: (D). This is just chain rule: $\frac{dz}{dt} = \frac{dz}{dx}\frac{dx}{dt} + \frac{dz}{dy}\frac{dy}{dt}$. □

Exercise 2.12. Let

$$z = f(x,y), \qquad x = u^2 - v^2, \qquad y = 2uv.$$

Which expression gives $\dfrac{\partial z}{\partial u}$?

(A) $2u\frac{\partial f}{\partial x} + 2v\frac{\partial f}{\partial y}$

(B) $2u\frac{\partial f}{\partial x} - 2v\frac{\partial f}{\partial y}$

(C) $2v\frac{\partial f}{\partial x} + 2u\frac{\partial f}{\partial y}$

(D) $2u\frac{\partial f}{\partial x} + 2u\frac{\partial f}{\partial y}$

(E) $(u^2 - v^2)\frac{\partial f}{\partial x} + 2uv\frac{\partial f}{\partial y}$

Solution: (A). Again, chain rule. □

Exercise 2.13. Let

$$z = f(x,y), \qquad x = s^2 + t, \qquad y = st.$$

Which expression gives $\dfrac{\partial^2 z}{\partial s^2}$?

(A) $4s^2 f_{xx} + 4st f_{xy} + t^2 f_{yy}$

(B) $4s^2 f_{xx} + 4st f_{xy} + t^2 f_{yy} + 2f_x$

(C) $2f_x + 4s^2 f_{xx} + 4st f_{xy} + 2t f_{yy}$

(D) $4s^2 f_{xx} + 2st f_{xy} + t^2 f_{yy}$

Solution: (B). First, we find

$$z_s = f_x \cdot x_s + f_y \cdot y_s = 2s f_x + t f_y.$$

Next, we have

$$z_{ss} = 2f_x + 2s\frac{\partial f_x}{\partial s} + t\frac{\partial f_y}{\partial s}.$$

We have that

$$\frac{\partial f_x}{\partial s} = f_{xx}x_s + f_{xy}y_s = 2s f_{xx} + t f_{xy}$$

and

$$\frac{\partial f_y}{\partial s} = f_{yx}x_s + f_{yy}y_s = 2s f_{xy} + t f_{yy}.$$

Therefore, we have

$$z_{ss} = 2f_x + 2s(2s f_{xx} + t f_{xy}) + t(2s f_{xy} + t f_{yy}).$$

This is just (B). □

2.4 Hessian

Let $f(x, y)$ be a twice-differntiable function on $\mathbb{R}^2$. Then, recall that the critical values occur when f_x and f_y are zero. To find the type, we use the Hessian matrix, H.

- If $\det H < 0$, then it is a saddle.
- If $\det H > 0$ and $f_{xx} > 0$, then it is a relative minima.
- If $\det H > 0$ and $f_{xx} < 0$, then it is a relative maxima.

2.5 Fourier Transformation

Recall that how to find the Fourier series of a periodic function. Let $f(x)$ be defined on the interval $(-L, L)$ and extended outside this interval by $f(x + 2L) = f(x)$, so that f is $2L$-periodic. The Fourier series of $f(x)$ is

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right),$$

where the Fourier coefficients are

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx, \qquad n = 0, 1, 2, \ldots,$$

and

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx, \qquad n = 1, 2, \ldots.$$

Remark 2.14. Note that in the Fourier series corresponding to an odd function, only sine terms can be present. In the Fourier series corresponding to an even function, only cosine terms (and possibly a constant) can be present.

Note that we can also formulate a complex version:

$$f(x) \sim \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L},$$

where the complex Fourier coefficients are

$$c_n = \frac{1}{2L}\int_{-L}^{L} f(x)e^{\frac{-in\pi x}{L}}dx.$$

Exercise 2.15. What is the Fourier series corresponds to the sawtooth wave $s(x)$:

$$s(x) = \begin{cases} x, & x \in [-\pi, \pi), \\ s(x \pm 2\pi), & \forall x \in \mathbb{R}. \end{cases}$$

Solution: Note that $s(-x) = -x = -s(x)$, so is odd. Thus, $a_0 = 0$ and $a_n = 0$, so $s(x) = \sum_{n=1}^{\infty} b_n \sin(nx)$. Recall that:

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x\sin(nx)dx.$$

The integrand is even, so

$$b_n = \frac{2}{\pi}\int_{0}^{\pi} x\sin(nx)dx,$$

and integration by parts gives

$$b_n = \frac{2(-1)^{n+1}}{n}.$$

□

2.6 Grad, Div, Curl

Recall that the gradient satisfies the following:

Proposition 2.16.

$$\nabla\phi = \left\langle \frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z} \right\rangle$$

Further, if we want the directional derivative of ϕ in the deriction of a unit vector $\hat{s}$, denoted $\frac{\partial\phi}{\partial s}$, we have

$$\frac{\partial\phi}{\partial s} = \hat{s}\cdot\nabla\phi.$$

Exercise 2.17. If $\phi = axy^2 + byz + cx^3z^2$, where a, b, c are constants, find $\nabla\phi$. Further, find the directional derivative of ϕ at the point $(1, 4, 2)$ in the direction towards the point $(2, 0 - 1)$.

Solution: First, we compute the gradient:

$$\nabla\phi = \left\langle ay^2 + 3cz^2z^2, 2axy + bz, by + 2cx^3z \right\rangle$$

Therefore, the gradient at $P = (1, 4, 2)$ is:

$$(\nabla\phi)_P = \langle 16a + 12c, 8a + 2b, 4b + 4c \rangle$$

Now, s is the difference between the two points, which is just $s = (2, 0, -1) - (1, 4, 2) = (1, -4, -3)$. Thus, $\hat{s}$ is

$$\hat{s} = \frac{\langle 1, -4, -3 \rangle}{\sqrt{26}}$$

Finally, the directional derivative $\frac{\partial\phi}{\partial s}$, is as follows:

$$\frac{\partial\phi}{\partial s} = (\nabla\phi \cdot \hat{s})_P = \frac{(16a + 12c) - 4(8a + 2b) - 3(4b + 4c)}{\sqrt{26}} = \frac{-16a - 20b}{\sqrt{26}}$$

□

Definition 2.18. Let F be a vector field.

- The *divergence* of F is the scalar product $\nabla \cdot F$, writing $\operatorname{div} F = \nabla \cdot F$.
- The *curl* of F is the vector product $\nabla \times F$, writing $\operatorname{curl} F = \nabla \times F$.

We recall some useful identities relating these different notions.

Theorem 2.19. *Let $f, g \colon \mathbb{R}^3 \to \mathbb{R}$ be a smooth scalar function and A, B be a vector field (or more simply, a vector valued function).*

- $\operatorname{curl}(\nabla f) = 0$
- $\operatorname{div}(\operatorname{curl} A) = 0$
- $\operatorname{curl}(\operatorname{curl}(A)) = \nabla(\operatorname{div} A) - \nabla^2 A$
- $\nabla(fg) = f\nabla g + g\nabla f$
- $\operatorname{div}(fA) = \nabla f \cdot A + f \operatorname{div} \cdot A$
- $\operatorname{curl}(fA) = \nabla f \times A + f \operatorname{curl} A$
- $\operatorname{div}(A \times B) = B \cdot \operatorname{curl} A - A \cdot \operatorname{curl} B$

Exercise 2.20. Let $f, g \colon \mathbb{R}^3 \to \mathbb{R}$ be smooth scalar functions. Show that $\nabla \cdot (\nabla f \times \nabla g) = 0$.

Solution: $\operatorname{div}(\nabla f \times \nabla g) = \nabla g(\operatorname{curl}(\nabla f)) - \nabla f(\operatorname{curl}(\nabla g))$. However, curl of grad is 0, so each of the parts inside the brackets are zero. □

Exercise 2.21. Let $f, g \colon \mathbb{R}^3 \to \mathbb{R}$ be smooth scalar functions. Show that $\operatorname{curl}(f\nabla g) = \nabla f \times \nabla g$.

Solution: We just apply one of the identities above:

$$\operatorname{curl}(f\nabla g) = \nabla f \times \nabla g + f \operatorname{curl}(\nabla g),$$

but curl of grad is zero, so the term on the right vanishes. □

2.7 Partial Derivatives and Jacobians

Exercise 2.22. Let $z = e^{xy^2}$, $x = t\cos t$, $y = t\sin t$. Compute $\frac{dz}{dt}$.

Solution: We use that $\frac{dz}{dt} = \frac{dz}{dx}\frac{dx}{dt} + \frac{dz}{dy}\frac{dy}{dt}$.

$$\begin{aligned}\frac{dz}{dt} &= y^2 e^{xy^2}\cdot(\cos t - t\sin t) + 2xye^{xy^2}\cdot(\sin t + t\cos t)\\ &= t^2\sin^2 t e^{t^3\cos t\sin^2 t}(\cos t - t\sin t) + 2xye^{t^3\cos t\sin^2 t}\cdot(\sin t + t\cos t)\end{aligned}$$

□

Exercise 2.23. If $x = 2r - s$ and $y = r + 2s$, find $\frac{\partial^2 U}{\partial y\partial x}$ in terms of derivatives with respect to r and s.

Solution: Use the multivariable chain rule, we should really solve for r and s in terms of x and y. We have $5r = 2x + y$ and $5s = 2y - x$. Therefore

$$\frac{\partial U}{\partial x} = \frac{\partial U}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial U}{\partial s}\frac{\partial s}{\partial x} = \frac{2}{5}\frac{\partial U}{\partial r} - \frac{1}{5}\frac{\partial U}{\partial s}.$$

We now differentiate by y.

$$\begin{aligned}\frac{\partial^2 U}{\partial y\partial x} &= \frac{\partial}{\partial y}\left(\frac{2}{5}\frac{\partial U}{\partial r} - \frac{1}{5}\frac{\partial U}{\partial s}\right)\\ &= \frac{\partial}{\partial r}\left(\frac{2}{5}\frac{\partial^2 U}{\partial r^2} - \frac{1}{5}\frac{\partial^2 U}{\partial rs}\right)\cdot\frac{\partial r}{\partial y} + \frac{\partial}{\partial s}\left(\frac{2}{5}\frac{\partial^2 U}{\partial sr} - \frac{1}{5}\frac{\partial^2 U}{\partial s^2}\right)\cdot\frac{\partial s}{\partial y}\\ &= \left(\frac{2}{5}\frac{\partial^2 U}{\partial r^2} - \frac{1}{5}\frac{\partial^2 U}{\partial rs}\right)\cdot\frac{1}{5} + \left(\frac{2}{5}\frac{\partial^2 U}{\partial sr} - \frac{1}{5}\frac{\partial^2 U}{\partial s^2}\right)\cdot\frac{2}{5}\\ &= \frac{1}{25}\left(2\frac{\partial^2 U}{\partial r^2} + 3\frac{\partial^2 U}{\partial rs} - 2\frac{\partial^2 U}{\partial s^2}\right)\end{aligned}$$

□

2.8 Line and Surface Integrals

Definition 2.24. Given a vector field F, if $\oint_C F\cdot d\mathbf{r} = 0$ for any closed curve C, then we say that **F** is a **conservative field**.

We recall some nice theorems that help computation.

Theorem 2.25 (Green's Theorem in the Plane)**.** *Suppose R is a closed plane region bounded by a simple plane closed convex curve in the x-y plane. Let L, M be continuous functions of x, y having continuous derivatives throughout R. Then:*

$$\oint_C (Ldx + Mdy) = \int_R \left(\frac{\partial M}{\partial x} - \frac{\partial L}{\partial y}\right)dxdy,$$

where C is the boundary of R described in the counter-clockwise (positive) sense.

Theorem 2.26. *Let* **A** *be a vector field (that is continously differentiable on an open, simply connected domain $D \subseteq \mathbb{R}^3$). Then, the following conditions are equivalent:*

- $\int_C \mathbf{A} \cdot d\mathbf{r}$ *is independent of the path.*
- *There exists a scalar function* θ *such that* $\mathbf{A} = \nabla\theta$*. In this case, the integral is equal to* $\theta(P) - \theta(Q)$*, where* P *and* Q *denote the finishing and starting points.*
- $\nabla \times \mathbf{A} = 0$ *(*$\operatorname{curl} \mathbf{A} = 0$*).*
- $\mathbf{A}$ *is conservative.*

Exercise 2.27. Which of the following apply to the vector field

$$\vec{F} = \langle x^2 - y^2, -2xy, z^2 \rangle$$

defined on all of $\mathbb{R}^3$?

(A) $\operatorname{div} \vec{F} = 0$

(B) Conservative and curl vanishes.

(C) Conservative, but curl does not vanish.

(D) Curl vanishes, but not conservative.

Solution: (B). Note that the curl vanishes, and $\mathbb{R}^3$ is simply connected, so (B). □

Definition 2.28 (Differential Element of Surface Area)**.** The differential element of surface area is

$$dS = \left| \frac{\partial \mathbf{r}}{\partial v_1} \times \frac{\partial \mathbf{r}}{\partial v_2} \right| dv_1\, dv_2.$$

Definition 2.29. If S is a surface then the *flux* of a vector field $\mathbf{A}$ across S is defined as

$$\int_S \mathbf{A} \cdot \hat{\mathbf{n}} dS,$$

where $\hat{\mathbf{n}}$ is a unit normal pointing out S.

Theorem 2.30 (Divergence Theorem)**.** *If* τ *is the volume enclosed by a closed surface* S *with unit outward normal* $\mathbf{n}$ *and* $\mathbf{A}$ *is a continuously differentiable vector field throughout* τ*, then*

$$\int_S \mathbf{A} \cdot \mathbf{n} dS = \int_\tau \operatorname{div} \mathbf{A} d\tau.$$

Theorem 2.31 (Stokes' Theorem)**.** *Suppose* S *is an open surface with a simple closed curve* C *forming its boundary, and let* $\boldsymbol{A}$ *be a vector field with continuous partial derivatives. Then:*

$$\oint_C \mathbf{A} \cdot d\mathbf{r} = \int_S \operatorname{curl} \mathbf{A} \cdot \hat{\mathbf{n}} dS$$

where the direction of the unit normal to S *and the sense of* C *are related by a right-hand rule. That is,* $\hat{\mathbf{n}}$ *is in the direction a right-handed screw moves when turned in the direction of* C*.*

Theorem 2.32 (Gauss's Flux Theorem)**.** *Let* S *be a closed surface with outward unit normal* $\hat{\boldsymbol{n}}$*, and let* O *be the origin of the coordinate system. Then:*

$$\int_S \frac{\hat{\boldsymbol{n}} \cdot \boldsymbol{r}}{r^3} = \begin{cases} 0, & \text{if } O \text{ is exterior to } S \\ 4\pi, & \text{if } O \text{ is interior to } S \end{cases}$$

Exercise 2.33. Evaluate

$$\int_{(0,1)}^{(1,2)} (x^2 - y)dx + (y^2 + x)dy$$

along:

1. a straight line from $(0, 1)$ to $(1, 2)$.
2. straight lines from $(0, 1)$ to $(1, 1)$ and then from $(1, 1)$ to $(1, 2)$.
3. the parabola $x = t$, $y = t^2 + 1$.

Solution: 1. The line adjoining these is $y = x + 1$. Then, $dy = dx$, and we just plug in:

$$\int_0^1 (x^2 - (x + 1))dx + ((x + 1)^2 + x)dx = \int_0^1 2x^2 + 2x = \frac{5}{3}.$$

2. We do a similar thing, but split the integral piecewise. The first part is

$$\int_0^1 (x^2 - 1)dx + (1 + x) \cdot 0 = -\frac{2}{3}.$$

The next part is

$$\int_1^2 (1 - y) \cdot 0 + (y^2 + 1)dy = \frac{10}{3}.$$

Thus, the integral is equal to $\frac{8}{3}$ by linearity.

3. For this, we note that $dx = dt$ and $dy = 2tdt$. The time is from 0 to 1.

$$\int_0^1 (t^2 - (t^2 + 1))dt + ((t^2 + 1)^2 + t)2tdt = \int_0^1 2t^5 + 4t^2 + 2t^2 + 2t - 1dt = 2.$$

□

Exercise 2.34. Evaluate $\int_C y ds$ along the curve C given by $y = 2\sqrt{x}$ from $x = 3$ to $x = 24$.

Solution: Recall that ds is the arclength element. It is probably easiest to use that

$$ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

We have $\frac{dy}{dx} = \frac{1}{\sqrt{x}}$, so $ds = \sqrt{1 + \frac{1}{x}}dx$. Thus, the integral is as follows:

$$\int_3^{24} 2\sqrt{x} \cdot \sqrt{1 + \frac{1}{x}}dx = 2\int_3^{24} \sqrt{x + 1}dx = \frac{4}{3}(x + 1)^{\frac{3}{2}}\Big|_3^{24} = 156$$

□

Exercise 2.35. Verify Green's theorem in the plane for

$$\oint_C (2xy - x^2)dx + (x + y^2)dy,$$

where C is the closed curve of the region bounded by $y = x^2$ and $y^2 = x$.

Solution: Note that this is just from the origin to $(1,1) =: P$ and back. From the origin to P, we have:

$$\int_0^1 (2x^3 - x^2)dx + (x + x^4)2xdx = \int_0^1 2x^5 + 2x^3 + x^2 dx = \frac{7}{6}.$$

From P to the origin, we have as follows:

$$\int_1^0 (2y^3 - y^4)2ydy + (y^2 + y^2)dy = \int_1^0 -2y^5 + 4y^4 + 2y^2 dy = -\frac{17}{15}$$

Thus, the original integeral is $\frac{1}{30}$.

On the other hand, the statement of Green's theorem gives us that the integral is equal to the following:

$$\int_R (1 - 2x)dxdy.$$

Now we just need to find the appropriate bounds. We have as follows:

$$\int_{y=0}^1 \int_{x=y^2}^{\sqrt{y}} (1 - 2x)dxdy = \int_0^1 \sqrt{y} - y - y^2 + y^4 dy = \frac{1}{30}.$$

Thus, true! □

Exercise 2.36. Prove that

$$\mathbf{F} = (2xz^3 + 6y)\mathbf{i} + (6x - 2yz)\mathbf{j} + (3x^2z^2 - y^2)\mathbf{k}$$

is a conservative force field. Further, evaluate

$$\int_C \mathbf{F} \cdot d\mathbf{r},$$

where C is any path from $(1, -1, 1)$ to $(2, 1, -1)$.

Solution: To show it is conservative, we show that the curl of $\mathbf{F}$ is zero.

$$\nabla \times \mathbf{F} = 0$$

The easiest way to do this is to find some potential function, where $\mathbf{F} = \nabla\theta$. Integrating variable by variable, we have that $\phi(x, y, z) = x^2z^3 + 6xy - y^2z + c$, where c is some constant, works. We should really find the constant c, but they cancel anyway. The integral is equal to

$$\phi(2, 1, -1) - \phi(1, -1, 1) = 15.$$

□

Exercise 2.37. Evaluate $\oint_C (2x - y + 4)dx + (5y + 3x - 6)dy$ around the triangle in the xy-plane with vertices $(0, 0)$, $(3, 0)$, and $(3, 2)$, traversed in the counterclockwise direction.

Solution: Use Green's theorem in the plane. The desired integral becomes the following.

$$\int_{x=0}^3 \int_{y=0}^{\frac{2}{3}x} (3 - (-1))dydx = 12.$$

□

Exercise 2.38. Evaluate

$$\int_{(1,1)}^{(4,2)} (x+y)\,dx + (y-x)\,dy$$

along:

1. the parabola $y^2 = x$
2. a straight line
3. straight lines from $(1,1)$ to $(1,2)$ and then to $(4,2)$
4. the curve $x = 2t^2 + t + 1$, $y = t^2 + 1$

Solution: 1. We just plug in simply, using $dx = 2ydy$.

$$\int_{y=1}^{2} (y^2+y)2ydy + (y-y^2)dy = \int_1^2 2y^3 + y^2 + ydy = \frac{34}{3}$$

2. The line connecting the two is $y = \frac{1}{3}(x-1)+1 = \frac{1}{3}x + \frac{2}{3}$. Therefore, $dy = \frac{1}{3}dx$, and the integral is as follows:

$$\int_{x=1}^{4} (x + \frac{1}{3}x + \frac{2}{3})dx + (\frac{1}{3}x + \frac{2}{3} - x)\frac{1}{3}dx = \int_1^4 \frac{10}{9}x + \frac{8}{9}dx = 11$$

3. We split up the integral. From $(1,1)$ to $(1,2)$, $dx = 0$ and from $(1,2)$ to $(4,2)$, $dy = 0$.

$$\int_{y=1}^{2} (y-1)dy = \frac{1}{2}$$
$$\int_{x=1}^{4} (x+2)dx = \frac{27}{2}$$

Thus, the total is 14.

4. We parametrize in terms of t. Note that the bound is $t = 0$ to $t = 1$. Also, $dx = (4t+1)dt$, $dy = 2tdt$.

$$\int_{t=0}^{1} (3t^2+t+2)(4t+1)dt + (-t^2-t)(2t)dt = \int_{t=0}^{1} 10t^3 + 5t^2 + 9t + 2 = \frac{32}{3}.$$

□

Exercise 2.39. Calculate

$$\oint_C (x^2 - xy^3)dx + (y^2 - 2xy)dy,$$

where C is the square with vertices $(0,0)$, $(2,0)$, $(2,2)$, and $(0,2)$, traversed counterclockwise.

Solution: Use Green's theorem.

$$\int_{x=0}^{2}\int_{y=0^2} (-2y - (-3xy^2))dydx = \int_{x=0}^{2} (xy^3 - y^2)\Big|_0^2 dx = \int_0^2 8x - 4dx = 8.$$

□

Exercise 2.40. Evaluate

$$\oint_C (2x - y + 4)dx + (5y + 3x - 6)dy,$$

where C is the circle $x^2 + y^2 = 16$, traversed counterclockwise.

Solution:

□

2.9 Jacobians and Volume/ Area Elements

We include a quick cheat sheet for the volume/ area elements for calculating integrals with Euclidean, polar, or spherical coordinates.

2.9.1 Euclidean (Cartesian) Coordinates

Example 2.41 ($\mathbb{R}^2$)**.** Here, $x = x$ and $y = y$, so $dA = dxdy$.

Example 2.42 ($\mathbb{R}^3$)**.** Here, $x = x$, $y = y$, and $z = z$, so $dV = dxdydz$.

2.9.2 Polar Coordinates

Example 2.43 ($\mathbb{R}^2$)**.** Here, $x = r\cos\theta$ and $y = r\sin\theta$, subject to

$$x^2 + y^2 = r^2, \qquad \theta = \arctan\left(\frac{y}{x}\right).$$

In this case, $dA = rdrd\theta$.

Example 2.44 ($\mathbb{R}^3$)**.** Here, $x = r\cos\theta, y = r\sin\theta, z = z$, subject to

$$x^2 + y^2 = r^2, \qquad \theta = \arctan\left(\frac{y}{x}\right).$$

Here, $dV = rdrd\theta dz$.

2.9.3 Spherical Coordinates ($\mathbb{R}^3$)

Here ϕ is measured from the positive z-axis and θ is measured from positive x-axis to y-axis. We set

$$\begin{aligned} x &= r\sin\phi\cos\theta, \\ y &= r\sin\phi\sin\theta, \\ z &= r\cos\phi, \\ x^2 + y^2 + z^2 &= r^2. \end{aligned}$$

In this case,

$$dV = r^2\sin\phi drd\phi d\theta$$

3 Linear Algebra

3.1 Dot Products and Cross Products

Proposition 3.1. *We have the following identities:*

$A \cdot B = |A||B|\cos\theta$*, where* $\theta \in [0, \pi]$ *is the angle between vecotrs* A *and* B*.*

$A \times B = |A||B|\sin\theta\vec{u}$*, where* $\theta \in [0, \pi]$ *is the angle between vecotrs* A *and* B *and* $\vec{u}$ *is a unit vector in the direction of* $A \times B$*.*

Proposition 3.2. $|A \times B|$ *is the area of a parallelogram with sides* A, B*.*

Proposition 3.3. $A \cdot (B \times C) = B \cdot (C \times A) = C \cdot (A \times B)$ *is in absolute value equal to the volume of a parallelepiped with sides of vectors* A, B, C*.*

Proposition 3.4. *We have the following identities:*

- $A \times B = -B \times A$.
- $A \cdot (B \times C) = (A \times B) \cdot C$
- $A \times (B \times C) = (A \cdot C)B - (A \cdot B)C$.

3.2 Some Facts About Adjugate Matrices

Theorem 3.5 (Adjugate Matrices)**.** *Let A be a square matrix. Then, the following hold.*

- *If A is invertible, then $A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$.*
- *The rank of* $\operatorname{adj}(A)$ *depends on A as follows:*

$$\operatorname{Rank}(\operatorname{adj}(A)) = \begin{cases} n & \operatorname{Rank}(A) = n, \\ 1 & \operatorname{Rank}(A) = n-1, \\ 0 & \operatorname{Rank}(A) \leq n-2 \end{cases}$$

- $A \operatorname{adj}(A) = \det(A) I$
- $\det(\operatorname{adj}(A)) = (\det A)^{n-1}$.
- $\operatorname{adj}(cA) = c^{n-1} \operatorname{adj}(A)$.
- $\operatorname{adj}(AB) = \operatorname{adj}(BA)$.

3.3 Miscellanious

Here, we just include many problems that will be useful. Some exercises emphasize how some wishful statements do not hold.

Exercise 3.6. Show that $\{1, (x-1), (x-1)(x-2)\}$ is a basis of $\mathbb{P}_3[x]$ and that $W = \{p(x) \in \mathbb{P}_3[x] \colon p(1) = 0\}$ is a subspace of $\mathbb{P}_3[x]$. Find $\dim W$. Here, $\mathbb{P}_3[x]$ denotes the space of real polynomials of degree strictly less than 3.

Solution: The fact that this is a basis is clear. We find $\dim W$ using rank-nulity. The image has dimension 1, since two of the basis elements vanish. Thus, $\dim W = 2$. □

Exercise 3.7. Let $\alpha_1, \alpha_2, \alpha_3$, and α_4 be linearly independent. Answer true or false.

1. $\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_4, \alpha_4 + \alpha_1$ are linearly independent.
2. $\alpha_1 - \alpha_2, \alpha_2 - \alpha_3, \alpha_3 - \alpha_4, \alpha_4 - \alpha_1$ are linearly independent.
3. $\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_4, \alpha_4 - \alpha_1$ are linearly independent.
4. $\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 - \alpha_4, \alpha_4 - \alpha_1$ are linearly independent.

Solution: 1. False. We can make an easy relation among these elements.

2. False. Just sum them all.

3. True. Just expand.

4. False. Sum of last two is equal to difference of second and first. □

Exercise 3.8. Let $t \in \mathbb{R}$. Discuss the linear independence of the vectors over $\mathbb{R}$:

$$\alpha_1 = (1, 1, 0), \qquad \alpha_2 = (1, 3, -1), \qquad \alpha_3 = (5, 3, t).$$

Solution: We have the relations:

$$a + b + 5c = 0, \qquad a + 3b + 3c = 0, \qquad -b + tc = 0$$

The first two equations provide that $2b - 2c = 0$, or $b = c$. Thus, so long as $t \neq 1$, these are LI. More systematically, use the determinant condition. □

Exercise 3.9. Let W_1 and W_2 be nontrivial subspaces of a vector space V. The sum $W_1 + W_2$ is called a *direct sum*, denoted as $W_1 \oplus W_2$, if every element $\alpha \in W_1 + W_2$ can be uniquely written as

$$\alpha = w_1 + w_2,$$

where $w_1 \in W_1$ and $w_2 \in W_2$. Show that the following statements are equivalent:

1. $W_1 + W_2$ is a direct sum.
2. If $w_1 + w_2 = 0$ and $w_1 \in W_1$, $w_2 \in W_2$, then $w_1 = w_2 = 0$.
3. $W_1 \cap W_2 = \{0\}$.
4. $\dim(W_1 + W_2) = \dim W_1 + \dim W_2$.

How can the direct sum be extended to more than two subspaces?

Solution: Standard. □

Exercise 3.10. Let A, B, C be matrices. Prove if $r(A)$ denotes the rank of A, then prove the following:

1. $r(A + B) \leq r(A) + r(B)$.
2. $r(AB) \leq \min\{r(A), r(B)\}$.
3. $r(AB) + r(BC) - r(B) \leq r(ABC)$.

Solution:

□

Exercise 3.11. Let $A \in M_n(\mathbb{R})$.

1. Show that if $A^t = -A$ and n is odd, then $\det(A) = 0$.
2. Show that if $A^2 + I = 0$, then n is even. What is $A \in M_n(\mathbb{C})$ instead?

Solution: 1. We have $\det(A) = -\det(A)$, becaause n is odd, so $\det(A) = 0$.

2. We have $\det(A)^2 = (-1)^n$. Therefore, n must be even. If A is complex, then n can be even, because $\det(A) \in \mathbb{C}$ now. □

Exercise 3.12. Let $A, B \in M_n(\mathbb{R})$. Show that

$$\det \begin{pmatrix} A & B \\ -B & A \end{pmatrix} \geq 0.$$

State the analogue for complex matrices.

Solution: $\det(A)^2 + \det(B)^2$ is a sum of nonnegative reals, so is nonnegative. If we want a similar statement for complex matrices, then we just need to conjugate the bottom two:

$$\det \begin{pmatrix} A & B \\ \overline{-B} & \overline{A} \end{pmatrix} \geq 0$$

□

Exercise 3.13. Let $A, B \in M_n(\mathbb{C})$. Show that

$$\det \begin{pmatrix} A & B \\ B & A \end{pmatrix} = \det(A+B)\det(A-B)$$

Solution: Just use the fact that $\det(A^2 - B^2) = \det((A-B)(A+B)) = \det(A - B)\det(A+B)$. □

Exercise 3.14. For matrices of appropriate sizes, answer true or false:

1. If $A^2 = B^2$, then $A = B$ or $A = -B$.
2. If $r(A) = r(B)$, then $r(A^2) = r(B^2)$.
3. $r(A + kB) \leq r(A) + kr(B)$, where k is a positive scalar.
4. $r(A - B) \leq r(A) - r(B)$.
5. If $r(AB) = 0$, then $r(BA) = 0$.
6. If $r(AB) = 0$, then $r(A) = 0$ or $r(B) = 0$.

Solution: 1. False. Take $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, B = I$.

2. False. Take $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.

3. False, take $A - 0$, $B \neq 0$, and $k = \frac{1}{2}$. The correct statement instead

$$r(A + kB) \leq r(A) + kr(B).$$

4. Take $A = 0$, $B \neq 0$. Then, $r(-B)$ is positive, while $-r(B)$ is negative.

5. Take $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then, $AB = 0$, while $BA = B \neq 0$.

6. Use the same example as 2. □

Proposition 3.15. *Show that $AB = I$ implies $BA = I$, where A, B are matrices over a field.*

Solution: This relies on some knowledge of algebra; particularly, about left or right Artinian rings. Note that the ring of matrices (over a field) form a finite dimensional algbera, and so they form an Artinian ring. We recall that left or right Artinian rings are rings where the DCC holds for left or right ideals. Let us use the left version. The claim we prove is that $ab = 1$ implies $ba = 1$, for any elements a, b (which satisfies the desired condition). We let the Artinian ring be called R. We note that $R \subseteq bR \subseteq b^2R \subseteq \dots$ is a descending chain of left ideals, and so $b^kR = b^{k+1}R$ for some positive integer k. In other words, $b^k = b^{k+1}c$, for some $c \in R$. Then, if we multiply by a^k on the left, we get that $bc = 1$. Then, $ba = babc = bc = 1$. □

Exercise 3.16. Let v_1, v_2, v_3, v_4 be vectors in $\mathbb{R}^2$, and consider the following six dot products:

$$v_1 \cdot v_2, \qquad v_1 \cdot v_3, \qquad v_1 \cdot v_4, \qquad v_2 \cdot v_3, \qquad v_2 \cdot v_4, \qquad v_3 \cdot v_4.$$

Which of the following statements about the six dot products could be true?

1. All six are negative.
2. Two are negative, and four are equal to 0.
3. Two are positive, and four are equal to 0.

Solution: Testing with some random vector might be good, but takes some clever guesses which I don't have the time to be searching for. The key is the dot product identity, which measures the angle depending on the sign. If positive, then acute, if 0, then perpendicular, if negative, then obtuse. Given four vectors in $\mathbb{R}^2$, then all six are negative if all vectors are obtuse with respect to each other. This is not possible. The remaining cases are possible. In the case 2, we situate them all perpendicularly to each other. In the case 3, we let $v_1 = v_2$, $v_3 = v_4$, and make these two vectors perpendicular to each other. □

Proposition 3.17 (Sylvester's Rank Inequality)**.** *Let $A \in M_{m\times n}, B \in M_{n\times p}$. Then,*

$$\text{Rank}(A) + \text{Rank}(B) - n \leq \text{Rank}(AB) \leq \min\{\text{Rank}(A), \text{Rank}(B)\}.$$

3.4 Special Forms

Theorem 3.18 (Diagonalizability)**.** *$A \in M_n(K)$, where K is a field, is diagonalizable (to say D) if and only if A has n linearly independent eigenvectors. In this case, the diagonal elements of D are the corresponding eigenvalues and $D = P^{-1}AP$, where P is the matrix whose columns are the eigenvectors.*

Theorem 3.19 (Spectral Theorem (for $\mathbb{R}$))**.** *Every eigenvalue of a real symmetric matrix is real. Furthermore, eveyr real symmetric matrix is orthogonally diagonalizable, meaning there is an orthogonal matrix P (so $PP^T = I$) and diagonal D such that $PDP^{-1} = A$ (A is the real symmetric matrix).*

Theorem 3.20. *A matrix is diagonalizable if and only if its minimal polynomial is a product of distinct linear factors (over the base field).*

Exercise 3.21. Let $A = \begin{pmatrix} 4 & 1 & -1 \\ 2 & 5 & -2 \\ 1 & 1 & 2 \end{pmatrix}$. Is it diagonalizable? If so, find the transformation matrix and the diagonal.

Solution: The characteristic polynomial is $(5-x)(x-3)^2$, so its eigenvalues are 5 and 3. We want to find that the total sum of each eigenspaces is equal to 3. The eigenspace corresponding to the eigenvalue 5 is spanned by $\begin{pmatrix} 1 & 2 & 1 \end{pmatrix}^T$. The eigenspace corresponding to the eigenvalue 3 is spanned by the vectors $\begin{pmatrix} -1 & 1 & 0 \end{pmatrix}^T, \begin{pmatrix} 1 & 0 & 1 \end{pmatrix}^T$. Thus, it's diagonalizable. In particular, we have

$$A = PDP^{-1}, \qquad P = \begin{pmatrix} -1 & 1 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

□

Exercise 3.22. Let A be an $m \times m$ matrix over $\mathbb{C}$ and $m > 1$. Suppose that $A^n = I$ for some $n > 0$, where I is the $m \times m$ identity matrix. Which of the following are true?

(I) A can be diagonalized.

(II) If λ is an eigenvalue that is not real, then

$$1 + \lambda + \cdots + \lambda^{n-1} = 0.$$

(III) The characteristic polynomial has coefficients in $\mathbb{R}$.

(A) 1 only

(B) 2 only

(C) 3 only

(D) 1 and 2 only

(E) 1, 2, and 3

Solution: (D). The minimal polynomial divides $x^n - 1$, so it doesn't have repeated roots. Thus, diagonalizable. If λ is a nonreal eigenvalue, then it is a complx n-th root of unity. 3 is not true if we consider say a 2×2 matrix with diagonal entries i. □

Theorem 3.23 (Spectral Theorem (for $\mathbb{C}$))**.** *Let $A \in M_n(\mathbb{C})$ be Hermitian, so $A^* = \overline{A}^T = A$. Then, there exists a unitary matrix U (so $UU^* = I$) such that $U^*AU = D$, where D is a diagonal matrix with real entries. Equivalently, $A = UDU^*$. In particular, A is unitarily diagonalizable and all of its eigenvalues are real. Moreover, $\mathbb{C}^n$ has an orthonormal basis consisting of eigenvectors of A.*

Theorem 3.24 (Triangulizability)**.** *Let A be a square matrix whose characteristic polynomial factors into linear polynomials. Then A is similar to a triangular matrix, that is, there exists an invertible matrix P such that $P^{-1}AP$ is triangular.*

Theorem 3.25 (Primary Decomposition Theorem)**.** *Let $T\colon V \to V$ be a linear operator with minimal polynomial*

$$m(t) = f_1(t)^{n_1} f_2(t)^{n_2} \cdots f_r(t)^{n_r},$$

where the $f_i(t)$ are distinct monic irreducible polynomials. Then V is the direct sum of T-invariant subspaces $W_1, \ldots, W_r$, where

$$W_i = \ker f_i(T)^{n_i}.$$

Moreover, $f_i(t)^{n_i}$ is the minimal polynomial of the restriction of T to W_i.

Theorem 3.26. *A linear operator $T\colon V \to V$ is diagonalizable if and only if its minimal polynomial $m(t)$ is a product of distinct linear polynomials.*

Theorem 3.27 (Jordan Canonical Form)**.** *Let $T : V \to V$ be a linear operator whose characteristic and minimal polynomials are, respectively,*

$$\chi_T(t) = (t-\lambda_1)^{n_1} \cdots (t-\lambda_r)^{n_r} \qquad \text{and} \qquad m_T(t) = (t-\lambda_1)^{m_1} \cdots (t-\lambda_r)^{m_r},$$

where the λ_i are distinct scalars. Then T has a block diagonal matrix representation J in which each diagonal entry is a Jordan block $J_{ij} = J(\lambda_i)$. For each λ_{ij}, the corresponding J_{ij} have the following properties:

1. *There is at least one J_{ij} of order m_i (the algebraic multiplicity); all other J_{ij} are of order $\leq m_i$.*
2. *The sum of the orders of the J_{ij} is n_i.*
3. *The number of J_{ij} equals the geometric multiplicity of λ_i.*
4. *The number of J_{ij} of each possible order is uniquely determined by T.*

Remark 3.28. The geometric multiplicity associated to λ is the dimension of the eigenspace, or $\dim\ker(A - \lambda I)$.

3.4.1 Schur Decomposition

I don't know if this is standar covered content, but Schur decomposition gives a useful way to find some determinants of matrices that feel like should be manageable. The set up is as follows: Let $A \in M_{n\times n}$, $B \in M_{n\times m}$, $C \in M_{m\times n}$, and $D \in M_{m\times m}$. Then, consider the big block matrix:

$$M := \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in M_{(m+n)\times(m+n)}.$$

Theorem 3.29. *Continue with the setup above, and suppose A is invertible. Define the* Schur complement of A, $S := D - CA^{-1}B$. *Then,*

- $\det(M) = \det(A)\det(S)$
- $\text{Rank}(M) = \text{Rank}(A) + \text{Rank}(S)$.
- $\dim\ker(M) = \dim\ker(S)$.

- *M is invertible if and only if S is invertible.*

Remark 3.30. We can assume invertibility of D and say similar statesments for $S' = A - BD^{-1}C$.

Theorem 3.31 (Sylvester's Determinant Theorem:)**.** *Let $A \in M_{m\times n}$ and $B \in M_{n\times m}$. Then,*

$$\det(I_m + AB) = \det(I_n + BA)$$

Proof. Use the previous theorem with the block matrix

$$\begin{pmatrix} I_n & B \\ -A & I_m \end{pmatrix}.$$

□

4 Probability and Statistics

4.1 Probability via Area

Exercise 4.1. If x, y are randomly chosen in $[0, 1]$, what is the probability that $x \geq 2y$?

Solution: This probability can be determined by a ratio of areas. The total space has area 1. The area of the condition $x \geq 2y$ is as follows:

$$\int_{y=0}^{\frac{1}{2}} \int_{x=2y}^{1} dxdy = \int_{y=0}^{\frac{1}{2}} 1 - 2ydy = \frac{1}{2} - \left(\frac{1}{2}\right)^2 - \frac{1}{4}.$$

Thus, the probability is $\frac{1}{4}$. Note that in this case, it's better to just draw it out in $\mathbb{R}^2$ and use the triangle area formula. □

4.2 Basic Probability

Let Ω be the sample space and let $A, B \subseteq \Omega$ be events. Recall that the complement of A is $A^c = \Omega \setminus A$, and we have

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

We can also talk about unions and intersections of events. This gives us the principle of inclusion and exclusion (PIE):

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Remark 4.2. If A and B are disjoint, then $\mathbb{P}(A \cap B) = 0$, so the above reads

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$

4.3 Conditional Probability

Recall that we also had the notion of conditional probability. If $\mathbb{P}(B) > 0$, then

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Equivalently,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B).$$

Symmetrically, if $\mathbb{P}(A) > 0$, then

$$\mathbb{P}(A \cap B) = \mathbb{P}(B \mid A)\mathbb{P}(A).$$

What is nice is that we can now define the notion of independence of events. We say that events A and B are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

In the case that $\mathbb{P}(B) > 0$, conditional probability tells us that independence is equivalent to

$$\mathbb{P}(A \mid B) = \mathbb{P}(A).$$

Remark 4.3. Note that independence and disjointness are distinct. If A and B are disjoint and both have positive probability, then

$$\mathbb{P}(A \cap B) = 0,$$

while

$$\mathbb{P}(A)\mathbb{P}(B) > 0.$$

These lead to some very useful results:

Theorem 4.4 (Law of Total Probability)**.** *Suppose $B_1, \ldots, B_n$ form a partition of Ω. Then*

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

Theorem 4.5 (Bayes' Theorem)**.** *For events A and B with positive probability,*

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

Combining the two previous, we have Bayes's generalized theorem:

Theorem 4.6. *If $B_1, \ldots, B_n$ form a partition of Ω, then*

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.$$

Exercise 4.7. Suppose you have two cards: one is painted black on both sides and the other is painted black on one side and orange on the other. You select a card at random and view one side. You notice it is black. What is the probability the other side is orange?

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{1}{4}$

(D) $\frac{1}{5}$

(E) 1

Solution: (B). $P(O \cap B) = \frac{1}{4}$ and $P(B) = \frac{3}{4}$. □

4.4 Random Variables

Definition 4.8 (Discrete Random Variable)**.** A discrete random variable X assigns a real number to each outcome in the sample space. Its probability mass function (pmf) is

$$p_X(x) = \mathbb{P}(X = x),$$

where

$$\sum_x p_X(x) = 1.$$

Definition 4.9 (Continuous Random Variable)**.** A continuous random variable X has a probability density function (pdf) f_X satisfying

$$f_X(x) \geq 0$$

and

$$\int_{-\infty}^{\infty} f_X(x)dx = 1.$$

For $a < b$,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x)dx.$$

Remark 4.10. For a continuous random variable,

$$\mathbb{P}(X = a) = 0.$$

We also have the notion of the cumulative distributive function.

Definition 4.11 (Cumulative Distribution Function)**.** The cumulative distribution function of X is

$$F_X(x) = \mathbb{P}(X \leq x).$$

Remark 4.12. For a continuous random variable, this means

$$F_X(x) = \int_{-\infty}^{x} f_X(t)dt,$$

where f_X is the pdf. Therefore, whenever F_X is differentiable,

$$f_X(x) = F_X'(x).$$

4.4.1 Expected Values of Random Variables

If X is a discrete random variable,

$$\boxed{\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x)}.$$

If X is a continuous random variable,

$$\boxed{\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx}.$$

Remark 4.13. More generally, for a function g,

$$\boxed{\mathbb{E}[g(X)] = \sum_x g(x)\mathbb{P}(X = x)}$$

in the discrete case, and

$$\boxed{\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx}$$

in the continuous case.

Note that expectation is linear (for both discrete and continuous cases). That is, for any (real) constants a, b,

$$\boxed{\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]}.$$

Of course, more generally, this says

$$\mathbb{E}\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i \mathbb{E}[X_i],$$

where the a_i are constants.

Remark 4.14. Note that independence is not required for the lienarity of expectation.

4.4.2 Variance

There is also the notion of variance. If we denote $\mu = \mathbb{E}[X]$, then

$$\boxed{\mathrm{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2}.$$

The second equality is justified by linearity. With this, we can define the standard deviation:

$$\boxed{\sqrt{\mathrm{Var}(X)}}.$$

Clearly, for constants a, b,

$$\boxed{\mathrm{Var}(aX + b) = a^2 \,\mathrm{Var}(X)}.$$

Particularly,

$$\mathrm{Var}(X + b) = \mathrm{Var}(X).$$

We can also define covariance of two random variables X and Y:

$$\boxed{\operatorname{Cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]}.$$

For arbitrary X and Y,

$$\boxed{\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y).}$$

If X and Y are independent, then we note that

$$\operatorname{Cov}(X,Y) = 0,$$

so

$$\boxed{\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y).}$$

Remark 4.15. Note that $\operatorname{Cov}(X,Y) = 0$ does not in general imply that X and Y are independent.

4.5 Common Distributions

Example 4.16 (Indicator Random Variable)**.** For an event A, define

$$I_A = \begin{cases} 1, & \text{if } A \text{ occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$\mathbb{E}[I_A] = 1 \cdot \mathbb{P}(A) + 0 \cdot \mathbb{P}(A^c) = \mathbb{P}(A).$$

If

$$X = I_1 + \cdots + I_n$$

counts how many of a collection of events occur, then linaerity implies

$$\mathbb{E}[X] = \mathbb{E}[I_1] + \cdots + \mathbb{E}[I_n].$$

Example 4.17. If

$$X = \begin{cases} 1, & \text{with probability } p, \\ 0, & \text{with probability } 1-p, \end{cases}$$

then

$$\mathbb{E}[X] = p, \qquad \operatorname{Var}(X) = p(1-p).$$

Example 4.18 (Binomial Distribution)**.** If X counts the number of successes in n independent trials, each with success probability p, then $X \sim \operatorname{Bin}(n,p)$. Its pmf is

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Moreover,

$$\mathbb{E}[X] = np, \qquad \operatorname{Var}(X) = np(1-p).$$

Example 4.19 (Geometric Distribution)**.** If X is the number of trials required to obtain the first success, then

$$\mathbb{P}(X = k) = (1-p)^{k-1}p, \qquad k = 1, 2, \ldots$$

and

$$\mathbb{E}[X] = \frac{1}{p}, \qquad \mathrm{Var}(X) = \frac{1-p}{p^2}.$$

The geometric distribution has the interesting property

$$\mathbb{P}(X > m + n \mid X > m) = \mathbb{P}(X > n).$$

Example 4.20 (Poisson Distribution)**.** If $X \sim \mathrm{Poisson}(\lambda)$, then

$$\mathbb{P}(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

and

$$\mathbb{E}[X] = \lambda, \qquad \mathrm{Var}(X) = \lambda.$$

Example 4.21 (Uniform Distribution)**.** If X is uniformly distributed on $[a, b]$, then

$$f_X(x) = \frac{1}{b-a}, \qquad a \le x \le b.$$

Moreover,

$$\mathbb{E}[X] = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}.$$

4.6 Sampling

Definition 4.22. Given the random variables $X_1, X_2, \ldots, X_n$, the *sample mean* $\overline{X}$ is the statistic defined as the arithmetic mean of these variables,

$$\overline{X} = \frac{X_1 + X_2 + \cdots + X_n}{n} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Definition 4.23. Given the random variables $X_1, X_2, \ldots, X_n$, the *sample variance* S^2 is defined by

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2.$$

The **sample standard deviation** is the statistic defined by $S = \sqrt{S^2}$.

Proposition 4.24. *Suppose that the random variables $X_1, X_2, \ldots, X_n$ are independently sampled from a distribution F_X that has mean μ and finite variance σ^2. Then*

1. $\mathbb{E}\left(\overline{X}\right) = \mu$
2. $\mathrm{Var}\left(\overline{X}\right) = \frac{\sigma^2}{n}$
3. $\mathbb{E}\left(S^2\right) = \sigma^2$

Definition 4.25. For real values $x_1, x_2, \ldots, x_n$, the *sample variance* s^2 is defined as

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2, \qquad \text{where} \qquad \overline{x} = \frac{1}{n}\sum_{j=1}^{n} x_j.$$

Definition 4.26. For real observations $(x_1, y_1), \ldots, (x_n, y_n)$, the *sample covariance* is defined as

$$s_{XY} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})$$

Definition 4.27. For a random variable X with mean μ and standard deviation σ, the *z-score* of an observation x is

$$z = \frac{x - \mu}{\sigma}.$$

Remark 4.28. The z-score measures how many standard deviations an observation is from the mean. For instance:

- $z = 0 \to$ at the mean.
- $z = 1 \to$ one standard deviation above the mean.
- $z = -2 \to$ two standard deviations below the mean.

Theorem 4.29 (Linear Regression (Least Squares))**.** *Given data points (x_i, y_i), $i = 1, \ldots, n$, the least-squares regression line of y on x is $\hat{y} = b_0 + b_1 x$, where the coefficients minimize the squared residuals*

$$\sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)^2.$$

Explicitly,

$$b_1 = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^{n}(x_i - \overline{x})^2}$$
$$b_0 = \overline{y} - b_1\overline{x}$$

Remark 4.30. Note that an equivalently, we have

$$b_1 = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)} = r\frac{s_y}{s_x}$$

Here r is the sample Pearson correlation coefficient, and s_x, s_y are the sample standard deviations.

Definition 4.31. The *Pearson correlation coefficient* r is $r = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$. The sample Pearson correlation coefficient is the same, with everything sampled version.

Remark 4.32. Here are some useful properties of the linear regression by least squared residuals.

- The regression line passes through $(\overline{x}, \overline{y})$.
- Residuals are $e_i = y_i - \widehat{y}_i$.

- For regression with an intercept, $\sum_i e_i = 0$.
- The sign of b_1 equals the sign of the correlation coefficient r.
- In simple linear regression with an intercept, $R^2 = r^2$.

Exercise 4.33. A random sample of size n is taken from a population with finite variance. If the sample size is increased from n to $4n$, by what factor does the standard deviation of the sample mean change?

(A) 4

(B) 2

(C) 1

(D) $\frac{1}{2}$

(E) $\frac{1}{4}$

Solution: (D). From the proposition above. □

Exercise 4.34. Suppose that $X_1, \ldots, X_n$ are independent random variables with common variance σ^2. If the sample size is increased by a factor of 9, what happens to the variance of the sample mean?

(A) It becomes 9 times as large.

(B) It becomes 3 times as large.

(C) It remains unchanged.

(D) It becomes $\frac{1}{3}$ as large.

(E) It becomes $\frac{1}{9}$ as large.

Solution: (E). □

Exercise 4.35. A population has mean 20 and standard deviation 12. A random sample of size 36 is selected. What are the mean and standard deviation of the sample mean?

(A) 20, 12

(B) 20, 2

(C) 20, $\frac{1}{2}$

(D) 3, 12

(E) 3, 2

Solution: (B). Note the mean doesn't change. The sampled variance is 4, so the sample sd is 2. □

Exercise 4.36. Suppose that the variance of a population is 100. For a sample of size 25, what is the variance of the sample mean?

(A) 0.16

(B) 0.4

(C) 4

(D) 25

(E) 2500

Solution: (C). □

Exercise 4.37. A population has mean μ and variance σ^2. Which of the following statements must be true for the sample mean $\overline{X}$ of a random sample of size n?

(A) $E[\overline{X}] = \frac{\mu}{n}$

(B) $E[\overline{X}] = \mu$

(C) $\text{Var}(\overline{X}) = \sigma^2$

(D) $\text{SD}(\overline{X}) = \frac{\sigma}{n}$

(E) $\text{Var}(\overline{X}) = \frac{\sigma^2}{\sqrt{n}}$

Solution: (B). □

Exercise 4.38. A population has mean 70 and standard deviation 8. What is the z-score of an observation with value 86?

(A) $\frac{1}{2}$

(B) 1

(C) $\frac{3}{2}$

(D) 2

(E) $\frac{5}{2}$

Solution: (D). $z = \frac{86-70}{8}$. □

Exercise 4.39. Suppose X is normally distributed with mean 100 and standard deviation 15. What is $P(X \leq 115)$, approximately?

(A) 0.1587

(B) 0.3413

(C) 0.5000

(D) 0.6826

(E) 0.8413

Solution: (D). We are +1 standard deviation. Because X is normally distributed, this is (D). □

Exercise 4.40. Suppose $E[X] = 12$ and $\mathrm{Var}(X) = 9$. Let $Y = -2X + 5$. If an observation of X has value 15, what is the corresponding z-score of Y?

(A) -2

(B) -1

(C) 0

(D) 1

(E) 2

Solution: (B). Note that the observation of Y has value -25. Also, $E[Y] = -19$. Further, $\mathrm{Var}(Y) = 36$, so the standard deivation is 6. Therefore,

$$z = \frac{-25 + 19}{6} = -1.$$

□

Exercise 4.41. The four values

$$2, \quad 4, \quad 4, \quad 6$$

constitute an entire population. What is the population standard deviation?

(A) $\sqrt{2}$

(B) 2

(C) $\sqrt{\frac{8}{3}}$

(D) 4

(E) 8

Solution: (A) . We note that the average is $\frac{1}{4}(2 + 4 + 4 + 6) = 4$. Thus, the population standard deviation is

$$\sqrt{\frac{1}{4}(2^2 + 0^2 + 0^2 + 2^2)} = \sqrt{2}.$$

□

Exercise 4.42. The observations

$$1, \quad 3, \quad 5$$

form a sample. What is their sample standard deviation, using the usual denominator $n - 1$?

(A) $\sqrt{\frac{8}{3}}$

(B) $\sqrt{2}$

(C) 2

(D) $\sqrt{8}$

(E) 4

Solution: (C). We calculate the sample mean is 3. Thus, the sample standard deviation is

$$\sqrt{\frac{1}{2}(2^2 + 2^2)} = 2$$

□

Exercise 4.43. Suppose $E[X] = 12$ and $\text{Var}(X) = 9$. Define

$$Z = \frac{X - 12}{3}.$$

What is the standard deviation of Z?

(A) 0

(B) 1

(C) 3

(D) 9

(E) Cannot be determined

Solution: (B). The variance of Z is 1. Thus, the standard deviation is 1. □

Exercise 4.44. Let $X_1, \ldots, X_{36}$ be independent, identically distributed normal random variables, each having mean 80 and standard deviation 12. Let

$$\overline{X} = \frac{1}{36} \sum_{i=1}^{36} X_i.$$

What is $P(\overline{X} \geq 84)$, approximately?

(A) 0.0228

(B) 0.1587

(C) 0.3413

(D) 0.8413

(E) 0.9772

Solution: (A). The z-score is 2. □

Exercise 4.45. Consider the data points $(1, 2), (2, 3), (3, 5)$. What is the least-squares regression line $\widehat{y} = b_0 + b_1 x$?

(A) $\widehat{y} = 1 + \frac{3}{2}x$

(B) $\widehat{y} = \frac{1}{3} + \frac{3}{2}x$

(C) $\widehat{y} = -\frac{1}{3} + \frac{3}{2}x$

(D) $\widehat{y} = \frac{1}{3} + 2x$

(E) $\widehat{y} = \frac{3}{2} + \frac{1}{3}x$

Solution: (B). We have $\overline{x} = 2$ and $\overline{y} = \frac{10}{3}$. Thus,

$$b_1 = \frac{(-1)(-\frac{4}{3}) + (0)(-\frac{1}{3}) + (1)(\frac{5}{3})}{1^2 + 0^2 + 1^2} = \frac{3}{2}.$$

Next,

$$b_0 = \frac{10}{3} - \frac{3}{2} \cdot 2 = \frac{1}{3}.$$

□

Exercise 4.46. For a dataset (x_i, y_i), suppose

$$\overline{x} = 3, \qquad \overline{y} = 8,$$

and

$$\sum_i (x_i - \overline{x})^2 = 20,$$

$$\sum_i (x_i - \overline{x})(y_i - \overline{y}) = -10.$$

What is the least-squares regression line of y on x?

(A) $\widehat{y} = 8 - \frac{1}{2}x$

(B) $\widehat{y} = \frac{13}{2} - \frac{1}{2}x$

(C) $\widehat{y} = \frac{19}{2} + \frac{1}{2}x$

(D) $\widehat{y} = \frac{19}{2} - \frac{1}{2}x$

(E) $\widehat{y} = 8 - 2x$

Exercise 4.47. The sample correlation coefficient between X and Y is $r = -0.6$. Their sample standard deviations are

$$s_X = 5, \qquad s_Y = 10.$$

What is the slope of the least-squares regression line of Y on X?

(A) $-\frac{12}{5}$

(B) $-\frac{6}{5}$

(C) $-\frac{3}{5}$

(D) $\frac{3}{5}$

(E) $\frac{6}{5}$

Exercise 4.48. A least-squares regression line is given by

$$\widehat{y} = 2 + 3x.$$

One observed data point is $(4, 11)$.
What is the residual at this point?

(A) -3

(B) -1

(C) 1

(D) 3

(E) 14

Exercise 4.49. In a simple linear regression model with an intercept, the sample correlation coefficient between X and Y is 0.8.
What percentage of the total variation in Y is explained by the fitted regression?

(A) 16%

(B) 36%

(C) 64%

(D) 80%

(E) 160%

Exercise 4.50. A least-squares regression line of Y on X has slope -2.
The sample means are

$$\overline{x} = 10, \qquad \overline{y} = 20.$$

What is the predicted value of Y when $X = 13$?

(A) 8

(B) 12

(C) 14

(D) 18

(E) 26

Exercise 4.51. A distribution has mean μ and positive standard deviation σ.
An observation of 53 has z-score -1, while an observation of 77 has z-score 2.
What is $\mu + \sigma$?

(A) 61

(B) 65

(C) 69

(D) 73

(E) 77

4.7 Chebyshev and Markov

Just in case.

Theorem 4.52 (Markov's Inequality)**.** *If a random variable X can only take nonnegative values, then*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}(X)}{a}, \qquad \text{for all } a > 0.$$

Theorem 4.53 (Chebyshev's Inequality)**.** *If X is a random variable with mean μ and variance σ^2, then for all $c > 0$,*

$$\mathbb{P}(|X - \mu| \geq c) \leq \frac{\sigma^2}{c^2}.$$

5 ODEs

Here, we have some basic ways to solve ODEs. We first go through some basic linear ODEs, which I wish to be familiar with and make it just an easy computational exercise.

Example 5.1 (First-order linear ODE; Integrating Factors)**.** Consider

$$y' + p(x)y = q(x).$$

Define the integrating factor

$$I(x) = e^{\int p(x)dx}.$$

This gives

$$y(x) = e^{-\int p(x)dx}\left(\int e^{\int p(x)dx} q(x)dx + C\right) = \frac{1}{I(x)}\left(\int I(x)q(x)dx + C\right).$$

Note that this comes from multiplying the ODE by $I(x)$ and integrating the differentiated chain rule.

Exercise 5.2. Let $y(t)$ be the solution to the differential equation

$$y' + 2ty = e^{-t^2}\sin t$$

such that $y(0) = 0$. What is $y(\pi)$?

Solution: This is first order linear. The integrating factor is e^{t^2}. Thus,

$$y(t) = e^{-t^2}\left(\int e^{t^2} \cdot e^{-t^2}\sin t dt + C\right) = e^{-t^2}(-\cos(t) + C).$$

$y(0) = 0$, so $C = 1$. Thus, $y(\pi) = 2e^{-\pi^2}$. □

Example 5.3 (Second-order linear ODE with Constant Coefficients)**.** Consider the homogeneous equation

$$a_2 y'' + a_1 y' + a_0 y = 0.$$

We consider $y = e^{\lambda x}$. Substitution gives the characteristic equation

$$a_2\lambda^2 + a_1\lambda + a_0 = 0.$$

If the characteristic roots are distinct $\lambda_1 \neq \lambda_2$, then

$$y(x) = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}.$$

If the roots coincide, say to λ, then

$$y(x) = c_1 e^{\lambda x} + c_2 x e^{\lambda x}.$$

If the roots are complex, so $\lambda = \alpha \pm i\beta$, then the real-valued general solution is

$$y(x) = e^{\alpha x} \left(c_1 \cos(\beta x) + c_2 \sin(\beta x) \right).$$

Example 5.4 (Euler-Cauchy Equation)**.** A kth-order Euler-Cauchy equation has the form

$$\beta_k x^k \frac{d^k y}{dx^k} + \beta_{k-1} x^{k-1} \frac{d^{k-1} y}{dx^{k-1}} + \cdots + \beta_1 x \frac{dy}{dx} + \beta_0 y = f(x),$$

where $\beta_0, \ldots, \beta_k$ are constants. The key feature is that the coefficient of the jth derivative is proportional to x^j. For this, use $x = e^z$. This makes

$$x^j \frac{d^j y}{dx^j}$$

to be expressed as a linear combination of

$$\frac{d^j y}{dz^j}, \frac{d^{j-1} y}{dz^{j-1}}, \ldots, \frac{dy}{dz},$$

with constant coefficients, so the substitution $x = e^z$ transforms the Euler–Cauchy equation into a linear ODE with constant coefficients in the variable z. After solving the resulting constant-coefficient ODE, substitute $z = \ln x$ to write the solution in terms of x.

We see how this plays out below.

Exercise 5.5. Find a solution to the ODE

$$x^2 y'' + 3xy' + y = x^3.$$

Solution: We use $x = e^z$. We calculate the differentials first.

$$\frac{dy}{dx} = \frac{dy}{dz}\frac{dz}{dx} = \frac{1}{x}\frac{dy}{dz}$$

Next, we differentiate this to find $\frac{d^2y}{dx^2}$.

$$\frac{d^2 y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dz}\right) = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x^2}\frac{d^2 y}{dz^2}.$$

Here, we used the chain rule:

$$\frac{d}{dx}\left(\frac{dy}{dz}\right) = \frac{d}{dz}\left(\frac{dy}{dz}\right)\frac{dz}{dx} = \frac{1}{x^2}\frac{d^2 y}{dz^2}$$

We now substitute back in, which gives the following ODE:

$$\frac{d^2 y}{dz^2} + 2\frac{dy}{dz} + y = e^{3z}.$$

This is in our realm now. We attain the following:

$$c_1 e^{-z} + c_2 z e^{-z} + \frac{1}{16} e^{3z}$$

We just substitute back $z = \ln(x)$ to get the desired form for $y(x)$. □

6 Complex Analysis

Here, we don't include proofs.

6.1 Basics of Holomorphic Functions

Theorem 6.1. *Let $f\colon \Omega \to \mathbb{C}$, where $\Omega \subseteq \mathbb{C}$ is open. The following statements are equivalent:*

- *f is holomorphic at $a \in \Omega$.*
- *f has a convergent power series $\sum_{n\geq 0} c_n(z-a)^n$ in an open disc centered at $a \in \Omega$.*

Definition 6.2. A complex-valued function is entire if it is holomorphic over $\mathbb{C}$.

Definition 6.3. Consider $f(z) = u + iv$ be a complex-valued function. Writing $z = x + iy$, the *Cauchy-Riemann equations* refer to the set of differential equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Theorem 6.4. *Consider $f = u + iv\colon \Omega \to \mathbb{C}$, defomed on an open set $\Omega \subseteq \mathbb{C}$. If u and v are continuously differentiable and satisfy the Cauchy-Riemann equations on Ω, then f is holomorphic on Ω and $f'(z) = \frac{\partial f(z)}{\partial z}$.*

Example 6.5. $f(z) = \bar{z}$ is not holomorphic on $\mathbb{C}$. $\bar{z} = x - iy$, so $u'_x = 1 \neq -1 = v'_y$.

Exercise 6.6. Suppose that $f(x+iy) = u(x,y) + iv(x,y)$. If $u(x,y) = x^2 - y^2$ and $v(1,1) = 2$. Find $v(4,1)$.

Solution: The key is the FTC and the Cuachy-Riemann equations, because f is clearly holomorphic.

$$v(4,1) - v(1,1) = \int_1^4 v'_x(x,1)dx = \int_1^4 -u'_y(x,1)dx = \int_1^4 2dx = 6.$$

Therefore, $v(4,1) = 8$. Note the use of the Cauchy Riemann equations. □

6.2 Cauchy's Integral Formula

Theorem 6.7 (Cauchy)**.** *Let f be holomorphic in an open set Ω, then f has infinitely many complex derivatives in Ω. Moreover, for simple, closed, piecewise-smooth curve $\gamma \subset \Omega$ and any z lying inside γ we have*

$$\frac{d^n f(z)}{dz^n} = \frac{n!}{2\pi i} \oint_\gamma \frac{f(\eta)}{(\eta - z)^{n+1}} d\eta.$$

Exercise 6.8. Evaluate the following integral, for $\gamma = \{z\colon |z| = 2\}$:

$$\frac{1}{2\pi i} \oint_\gamma \frac{e^z}{(z+i)(z-i)} dz$$

Solution: We have as follows:

$$\frac{1}{2\pi i}\oint_\gamma \frac{e^z}{(z+i)(z-i)}dz = \frac{1}{2\pi i}\frac{1}{2i}\oint_{\{z:\,|z|=2\}}\left(\frac{e^z}{z+i}-\frac{e^z}{z-i}\right)dz = \frac{1}{2i}(e^i-e^{-i}) = \sin 1.$$

□

Exercise 6.9. Let $C = \{z = 1 + e^{i\theta} : \theta \in [0, 2\pi]\}$. Calculate

$$\frac{1}{2\pi i}\int_C \left(\frac{\sin z}{z-1}\right)^2 dz.$$

Solution: Use Cauchy's integral formula with $n = 1$. We want the first derivative of $\sin^2(z)$, and plug in 1 into it. We have $2\sin(1)\cos(1) = \sin(2)$. □

Corollary 6.10 (Cauchy-Goursat for a Disc)**.** *If f is holomorphic in a disc,*

$$\oint_\gamma f(z)dz = 0$$

for any closed curve γ in that disc.

Theorem 6.11 (Morera)**.** *Suppose f is a continuous function in the open disc D such that for any triangle T contained in D*

$$\int_T f(z)dz = 0,$$

then f is holomorphic.

Corollary 6.12 (Liouville)**.** *A bounded entire function is constant.*

6.3 Residue Theory

Definition 6.13. Let $f(z) = \sum_{-\infty}^{\infty} a_n(z-z_0)^n$, for $0 < |z - z_0| < R$ be the Laurent series for f at z_0. The *residue* of f at z_0 is $\mathrm{Res}[f, z_0] = a_{-1}$.

Theorem 6.14. *Let f be holomorphic function inside and on a simple, closed, piecewise-smooth curve γ except at the singularities $z_1, \ldots, z_n$ in its interior. Then,*

$$\oint_\gamma f(z)dz = 2\pi i\sum_{j=1}^{n}\mathrm{Res}[f, z_j]$$

Remark 6.15. Note that we can find the residue of f at a pole z_0 of order m as follows:

$$\mathrm{Res}[f, z_0] = \lim_{z\to z_0}\frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}((z-z_0)^m f(z)).$$

Exercise 6.16. Evaluate $\oint_\gamma \frac{1}{z^5-z^3}dz$, where $\gamma = \{z : |z| = \frac{1}{2}\}$.

Solution: Let $f(z) = \frac{1}{z^5-z^3}$. The only relevant singularity in γ is $z = 0$. Thus, the integral is equal to $\mathrm{Res}[f, 0]$, which is a pole of order 3:

$$\begin{aligned}\lim_{z\to 0}\frac{1}{2!}\frac{d^2}{dz^2}\left(z^3\cdot f(z)\right) &= \frac{1}{2}\lim_{z\to 0}\left(\frac{-2z}{(z^2-1)^2}\right)' \\ &= \frac{1}{2}\lim_{z\to 0}\frac{-2(z^2-1)^2+4z(z^2-1)\cdot 2z}{(z^2-1)^4} \\ &= -1\end{aligned}$$

Thus, the integral evaluates to $-2\pi i$. □

6.4 More Results and Problems

Theorem 6.17 (Maximum Modulus Principle)**.** *Let $\Omega \subseteq \mathbb{C}$ be a nonempty, connected, open subset of $\mathbb{C}$, and let $f\colon \Omega \to \mathbb{C}$ be holomorphic. If there exists $z_0 \in \Omega$ such that $|f|$ attains a local maximum at z_0, then f is constant on Ω. If Ω is actually bounded, f is actually defined on $\overline{\Omega}$ such that f is continuous on $\overline{\Omega}$ and holomorphic on Ω, then*

$$\max_{z\in\overline{\Omega}} |f(z)| = \max_{z\in\partial\Omega} |f(z)|.$$

In otherwords, it attains its maximum on the boundary.

Remark 6.18. In the setting above, if f is nonconstant on Ω, then in particular $|f(z)| < \max_{\omega\in\partial\Omega} |f(w)|$, for every $z \in \Omega$.

Remark 6.19. If in addition f is nonconstant and has no zeroes, we can say a similar result for minimas.

Theorem 6.20 (Polynomial Growth of Entire Functions)**.** *Let f be entire and suppose there exist constants $C > 0, \alpha > 0$ such that*

$$|f(z)| \leq C(1 + |z|^\alpha),$$

for all $z \in \Omega$. Then f is a polynomial of degree at most $\lfloor \alpha \rfloor$. In particular, if $0 \leq \alpha < 1$, then f is constant.

Theorem 6.21 (Open Mapping Theorem)**.** *Let $\Omega \subseteq \mathbb{C}$ be a nonempty, connected, open subset of $\mathbb{C}$, and $f : \Omega \to \mathbb{C}$ be a nonconstant holomorphic function. Then f is an open mapping.*

Theorem 6.22 (Argument Principle)**.** *Let $\Omega \subseteq \mathbb{C}$ be a nonempty, connected, open subset of $\mathbb{C}$. Let Γ be a positively oriented, piecewise C^1, simple closed curve in $\mathbb{C}$, and let D denote its bounded interior. Assume that $\overline{D} \subseteq \Omega$. Further, let $f\colon \Omega \to \widehat{\mathbb{C}}$ be a meromorphic function that is not identically zero. Suppose f has neither zeros nor poles on Γ. Let N denote the number of zeros of f in D, counted with multiplicity, and let P denote the number of poles of f in D, counted with order. Then*

$$\frac{1}{2\pi i} \oint_\Gamma \frac{f'(z)}{f(z)}\, dz = N - P.$$

Corollary 6.23 (Rouché's Theorem)**.** *Continue with the hypothesis above. If $f, g\colon \Omega \to \mathbb{C}$ are holomorphic such that $|g(z)| < |f(z)|$ for all $z \in \Gamma$, then f and $f + g$ have the same number of zeros in D, counted with multiplicity.*

Exercise 6.24. Let f be holomorphic on the open unit disk and continuous on its closure. Suppose $|f(z)| \leq 2$ for $|z| = 1$ and $f(0) = 2i$. Which of the following must be true?

(A) $f(z) = 0$ for all z.

(B) $f(z) = 2iz$ for all z.

(C) $f(z) = 2i$ for all z.

(D) f has exactly one zero in the unit disk.

(E) $|f(z)| < 2$ for every $|z| < 1$.

Solution: (C). Maximum-modulus principle. □

Exercise 6.25. What is

$$\max_{|z|=1} |z^4 - 3z^2|?$$

(A) 1

(B) 2

(C) 3

(D) 4

(E) 5

Solution: (D). Factor out the z^2, but maximum of $|z^2 - 3|$ is 4; take $z = -i$. □

Exercise 6.26. Let $f : \mathbb{C} \to \mathbb{C}$ be entire and suppose

$$|f(z)| \leq 2 + |z|^{3/2}$$

for every $z \in \mathbb{C}$. What is the largest possible degree of f?

(A) 0

(B) 1

(C) 2

(D) 3

(E) f need not be a polynomial.

Solution: (B). Use polynomial growth result of entire functions. □

Exercise 6.27. Suppose f is an entire function satisfying

$$\operatorname{Re}(f(z)) \leq 5$$

for every $z \in \mathbb{C}$. Which of the following must be true?

(A) $f(0)$ is real.

(B) f is identically zero.

(C) f is constant.

(D) f has at least one zero.

(E) f is real-valued.

Solution: (C). This implies e^f is bounded, which is entire, so e^f is constant. This implies f is constant, because $e^f f' = 0$ (by taking derivative), implying that $f' = 0$. □

Exercise 6.28. Let $D = \{z \in \mathbb{C} : |z| < 1\}$ and let $f : D \to \mathbb{C}$ be a nonconstant holomorphic function. Which of the following sets could be the image $f(D)$?

(A) $[0, 1]$

(B) $\{z : |z| = 1\}$

(C) $\{z : |z| < 1\}$

(D) $\mathbb{R}$

(E) $\{z : |z| \leq 1\}$

Solution: (C). Open mapping theorem (or just $f = z$). □

Exercise 6.29. Let

$$f(z) = \frac{z^2 + 1}{z(z - 2)}$$

and let C be the positively oriented circle $|z| = \frac{3}{2}$. Evaluate

$$\frac{1}{2\pi i} \oint_C \frac{f'(z)}{f(z)}\, dz.$$

(A) -1

(B) 0

(C) 1

(D) 2

(E) 3

Solution: (B). Use argument principle; there are two zeroes and two poles. Thus, $2 - 2$. □

Exercise 6.30. How many zeros, counted with multiplicity, does

$$p(z) = z^6 + 4z^2 + 1$$

have in the open unit disk $|z| < 1$?

(A) 0

(B) 1

(C) 2

(D) 4

(E) 6

Solution: (C). Let $f = 4z^2$ and $g = z^6 + 1$. Then, $|g| < |f|$, so p and f have the same number of zeroes. This is 2 (counted with multiplicity of course). □

Exercise 6.31. Let C be the positively oriented unit circle $|z| = 1$.

What is

$$\oint_C z^2 e^{1/z}\, dz?$$

(A) 0

(B) $\frac{\pi i}{6}$

(C) $\frac{\pi i}{3}$

(D) πi

(E) $2\pi i$

Solution: (C). Use residue theory. There's a pole at 0 from the $e^{\frac{1}{z}}$. The third term from the expansion of $e^{\frac{1}{z}}$ matters, which has coefficient $\frac{1}{3!} = \frac{1}{6}$. We need to multiply by $2\pi i$, so we get (C). □

7 Analysis and Point-Set Topology

We provide no proofs here.

7.1 Real Analysis

7.1.1 Continuity (on metric spaces)

Recall the following notions of continuity.

Definition 7.1. Let (X, d_X) and (Y, d_Y) be metric spaces, and $f\colon X \to Y$ be a map. Then,

- We say that f is *continuous* at $x \in X$, if for every $\epsilon > 0$ there is $\delta > 0$ such that for every $x' \in X$ satisfying $d_X(x, x') < \delta$, we have

  $$d_Y(f(x), f(x')) < \epsilon.$$

  f is *continuous* if f is continuous at every $x \in X$.

- We say that $f\colon X \to Y$ is *uniformly continuous*, if f is continuous at every $x \in X$ and $\delta = \delta(\epsilon)$ doesn't depend on x. Let (X, d_X) and (Y, d_Y) be metric spaces, and $f\colon X \to Y$.

- We say that f is *Lipschitz*, if there is a constant $M > 0$ such that for all $x_1, x_2 \in X$, we have

  $$d_Y(f(x_1), f(x_2)) \le M \cdot d_X(x_1, x_2).$$

- We say that f is *bi-Lipschitz*, if there are constants $M_1 > 0, M_2 > 0$ such that for all $x_1, x_2 \in X$, we have

  $$M_2 \cdot d_X(x_1, x_2) \le d_Y(f(x_1), f(x_2)) \le M_1 \cdot d_X(x_1, x_2).$$

- We say that f is an *isometry*, or *distance preserving*, if for every $x_1, x_2 \in X$, we have

$$d_Y(f(x_1), f(x_2)) = d_X(x_1, x_2).$$

Remark 7.2. It is immediate that we have the following implications:

$$\text{biLipschitz} \implies \text{Lipschitz} \implies \text{Uniform Continuity} \implies \text{Continuity}$$

Theorem 7.3. *A continuous function $f\colon S \to \mathbb{R}$, where $S \subseteq \mathbb{R}^n$ is compact, is uniformly continuous.*

Proposition 7.4. *BiLipschitz maps are injective.*

Proposition 7.5. *A surjective biLipschitz map is a homeomorphism.*

Theorem 7.6. *Let X be a complete metric space. If $A \subseteq X$ is closed, then A is also complete.*

7.1.2 Convergence of Functions, DCT

Recall the following different notions of convergence of functions:

Definition 7.7. A sequence of functions $f_n\colon [a,b] \to \mathbb{R}$ *converges point-wise* to $f\colon [a,b] \to \mathbb{R}$, if for every $x \in [a,b]$, the sequence of real numbers $f_n(x)$ converges to $f(x)$. That is, for every $x \in [a,b]$ and every $\epsilon > 0$, there exists $N_{x,\epsilon} \in \mathbb{N}$ such that for all $n \geq N_{x,\epsilon}$ we have $|f_n(x) - f(x)| < \epsilon$.

Definition 7.8. A sequence of functions $f_n\colon [a,b] \to \mathbb{R}$ *converges uniformly* to $f\colon [a,b] \to \mathbb{R}$, if for all $\epsilon > 0$, there exists $N_\epsilon > 0$ such that for all $n \geq N_\epsilon$ and for all $x \in [a,b]$ we have $|f_n(x) - f(x)| < \epsilon$. This is equivalent to

$$\sup_{x \in [a,b]} |f_n(x) - f(x)| \to 0, \text{ as } n \to \infty.$$

Remark 7.9. Note that it is clear that uniform convergence implies point-wise convergence.

Remark 7.10. If we have uniform convergence, we can interchange limits and integrals, so long as the domain of integration is finite. Point-wise convergence is generally not enough. For point-wise convergence, we can genrerally use DCT.

Theorem 7.11 (Dominated Convergence Theorem)**.** *Let $g\colon X \to [0,\infty]$ be integrable, and $f, f_n\colon X \to [-\infty,\infty]$, $n \geq 1$ be measurable functions such that*

$$|f_n(x)| \leq g(x), \forall x \in X \text{ and } f_n \overset{n\to\infty}{\to} f \mu\text{-a.e.}$$

Then,

$$\left| \int f_n d\mu - \int f d\mu \right| \leq \int |f_n - f| d\mu \overset{n\to\infty}{\to} 0.$$

Remark 7.12. $f_n \to f$ μ-a.e. means that f_n tends to f pointwise except for a measure zero set.

Proposition 7.13. *If a sequence of continuous functions (f_n) converge to f uniformly, then f is continuous.*

7.2 Supremum, Infimum, Limsup, and Liminf

Recall that for a real sequence (a_n), we can define

$$\limsup_{n\to\infty} a_n := \lim_{n\to\infty} \sup_{k\geq n} a_k$$

and

$$\liminf_{n\to\infty} a_n := \lim_{n\to\infty} \inf_{k\geq n} a_k.$$

We recall some useful properties regarding lim sup and lim inf.

Proposition 7.14 (Sequences of Reals)**.** *Let (a_n) be a real sequence. Then, the following hold:*

- $\liminf a_n \leq \limsup a_n$
- *a_n converges to L if and only if* $\liminf a_n = \limsup a_n = L$.
- $\limsup(-a_n) = -\liminf a_n$ *and* $\liminf(-a_n) = -\limsup a_n$.
- *If a_n is bounded, then*
  - $\limsup a_n$ *is the largest subsequential limit.*
  - $\liminf a_n$ *is the smallest subsequential limit.*

Proposition 7.15 (Sequences of Sets)**.** *Let (A_n) be a sequence of sets. Then,*

$$\limsup_{n\to\infty} A_n = \bigcap_{N=1}^{\infty} \bigcup_{n\geq N} A_n,$$

which is the set of points belonging to infinitely many A_n. We also have

$$\liminf_{n\to\infty} A_n = \bigcup_{N=1}^{\infty} \bigcap_{n\geq N} A_n,$$

which is the set of points belonding to all but finitely many A_n.

7.3 Topology

7.3.1 Open and Closed Sets

Lemma 7.16. *Let Y be a subspace of X. If U is open in Y and Y is open in X, then U is open in X.*

Remark 7.17. Note that if Y is not open in X, then there is no reason U should be open in X. As a counter example, consider $X = \mathbb{R}$ with the standard topology, $Y = [0, 1]$ (with the subpace topology), and $U = [0, \frac{1}{2})$. $U = (-\frac{1}{2}, \frac{1}{2}) \cap Y$, so is open in Y. However, clearly, U is not open in X, since opens in X are unions of open intervals.

Example 7.18. In the discrete topology on a set X, every subset of X is open. It follows that every set is closedE as well. It follows that every function from a discrete space is continuous.

Example 7.19. In any topology space X, $\emptyset$ and X are both closed and open.

A nice way to test if a set is closed in the subspace topology of some larger set is from the following:

Theorem 7.20. *Let $Y \subseteq X$. A is closed in Y if and only if it equals the intersection of a closed set of X with Y.*

It also follows then, from the lemma earlier, that:

Corollary 7.21. *Let $Y \subseteq X$. If A is closed in Y and Y is closed in X, then A is closed in X.*

There was also the notion of a closure, which is the intersection of all closed sets containing a set. Note that the resulting set is closed because arbitrary intersections of closed sets are closed.

Theorem 7.22. *Let $A \subseteq X$ be (only) a subset.*

- *$x \in \overline{A}$ if and only if every open set U containing x intersects A.*
- *If the topology of X is given by a basis, then $x \in \overline{A}$ if and only if every basis element containing x intersects A.*

Exercise 7.23. Let $X = \{n \in \mathbb{Z} \colon n \geq 2\}$ be the topoological space X with the topology generated by sets of the form $U_k := \{n \in X \colon n \mid k\}$, for $k \geq 2$, so that open sets of X are the empty set and arbitrary unions of the sets U_k. For any $n \in X$, determine the closure of the one point set $\{n\}$ in X.

Solution: Note that $m \in \overline{\{n\}}$ if and only if every open neighborhood of m meets $\{n\}$. In other words, we mean that if $m \in U_k$, then $\{n\} \cap U_k \neq \emptyset$, so $n \in U_k$. That is to say, if $m \mid k$, then we need $n \mid k$. Therefore, $m = ns$. Therefore, the closure is of the form $\{sn \colon s \in \mathbb{Z}_{>0}\}$. □

Definition 7.24. If A is a subset of the topological space X, then $x \in X$ is called a *limit point* of A if every neighborhood of x intersects A in some point other than x itself.

Remark 7.25. In other words, x is a limit point of A if x is in the closure of $A \setminus \{x\}$.

Theorem 7.26. *The closure of a set A in a topological space X is precisely the set A union with the set of limit points of A in X.*

Remark 7.27. A useful thing to note is that $\overline{A} = A \cup \{\text{limit points}\}$ and $\partial A = \overline{A} \setminus A^\circ$.

7.3.2 Hausdorff

Recall the notion of Hausdorff:

Definition 7.28. A topological space X is *Hausdorff* if for each pair of distinct points $x, y \in X$, there are neighborhoods of x, y, respectively, which are disjoint.

Theorem 7.29. *Every finite point set in a Hausdorff space is closed.*

Theorem 7.30. *If X is a Hausdorff space, then a sequence of points of X converges to at most one point of X.*

Proposition 7.31. *Products and subspaces of Hausdorff spaces are Hausdorff.*

Proposition 7.32. *Metric spaces are Hausdorff.*

7.4 Connectedness

Definition 7.33. A topological space X is disconnected if there exists a pair of open sets U and V in X such that $U \cap V = \emptyset$, $U \cup V = X$, and $U, V \neq \emptyset$. X is connected if it is not disconnected. That is, for all nonempty opens such that $U \cap V = \emptyset$ and $U \cup V = X$, $U = \emptyset$ or $V = \emptyset$.

Proposition 7.34. *The following are equivalent:*

- *X is connected.*
- *The only subsets of X which are both open and closed are X and $\emptyset$.*
- *X is nonempty and it can't be written as $X = V_1 \cup V_2$, of two disjoint nonempty closed subset.*

Definition 7.35. A topological space X is *path connected* if there exists a path between any two points in X.

Theorem 7.36. *Every path connected space is connected.*

Remark 7.37. If the space is locally path connected, then the converse to the above holds.

Proposition 7.38. *The closure of a connected set is connected.*

Corollary 7.39. *Let X be a connected set, and $\overline{X}$ be its closure. If $X \subseteq A \subseteq \overline{X}$, then A is also connected.*

Example 7.40. The Topologist's sine curve is a counter example to the converse statement. That is, it is connected, but not path connected nor locally path connected. Here, we are talking about the set $\{(x, \sin(\frac{1}{x})) \colon x \in (0, 1]\} \cup \{(0, 0)\}$. If we consider the set

$$\{(x, \sin(\frac{1}{x})) \colon x \in (0, 1]\} \cup \{(0, y) \colon y \in [-1, 1]\},$$

then we have taken the closure. It remains connected.

7.5 Compactness

Definition 7.41. A topological space X is *sequentially compact*, if every sequence in X has a convergent subsequence in X. That is, for every sequence $(x_n)_{n \geq 1}$ in X, there is a sub-sequence $(x_{n_k})_{k \geq 1}$ and a point $x \in X$ such that $x_{n_k} \to x$ as $k \to \infty$.

Theorem 7.42. *If X is a metric space, then sequentially compactness is equivalent to compactness.*

Corollary 7.43. *Every compact metric space is complete.*

8 Group Theory

8.1 Group Actions and Orbit-Stabilizers

Let G be a group acting on a set X.

Definition 8.1. The *stabilizer* of $x \in X$ of a group action G, G_x, is the set $\{g \in G \colon gx = x\}$. We denote the *orbit* of which contains x as $\mathcal{O}_x$.

Proposition 8.2. *G_x is a subgroup of G.*

Proof. Clear. □

Proposition 8.3. *$\mathcal{O}_x \cong G/G_x$. If X is finite, this says $|\mathcal{O}_x| = [G : G_x]$.*

Proof. We define a bijective function between the cosets G/G_x and $\mathcal{O}_x$. We define $\phi \colon \mathcal{O}_x \to G/G_x$ as $y \mapsto gG_x$, where $gx = y$. Note that such a $g \in G$ exists because $y \in \mathcal{O}_x$. This is also well-defined in the sense that $\phi(y)$ is not dependent on g (easy check). The claim is that this is bijective, but this is, again, a direct check. □

Proposition 8.4 (The Class Equation)**.** *Let X be finite. Let $X_G = \{x \in X \colon gx = x, \forall g \in G\}$ be the fixed points of X. Then,*

$$|X| = |X_G| + \sum |\mathcal{O}_{x_i}|,$$

where the x_i are the representatives of each orbit.

Proof. This is clear. Elements not fixed by G are in some orbit, while those fixed by G are in X_G. □

Corollary 8.5. *Let G be a finite group and act on itself via conjugation. Then,*

$$|G| = |Z(G)| + \sum n_i,$$

where each n_i denotes the cardinality of each nontrivial conjugacy class. Further, n_i divides $|G|$ for all i.

Proof. The summation follows from the theorem above. The size of each orbit is also equal to the size of the cosets, which divide $|G|$ by Lagrange. □

Theorem 8.6. *Let $|G| = p^n$, for p prime. Then, G has nontrivial center.*

Proof. We let G act on itself via conjugation. Then, the previous proposition says

$$|G| = |Z(G)| + \sum n_i,$$

where each n_i is the size of the conjugacy classes. Then, $p \mid n_i$ and solution $p \mid |Z(G)|$. The identity is always in the center, so $|Z(G)| \geq 1$. Therefore, $|Z(G)| \geq p$ and so there is some nontrivial element in the center. □

Corollary 8.7. *Every group of order p^2 is abelian.*

Proof. By the previous theorem, if the center has order p^2, then clearly the group is abelian. Suppose not then. It follows that $Z(G)$ and $G/Z(G)$ both have order p, which are cyclic. Let $aZ(G)$ generate $G/Z(G)$. Then, we can write $gZ(G), hZ(G)$ as $g = a^m x, h = a^m y$ for some $x, y \in Z(G)$ and m, n. We just check for commuting of h and g, but this follow directly. □

Notation 8.8. We write $x \sim y$ if x and y belond to the same orbit.

Proposition 8.9. *If $x \sim y$, for $x, y \in X$, then G_x and G_y are iomorphic (as groups).*

Proof. Let $y = gx$, for some $g \in G$. Then, the map $\phi\colon G_x \to G_y$ defined as $a \mapsto gag^{-1}$ defines a bijective group homomorphism (thus, a group isomorphim). □

Theorem 8.10 (Burnide's Counting Lemma)**.** *Let k denote the number of orbits of X. We have as follows:*

$$k = \frac{1}{|G|} \cdot \sum_{g \in G} |X_g|.$$

Proof. The cardinality of elements of G which fix x is the sum $\sum_{g \in G} |X_g|$. Dually, viewed in terms of the stabilizer subgroups, this number is equal to the sum $\sum_{x \in X} |G_x|$. We also note that

$$\sum_{y \in \mathcal{O}_x} |G_y| = |\mathcal{O}_x| \cdot |G_x| = |G|.$$

Note that the first part follows by the previous lemma and the second equality follows by Lagrange. Now, we sum over the k distinct orbits to deduce that

$$\sum_{g \in G} |X_g| = \sum_{x \in X} X = k \cdot |G|.$$

□

8.2 Sylow's Theorems

Definition 8.11. A group G is a *p-group* if every element in G has as its order a power of p, for a prime p. A subgroup of a group is a *p-subgroup* if it is a p-group.

Recall Cauchy's theorem.

Theorem 8.12 (Cauchy)**.** *Let G be a finite group be a finite group with p dividing the order $|G|$. Then, G contains a subgroup of order p.*

Proof Sketch. Induct on the order of the group. The base case is immediate. Assume now that the result holds for all groups of order less than n (and p must divide n by hypothesis). We use the class equation and case work.

If the order of some conjugacy class is divisible by p, then we are done by the inductive hypotehsis.

Now, suppose that the order of no conjugacy class has order divisible by p. In this case, $[G : C_i]$ are all diviible by p (where C_i are the conjugacy classes), and the class equation implies that p divides the center. Now, $Z(G)$ is a finite abelian group, so the structure theorem implies that there is an element of order p in $Z(G)$. □

Corollary 8.13. *If G is finite, then G is a p-group if and only if $|G| = p^n$.*

The point of the Sylow theorem is to provide a partial converse to Cauchy's theorem.

Theorem 8.14 (Sylow I)**.** *Let G be finite and p be a prime such that p^r divides $|G|$. Then, G contains a subgroup of order p^r.*

Proof. Theorem 15.3 in [1]. □

Definition 8.15. A Sylow p-subgroup of a group G is a maximal p-subgroup of G. That is, if $p \mid |G|$, it is a subgroup with order p^a, where no higher order of p divides $|G|$.

Theorem 8.16 (Sylow II)**.** *Let G be a finite group and p be a prime dividing $|G|$. Then all Sylow p-subgroups of G are conjugate to each other.*

Proof. Theorem 15.6 in [1]. □

Theorem 8.17 (Sylow III)**.** *Let G be a finite group and let p be a prime dividing the order of G. Then the number of Sylow p-subgroups is congruent to* $1 \pmod{p}$ *and divides $|G|$.*

Definition 8.18. A *simple group* is one with no proper nontrivial normal subgroups.

8.3 Miscellaneous

Theorem 8.19 (Internal Direct Product Theorem)**.** *Let G be a group. If we have $H_1, \ldots, H_r \trianglelefteq G$ such that $G = H_1 H_2 \ldots H_r$ and $H_i \cap \prod_{j \neq i} H_j = \{e\}$, then $G \cong H_1 \times \cdots \times H_r$.*

9 Miscellanious

9.1 Logic

We let P and Q be propositions.

- $P \Rightarrow Q \iff \neg P$ or Q
- $\neg(P \Rightarrow Q) \iff P$ and $\neg Q$
- Contrapositive: $P \Rightarrow Q \iff \neg Q \Rightarrow \neg P$

9.2 Fundamental Groups

Some random fundamental groups to remember.

- $\pi_1(\mathbb{R}^n) = \{e\}$
- $\pi_1((0,1)) = \{e\}$
- $\pi_1([0,1]) = \{e\}$
- $\pi_1(D^n) = \{e\}$
- $\pi_1(\mathbb{S}^1) = \mathbb{Z}$
- $\pi_1(\mathbb{S}^n) = \{e\}$ for $n \geq 2$
- $\pi_1(\mathbb{R}^2 \setminus \{0\}) = \mathbb{Z}$
- $\pi_1(\mathbb{R}^n \setminus \{0\}) = \{e\}$ for $n \geq 3$

- $\pi_1(S^1 \times S^1) = \mathbb{Z}^2$
- $\pi_1(\mathbb{R}^2 \setminus \{p_1, \dots, p_k\}) = F_k$
- $\pi_1(\mathbb{RP}^n) = \mathbb{Z}/2\mathbb{Z}$ for $n \geq 2$
- $\pi_1(\text{Klein bottle}) = \langle a, b \mid aba^{-1} = b^{-1} \rangle$
- $\pi_1\left(\bigvee_{i=1}^k S^1\right) = F_k$, the free group on k generators.

9.3 Some Specific Rings

Proposition 9.1. *Let R be an idempotent commutative ring, i.e. $a^2 = a$, for all $a \in R$. Then, the following hold:*

- $\text{char}(R) = 2$.
- *R is commutative.*
- *If R is an integral domain, then $R \cong \mathbb{F}_2$.*

Example 9.2. Let R be Noetherian. $R[x_1, x_2, \dots]$ is not Noetherian (as a ring).

10 Obscure Questions in MVC Format

Exercise 10.1. Let $X = \mathbb{R}^2 \setminus \{(0,0)\}$. What is $\pi_1(X)$?

(A) 0

(B) $\mathbb{Z}$

(C) $\mathbb{Z}/2\mathbb{Z}$

(D) $\mathbb{Z}^2$

(E) A free group on two generators

Solution: (B). $\mathbb{R}^2 \setminus \{(0,0)\}$ is homeomorphic to $\mathbb{S}^1$, which has first fundamental group $\mathbb{Z}$. □

Exercise 10.2. Let G be a group of order 56. Which of the following are the possible numbers of Sylow 7-subgroups?

(A) 1 only

(B) 7 only

(C) 1 or 7

(D) 1 or 8

(E) 7 or 8

Solution: (D). Use Sylow III. □

Exercise 10.3. What is $\pi_1(\mathbb{S}^1 \vee \mathbb{S}^1)$?

(A) $\mathbb{Z}$

(B) $\mathbb{Z}^2$

(C) $\mathbb{Z}/2\mathbb{Z}$

(D) The free group on two generators

(E) 0

Solution: (D). This really requires the Seifert Van Kampen, but just remember it. □

Exercise 10.4. How many conjugacy classes are there in S_5?

(A) 5

(B) 6

(C) 7

(D) 8

(E) 120

Solution: (C). This is just partitions of 5 elements. There is:

$$e, (12), (123), (1234), (12345), (12)(34), (123)(45)$$

□

Remark 10.5. Note that in the problem above, conjugacy classes of S_n biject with the number of partitions of n.

Exercise 10.6. How many idempotent elements are there in $\mathbb{Z}/12\mathbb{Z}$? That is, how many x satisfy $x^2 = x$?

(A) 1

(B) 2

(C) 3

(D) 4

(E) 6

Solution: (D). $0, 1, 4, 9$. More generally, this follows by the CRT. $\mathbb{Z}/3\mathbb{Z}$ and $\mathbb{Z}/4\mathbb{Z}$ have two idempotents each, so CRT tells us to multiply these quantities. □

Exercise 10.7. Let $X = \mathbb{RP}^2$. Then $\pi_1(X)$ is isomorphic to

(A) 0

(B) $\mathbb{Z}$

(C) $\mathbb{Z}/2\mathbb{Z}$

(D) $\mathbb{Z}^2$

(E) S_2

Solution: (C). Just remember it. □

Exercise 10.8. Suppose G has order 21. Which statement must be true?

(A) Every subgroup is normal.

(B) G has a unique Sylow 7-subgroup.

(C) G has a unique Sylow 3-subgroup.

(D) G is abelian.

(E) G is cyclic.

Solution: (B). Sylow I says that there are 3-subgroups and 7-subgroups. Sylow III for $p = 7$ says that there is a unique Sylow 7-subgroup. □

Exercise 10.9. Which space has fundamental group $\mathbb{Z}^2$?

(A) S^2

(B) S^1

(C) $S^1 \times S^1$

(D) $S^1 \vee S^1$

(E) D^2

Solution: (C). Products of spaces and first fundamental group commute. □

Exercise 10.10. How many units does $\mathbb{Z}/20\mathbb{Z}$ have?

(A) 4

(B) 6

(C) 8

(D) 10

(E) 12

Solution: (C). Units of $\mathbb{Z}/20\mathbb{Z}$ are the integers coprime to 20. Then, use Euler's ϕ-function. □

Exercise 10.11. Suppose X is compact, Y is Hausdorff, and $f : X \to Y$ is a continuous bijection. Which must be true?

(A) f^{-1} need not be continuous.

(B) f is a homeomorphism.

(C) Y must be compact but f^{-1} can be discontinuous.

(D) X must be connected.

(E) Y must be metric.

Solution: (B). This is a result; a continuous bijection from a compact to Hausdorff has a continuous inverse. The proof isn't bad, so we'll do this here. Note that the image is compact by continuity, and surjectivity implies Y is compact. Now, f is a closed map because if you take some closed $Z \subseteq X$, then it is compact (closed subset of compact is compact), so its image is compact, so closed in Y. Now, a closed bijeciton is equivalent to an open bijection, so the inverse is continuous. □

Exercise 10.12. Let G have order 12. The number n_3 of Sylow 3-subgroups can be

(A) only 1

(B) only 4

(C) 1 or 3

(D) 1 or 4

(E) 3 or 4

Solution: (D). Use Sylow III. □

Exercise 10.13. What is the Euler characteristic of the torus $\mathbb{T}^2$?

(A) -2

(B) -1

(C) 0

(D) 1

(E) 2

Solution: (A). The genus of the 2-torus (usual torus with one hole) is 1. This is 0. □

Remark 10.14. Note that the Euler characteristic is multiplicative across products.

Exercise 10.15. Over the field $\mathbb{F}_p$, how many roots does $x^p - x$ have?

(A) 1

(B) $p - 1$

(C) p

(D) $p + 1$

(E) It depends on p

Solution: (C). Use FLT. □

Exercise 10.16. The cylinder $\mathbb{S}^1 \times [0,1]$ has fundamental group isomorphic to

(A) 0

(B) $\mathbb{Z}$

(C) $\mathbb{Z}^2$

(D) $\mathbb{Z}/2\mathbb{Z}$

(E) F_2

Solution: (B). $[0,1]$ is contractible, so has trivial fundamental group. $\mathbb{S}^1$ has fundamental group $\mathbb{Z}$. □

Exercise 10.17. Which quotient ring is isomorphic to $\mathbb{C}$?

(A) $\mathbb{R}[x]/(x)$

(B) $\mathbb{R}[x]/(x^2-1)$

(C) $\mathbb{R}[x]/(x^2+1)$

(D) $\mathbb{Z}[x]/(x^2+1)$

(E) $\mathbb{C}[x]/(x^2+1)$

Solution: (C). □

Exercise 10.18. What is the closure of $\mathbb{Q}$ in $\mathbb{R}$ with its usual topology?

(A) $\mathbb{Q}$

(B) $\mathbb{R} \setminus \mathbb{Q}$

(C) $\mathbb{R}$

(D) $\varnothing$

(E) $\mathbb{Q} \cup \{0\}$

Solution: (C). $\mathbb{Q}$ is dense in $\mathbb{R}$ with respect to the usual Euclidean topology $d(x,y) = |x-y|$. □

Exercise 10.19. Which of the following spaces is contractible?

(A) S^1

(B) S^2

(C) T^2

(D) D^2

(E) $\mathbb{R}^2 \setminus \{0\}$

Solution: (D). □

Exercise 10.20. If G acts transitively on a set X, and for $x \in X$,

$$|G| = 24, \qquad |\operatorname{Stab}(x)| = 6,$$

then $|X|$ is

(A) 3

(B) 4

(C) 6

(D) 18

(E) 144

Solution: (B). We use the orbit-stabilizer theorem. The orbit of $x \in X$ has cardinality 4, but transitivity implies that there is only one orbit. Thus, $|X| = 4$. □

Exercise 10.21. Which statement is true?

(A) Every connected space is path connected.

(B) Every compact space is Hausdorff.

(C) Every path-connected space is connected.

(D) Every bounded subset of a metric space is compact.

(E) Every continuous bijection is a homeomorphism.

Solution: (C). The converse is generally not true. □

Exercise 10.22. Let

$$f(z) = \frac{e^z}{z^2(z-1)}.$$

What is the order of the pole at $z = 0$?

(A) Removable

(B) 1

(C) 2

(D) 3

(E) Essential

Solution: (C). Just remember it. □

Exercise 10.23. Newton's method for approximating a root of $f(x) = 0$ is

(A) $x_{n+1} = x_n - f'(x_n)$

(B) $x_{n+1} = \dfrac{f(x_n)}{f'(x_n)}$

(C) $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$

(D) $x_{n+1} = x_n + \dfrac{f'(x_n)}{f(x_n)}$

(E) $x_{n+1} = f(x_n)$

Solution: (C). □

Exercise 10.24. Let G be a finite group and let $H \leq G$ have index 2. Which must be true?

(A) H is cyclic.

(B) H is abelian.

(C) H is normal.

(D) G/H has four elements.

(E) G is abelian.

Solution: (C). □

Exercise 10.25. Which of the following spaces is simply connected?

(A) S^1

(B) $\mathbb{R}^2 \setminus \{0\}$

(C) S^2

(D) T^2

(E) An annulus

Solution: Note that any loop (a closed path) can contract to a point, so its fundamental group $\pi_1(\mathbb{S}^2)$ is trivial. □

Exercise 10.26. A tree has 17 vertices. How many edges does it have?

(A) 15

(B) 16

(C) 17

(D) 18

(E) Cannot be determined

Solution: (B). A tree with n vertices has $n - 1$ edges. □

Exercise 10.27. Suppose R is an integral domain. Which condition is guaranteed?

(A) Every nonzero element is invertible.

(B) R has no nonzero zero divisors.

(C) Every ideal is principal.

(D) R is finite.

(E) R is a field.

Solution: (B). □

Exercise 10.28. Let $\varphi : G \to H$ be a group homomorphism. Which is always a normal subgroup of G?

(A) $\operatorname{im} \varphi$

(B) $\ker \varphi$

(C) Every subgroup of $\operatorname{im} \varphi$

(D) Every cyclic subgroup

(E) The center of H

Solution: (B). □

Exercise 10.29. How many homomorphisms

$$\mathbb{Z}/6\mathbb{Z} \longrightarrow \mathbb{Z}/4\mathbb{Z}$$

are there?

(A) 1

(B) 2

(C) 3

(D) 4

(E) 6

Solution: (B). □

Exercise 10.30. Suppose $f : \mathbb{S}^1 \to \mathbb{S}^1$ is given by $f(z) = z^5$. Under the usual identification $\pi_1(S^1) \cong \mathbb{Z}$, the induced map

$$f_* : \pi_1(\mathbb{S}^1) \to \pi_1(\mathbb{S}^1)$$

is

(A) $n \mapsto 0$

(B) $n \mapsto n$

(C) $n \mapsto 5n$

(D) $n \mapsto n + 5$

(E) $n \mapsto n^5$

Solution: (C). This winds around the circle 5 times. □

Exercise 10.31. If F is a field, which ideals does F have?

(A) Only (0)

(B) Only F

(C) Exactly (0) and F

(D) Every subset is an ideal

(E) Infinitely many proper ideals

Solution: (C). □

Exercise 10.32. Which group is necessarily cyclic?

(A) Every group of order 4

(B) Every group of order 6

(C) Every group of prime order

(D) Every abelian group

(E) Every finite group

Solution: (C). □

Exercise 10.33. We call a group G "pleasant" if it is abelian and every non-identity element g in G has prime order. Up to isomorphism, how many pleasant groups are there of order at most 25?

(A) 0

(B) 9

(C) 15

(D) 25

(E) 31

(F) Infinitely many

Solution: (C). By the structure theorem for finitely generated abelian groups, G is a finite product of finite cyclic groups. If two primes are present, say C_p and C_q, then there is an element of C_{pq}, which is not good. Thus, G is of the form $(C_p)^k$. There are nine primes total, but for $p = 2, 3, 5$, we can have powers. For 2, we can have $k \leq 4$, For 3 and 5, $k \leq 2$. This gives 15 possibilities. □

Exercise 10.34. Given a set S, let $\mathcal{F}(S)$ denote the collection of all finite subsets of S, and $\mathcal{I}(S)$ denote the collection of all infinite subsets of S. Which of the following are true?

1. $\mathcal{F}(\mathbb{N})$ is a topology on the natural numbers.
2. $\mathcal{I}(\mathbb{N})$ is a topology on the natural numbers.
3. The union $\mathcal{F}(\mathbb{N}) \cup \mathcal{I}(\mathbb{N})$ is a topology on the natural numbers.

(A) 1 only.

(B) 2 only.

(C) 3 only.

(D) 2 and 3 only.

(E) 1, 2, and 3 only.

Solution: (C). Note that 1 doesn't contain the whole space, and 2 doesn't contain the emptyset. 3 is a topology because it contains both and contains arbitrary unions and finite intersections. □

Exercise 10.35. Find the limit as x goes to 0 of the following fraction:

$$\frac{6\sin(x) - 6x + x^3}{x^5}.$$

(a) 0

(b) 1

(c) $\frac{1}{20}$

(d) $\frac{6}{20}$

(e) $-\frac{1}{120}$

Solution: (C). The polynomial terms in the numereator are irrelevant. Then, use the Taylo expansion of $\sin(x)$, where the degree 5 is the only one which matters. Its coefficient is $\frac{6}{5!}$, so by l'Hôpital. □

Exercise 10.36. Suppose that A, B, C are three pairwise independent events. Which of the following is not necessarily true?

(A) $P(A) = P(A \mid B)$.

(B) $P(A) \cdot P(B) \cdot P(C) = P(A \cap B \cap C)$.

(C) $P(A) = \frac{P(A \cap C)}{P(C)}$.

(D) $(P(A) \cdot P(B) \cdot P(C))^2 = P(A \cap B) \cdot P(A \cap C) \cdot P(B \cap C)$.

(E) $P(B \cap C) = P(B) \cdot P(C)$.

Solution: (B). Pairwise independence doesn't tell you that $A \cap B$ and C (or any other permutation) are independent events. □

Exercise 10.37. Determine the radius of convergence of the following power series:

$$\sum_{n=1}^{\infty} \frac{\cos(\pi n) x^n}{n}.$$

(a) ∞

(b) 0

(c) 1

(d) π

(e) n

Solution: (C). The ratio test gives the term $|x|\frac{n}{n+1}$. Sending $n \to \infty$, the limit is $|x|$, which we want to be less than 1, so $|x| < 1$. □

Exercise 10.38. Let f and g be twice-differentiable real-valued functions defined on $\mathbb{R}$. If $f'(x) > g'(x)$ for all $x > 0$, which of the following inequalities must be true for all $x > 0$?

(A) $f(x) > g(x)$

(B) $f''(x) > g''(x)$

(C) $f(x) - f(0) > g(x) - g(0)$

(D) $f'(x) - f'(0) > g'(x) - g'(0)$

(E) $f''(x) - f''(0) > g''(x) - g''(0)$

Solution: (C). Consider $h(x) = f(x) - g(x)$. Note that $h'(x) > 0$ and is twice differentiable. h is strictly increasing for all $x > 0$, so continuity (and thus the mean value theorem) implies that $h(x) > h(0)$, for all $x > 0$. □

Exercise 10.39. Let $f(x, y) = x^2 - 2xy + y^3$ for all real x and y. Which of the following is true?

(A) f has all of its relative extrema on the line $x = y$.

(B) f has all of its relative extrema on the parabola $x = y^2$.

(C) f has a relative minimum at $(0, 0)$.

(D) f has an absolute minimum at $\left(\frac{2}{3}, \frac{2}{3}\right)$.

(E) f has an absolute minimum at $(1, 1)$.

Solution: (A). $f_x = 2x - 2y$ and $f_y = -2x + 3y^2$. Thus, critical points are $P := (0, 0)$ and $Q := \left(\frac{2}{3}, \frac{2}{3}\right)$. We now check the Hessian. The Hessian has determinant $12y - 4$. At P, we have a saddle. At Q, the determinant is positive and f_{xx} is positive, so we have a relative minima. □

Exercise 10.40. For all positive functions f and g of the real variable x, let $\sim$ be a relation defined by

$$f \sim g \quad \iff \quad \lim_{x\to\infty} \frac{f(x)}{g(x)} = 1.$$

Which of the following is not a consequence of $f \sim g$?

(A) $f^2 \sim g^2$

(B) $\sqrt{f} \sim \sqrt{g}$

(C) $e^f \sim e^g$

(D) $f + g \sim 2g$

(E) $g \sim f$

Solution: (C). This requires $\lim_{x\to\infty} f(x) - g(x) = 0$, which is not guaranteed. In particular, this limit of $\frac{f}{g}$ doesn't ensure that limit of f is equal to limit of g. The one fishy one is (B), but the square root is continuous for positive x, so the limit commutes. □

Exercise 10.41. At a banquet, 9 women and 6 men are to be seated in a row of 15 chairs. If the entire seating arrangement is to be chosen at random, what is the probability that all of the men will be seated next to each other in 6 consecutive positions?

(A) $\dfrac{1}{\binom{15}{6}}$

(B) $\dfrac{6!}{\binom{15}{6}}$

(C) $\dfrac{10!}{15!}$

(D) $\dfrac{6!\,9!}{14!}$

(E) $\dfrac{6!\,10!}{15!}$

Solution: (D). There are $\binom{15}{6}$ ways to assort M and W, and there are ten positions where we can place the six. Thus, it's $\frac{10}{\binom{15}{6}}$. □

Exercise 10.42. In the complex z-plane, the set of points satisfying the equation

$$z^2 = |z|^2$$

is a

(A) pair of points

(B) circle

(C) half-line

(D) line

(E) union of infinitely many different lines

Solution: (D). We get $e^{2i\theta} = 1$, after cancelling the radius. Then, the valid points are just on the real line (0 works too!). □

Exercise 10.43. Say we have a triangle with sides $1, r, s$, with the angle between sides with lengths 1 and r being 110°. Here, as r and s increase, the length of the third side of the triangle remains 1 and the measure of the obtuse angle remains 110°. What is of the triangle remains 1 and the measure of the obtuse angle remains 110°. What is

$$\lim_{\substack{s\to\infty \\ r\to\infty}} (s - r)?$$

(A) 0

(B) A positive number less than 1

(C) 1

(D) A finite number greater than 1

(E) ∞

Solution: (B). Use law of cosines to get

$$s^2 = 1 + r^2 - 2r\cos(110°).$$

From this, we can deduce that

$$s - r = \frac{\frac{1}{r} - 2\cos(110°)}{\frac{s}{r} + 1}.$$

Now, we just need to determine the limit of $\frac{s}{r}$, but this is 1, because the law of cosines tell us that the limit of $\frac{s^2}{r^2}$ is 1 (and the square root is continuous on positive reals). Precisely, the limits is $\cos(110°)$ then. □

Exercise 10.44. A fair coin is to be tossed 100 times, with each toss resulting in a head or a tail. If H is the total number of heads and T is the total number of tails, which of the following events has the greatest probability?

(A) $H = 50$

(B) $T \geq 60$

(C) $51 \leq H \leq 55$

(D) $H \geq 48$ and $T \geq 48$

(E) $H \leq 5$ or $H \geq 95$

Solution: (D). This contains events around the center, which are greatest probability. □

Exercise 10.45. Let $\mathbf{F}$ be a constant unit force that is parallel to the vector $(-1, 0, 1)$ in xyz-space. What is the work done by $\mathbf{F}$ on a particle that moves along the path given by (t, t^2, t^3) between time $t = 0$ and time $t = 1$?

(A) $-\dfrac{1}{4}$

(B) $-\dfrac{1}{4\sqrt{2}}$

(C) 0

(D) $\sqrt{2}$

(E) $3\sqrt{2}$

Solution: (C). Note that work is $\int \mathbf{F} \cdot d\mathbf{r}$. We use a parametrization, so this is actually $\int \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)dt$. We get

$$\int_0^1 \frac{1}{\sqrt{2}}(-1, 0, 1) \cdot (1, 2t, 3t^2)dt = \int_0^1 \frac{3t^2 - 1}{\sqrt{2}}dt = 0.$$

Note that in the case of constant force (which is our case), we can just do $\mathbf{F} \cdot \nabla\mathbf{r}$. □

Exercise 10.46. Consider the theorem: If f and f' are both strictly increasing real-valued functions on the interval $(0, \infty)$, then

$$\lim_{x\to\infty} f(x) = \infty.$$

The following argument is suggested as a proof of this theorem.

1. By the Mean Value Theorem, there is a c_1 in the interval $(1, 2)$ such that
$$f'(c_1) = \frac{f(2) - f(1)}{2 - 1} = f(2) - f(1) > 0.$$
2. For each $x > 2$, there is a c_x in $(2, x)$ such that
$$\frac{f(x) - f(2)}{x - 2} = f'(c_x).$$
3. For each $x > 2$,
$$\frac{f(x) - f(2)}{x - 2} = f'(c_x) > f'(c_1)$$
since f' is strictly increasing.
4. For each $x > 2$,
$$f(x) > f(2) + (x - 2)f'(c_1).$$
5. $$\lim_{x\to\infty} f(x) = \infty.$$

Which of the following statements is true?

(A) The argument is valid.

(B) The argument is not valid since the hypotheses of the Mean Value Theorem are not satisfied in (1) and (2).

(C) The argument is not valid since (3) is not valid.

(D) The argument is not valid since (4) cannot be deduced from the previous steps.

(E) The argument is not valid since (4) does not imply (5).

Solution: (A). Note that the only cautious step may be the usage of the MVT, but it is implicit in the problem that $f'(x)$ exists on $(0, \infty)$, so continuity and differentiability is assumed to be true on the intervals used. □

Exercise 10.47. Let A be a real 2×2 matrix. Which of the following statements must be true?

1. All of the entries of A^2 are nonnegative.
2. The determinant of A^2 is nonnegative.
3. If A has two distinct eigenvalues, then A^2 has two distinct eigenvalues.

(A) 1 only

(B) 2 only

(C) 3 only

(D) 2 and 3 only

(E) 1, 2, and 3

Solution: (B). Counter example for 1 is easy and 2 is clearly true. For 3, the counter example $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ is good. □

Exercise 10.48. For how many positive integers k does the ordinary decimal representation of the integer $k!$ end in exactly 99 zeros?

(A) None

(B) One

(C) Four

(D) Five

(E) Twenty-four

Solution: (D). 5 is the limiting factor here. The power of 5 dividing $k!$ is the sum

$$\sum_{n \geq 1} \left\lfloor \frac{k}{5^n} \right\rfloor .$$

It is obvious that this sum is the same for five consecutive numbers k. □

Exercise 10.49. Let K be a nonempty subset of $\mathbb{R}^n$, where $n > 1$. Which of the following statements must be true?

1. If K is compact, then every continuous real-valued function defined on K is bounded.
2. If every continuous real-valued function defined on K is bounded, then K is compact.
3. If K is compact, then K is connected.

(A) 1 only

(B) 2 only

(C) 3 only

(D) 1 and 2 only

(E) 1, 2, and 3

Solution: (D). 1 is clearly true. 2 requires more justification. It is true by Heine-Borel. K is bounded by considering the norm function, which is continuous. It is also closed: if not, take $a \in \overline{K} \setminus K$. Then, $\frac{1}{||x-a||}$ is continuous on K because $a \notin K$. However, because a is a limit point, $||x - a|| \to 0$, so the whole expression becomes unbounded, which is a contradiction. □

Exercise 10.50. If f is the function defined by

$$f(x) = \begin{cases} xe^{-x^2-x^{-2}}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

at how many values of x does the graph of f have a horizontal tangent line?

(A) None

(B) One

(C) Two

(D) Three

(E) Four

Solution: (D). Away from zero, we can differentiate $f(x)$ to find two zeroes of $f'(x)$. We consider the behavior at 0 now. We use the standard limit:

$$\lim_{h\to 0} \frac{f(h) - f(0)}{h - 0} = \lim_{h\to 0} e^{-h^2-h^{-2}} = 0.$$

□

Exercise 10.51. For each positive integer n, let f_n be the function defined on the interval $[0, 1]$ by

$$f_n(x) = \frac{x^n}{1 + x^n}.$$

Which of the following statements are true?

1. The sequence $\{f_n\}$ converges pointwise on $[0,1]$ to a limit function f.
2. The sequence $\{f_n\}$ converges uniformly on $[0,1]$ to a limit function f.
3. $$\lim_{n\to\infty} \int_0^1 f_n(x)\,dx = \int_0^1 \left(\lim_{n\to\infty} f_n(x)\right) dx.$$

(A) 1 only

(B) 3 only

(C) 1 and 2 only

(D) 1 and 3 only

(E) 1, 2, and 3

Solution: (D). The pointwise limit is

$$f(x) = \begin{cases} 0 & x \in [0,1) \\ \frac{1}{2} & x = 1 \end{cases}.$$

3 is true by dominated convergence, using f as its point-wise limit. As for the counter example for 2, take some sequence approaching 1, say $x_n = 1 - \frac{1}{n}$. Then, $(x_n)^n \to e^{-1}$ and so $f_n(x_n) \to \frac{1}{1+e} > 0$, while $f(x_n) = 0$. □

Exercise 10.52. Let R be a ring with a multiplicative identity. If U is an additive subgroup of R such that $ur \in U$ for all $u \in U$ and for all $r \in R$, then U is said to be a right ideal of R. If R has exactly two right ideals, which of the following must be true?

1. R is commutative.
2. R is a division ring (that is, all elements except the additive identity have multiplicative inverses).
3. R is infinite.

(A) 1 only

(B) 2 only

(C) 3 only

(D) 1 and 2 only

(E) 1, 2, and 3

Solution: (B). 0 and R are always ideal and are the only ones. Then, every element has an inverse (which is unique), so is a division ring. Note that the ring need not be commutative at all, with the quaternions (which is a central simple algebra over $\mathbb{R}$) being a good example. □

Exercise 10.53. Suppose that X and Y are independent random variables with standard deviations σ_X and σ_Y, respectively. If $Z = X + Y$, then $\sigma_Z =$

(A) $\sigma_X + \sigma_Y$

(B) $\sigma_X^2 + \sigma_Y^2$

(C) $(\sigma_X^2 + \sigma_Y^2)^{1/2}$

(D) $2\sigma_X + 2\sigma_Y$

(E) $\sigma_X \sigma_Y$

Solution: (C). Note that

$$\sigma_Z^2 = \mathrm{Var}(Z) = \mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) = \sigma_X^2 + \sigma_Y^2,$$

since variance is linear. □

Exercise 10.54. Let $S = \{a, b, c, d\}$ and $X = \{f : S \to S \mid f \text{ is bijective and } f(x) \neq x \text{ for each } x \in S\}$. Then $|X| =$

(A) 8

(B) 9

(C) 12

(D) 18

(E) 20

Solution: (B). Count the number of conjugacy classes of type 2-2 and 4. □

Exercise 10.55. Which of the following graphs has two connected components?

(A) $\dfrac{x^2}{5} + y^2 - \dfrac{z^2}{2} = 1$

(B) $\dfrac{x^2}{5} - \dfrac{y^2}{5} - \dfrac{z^2}{5} = 1$

(C) $x^2 - 2x + y^2 + z^2 = 1$

(D) $x = y^2 - z^2$

(E) $z = x - y + 2$

Solution: (B). (A) has two positive squares and one negative square, which is one-sheeted hyperboloid. (B) has one positive square and two negative squares, which is a two-sheeted hyperboloid. (C) is a sphere. (D) is a hyperbolic paraboloid. (E) is a plane. □

Exercise 10.56. The function

$$f(x,y) = \begin{cases} \dfrac{x^2 y}{x^4 + y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0), \end{cases}$$

Which of the following is true?

1. f is not continuous at $(0,0)$.
2. f is differentiable everywhere.
3. f has well-defined partial derivatives everywhere (i.e. $\dfrac{\partial f}{\partial x}, \dfrac{\partial f}{\partial y}$ are both defined).
4. f is continuous at $(0,0)$ but not differentiable at $(0,0)$.

(A) 1 only

(B) 2 only

(C) 3 only

(D) 4 only

(E) 1 and 3 only

Solution: (E). 1 is false because if we use the contour $y = x^2$, then $f(x, x^2) = \frac{1}{2}$. 3 is true because the limits

$$\lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} \qquad \lim_{h\to 0} \frac{f(0,h) - f(0,0)}{h}$$

exist (they are both zero). Away from zero, of course the derivative exists. We've basically shown that 2 and 4 are false here too. □

Exercise 10.57. Evaluate the limit $\lim_{n\to\infty} (3^n + 5^n)^{\frac{2}{n}}$.

(a) 0

(b) 1

(c) 9

(d) 16

(e) 25

Solution: (E). Use sandwich on the following:

$$5^2 \leq (3^n + 5^n)^{\frac{2}{n}} \leq (2 \cdot 5^n)^{\frac{2}{n}}.$$

The limit of the RHS is 25. □

Exercise 10.58. Evaluate:

$$\int_0^{\pi/2} \frac{1}{1 + \tan x} dx.$$

(A) $\dfrac{\pi}{2}$

(B) $\dfrac{\pi}{3}$

(C) $\dfrac{\pi}{4}$

(D) $\dfrac{\pi^2}{2}$

(E) $\dfrac{\pi^2}{4}$

Solution: (C). Write the integral as

$$\int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx$$

If we consider

$$\int_0^{\pi/2} \frac{\sin x}{\cos x + \sin x} dx$$

as well, then the sum of these two integrals is $\frac{\pi}{2}$, while the difference (second minus first) is equal to $\ln|\cos x + \sin x|$ evaluated from the bound. This is zero. Therefore, the integral is $\frac{\pi}{4}$. □

Exercise 10.59. The series

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \cdots =$$

(A) $\dfrac{\pi}{6}$

(B) $\dfrac{\pi}{5}$

(C) $\dfrac{\pi}{4}$

(D) $\dfrac{\pi^2}{2}$

Solution: (C). Recall that

$$\arctan(x) = \sum_{n \geq 0} \frac{(-1)^n x^{2n+1}}{2n+1}.$$

Thus, this is just $\arctan(1) = \frac{\pi}{4}$. □

Exercise 10.60. Which of the following are true of a continuous function $f : \mathbb{R} \to \mathbb{R}$ such that $f(x+1) = f(x)$ for all x?

1. f achieves its maximum and minimum.
2. f is uniformly continuous.
3. There is a x_0 such that
$$f(x_0 + \pi) = f(x_0).$$
4. There is a x_1 such that
$$f\left(x_1 + \frac{1}{2}\right) = f(x_1).$$

(A) 1 only

(B) 1, 2 only

(C) 1, 2, and 3 only

(D) 1, 2, 3, 4

Solution: (D). For 1, 2, we can restrict f to $[0,1]$, which is compact, so it achieves a maximum and minimum (by EVT) and is uniformly continuous. For 3 and 4, we consider $g(x) := f(x+\pi) - f(x)$. The integral over $[0,1]$ is zero, so the function $g(x) = 0$ for some $x \in [0,1]$ (by IVT). □

Exercise 10.61. Let S be a set with $|S| \geq 3$. How many non-surjective functions from $S \to \{1,2,3\}$ are there?

(A) $3 \cdot 2^{|S|}$

(B) $3 \cdot 2^{|S|} - 3$

(C) $3^{|S|} - 3$

(D) $|S|^3$

(E) $2|S|^2 - 3$

Solution: (B). We choose the two elements of the target which are hit. There are three possible elements we don't choose. However, we overcount by three possible maps (which are those sending everything to a single element), so we subtract off 3. □

Exercise 10.62. A tank contains 50 gal of pure water. Starting at time zero, a salt-water solution containing 0.4 lb/gal of salt enters the tank at a rate of 2 gal/min. At the same time a drain is opened at the bottom of the tank allowing the salt solution to leave at a rate of 2 gal/min. Assume that the solution is kept perfectly mixed at all times. How long will it take for the amount of salt (in lbs) in the tank to reach 10 lbs? Leave your answer in exact form (in terms of logarithms).

(A) $2\ln 25$

(B) $-2\ln\left(\frac{1}{2}\right)$

(C) $25\ln\left(\frac{1}{2}\right)$

(D) $25\ln 2$

(E) $25\ln 10$

Solution: (D). If S denotes the gal of salt, then

$$\frac{dS}{dt} = 2 \cdot 0.4 - 2 \cdot \frac{S}{50}.$$

We can solve this ODE using an integrating factor of $e^{\frac{1}{25}t}$. This gives $S(t) = 20 + Ce^{-\frac{1}{25}t}$. Since $S(0) = 0$, we have $C = -20$. We then solve

$$10 = 20 - 20e^{-\frac{1}{25}t} \implies t = 25\ln(2).$$

□

Exercise 10.63. Evaluate

$$\int_0^{2\pi} e^{e^{i\theta}} d\theta.$$

(A) $2\pi i$

(B) 2π

(C) $4\pi i$

(D) $4\pi e$

(E) $-2\pi i$

Solution: (B). We use the infinite series form of e^x.

$$\int_0^{2\pi} e^{e^{i\theta}} d\theta = \sum_{n\geq 0} \int_0^{2\pi} \frac{e^{i\theta n}}{n!} d\theta.$$

For $n > 0$, we have $\int_0^{2\pi} \frac{e^{i\theta n}}{n!} d\theta = 0$, so the $n = 0$ remains. That is, $\int_0^{2\pi} d\theta = 2\pi$. □

Exercise 10.64. Let $F : \mathbb{R}_{>0} \to \mathbb{R}$ be given by

$$F(t) = \int_{-\infty}^{\infty} e^{-tx^2} dx.$$

Given

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi},$$

then $F'(t) =$

(A) $\dfrac{\sqrt{\pi}}{2t^{3/2}}$

(B) $-\dfrac{\sqrt{\pi}}{t^{3/2}}$

(C) $-\dfrac{\sqrt{\pi}}{t^{1/2}}$

(D) $-\dfrac{\sqrt{\pi}}{2t^{3/2}}$

(E) $\dfrac{\sqrt{\pi}}{t^{1/2}}$

Solution: (D). Consider the substitution $u = x\sqrt{t}$, so $dx = \frac{du}{\sqrt{t}}$. The integral is now

$$\frac{1}{\sqrt{t}} \int_{-\infty}^{\infty} e^{-u^2} du,$$

so $F(t) = \frac{1}{\sqrt{t}}\sqrt{\pi}$. Therefore, $F'(t) = -\frac{\sqrt{\pi}}{2t^{\frac{3}{2}}}$. □

Exercise 10.65. Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable. Then

$$\lim_{\epsilon \to 0} \frac{3}{\epsilon^3} \int_{-\epsilon}^{\epsilon} t f(x+t)\, dt =$$

(A) $\frac{1}{3}f(x)$

(B) $\frac{1}{2}f'(x)$

(C) $\frac{1}{2}f''(x)$

(D) $2f'(x)$

(E) $3f''(x)$

Solution: (D). Use the Taylor expansion of $f(x+t)$ around x, so

$$tf(x+t) = tf(x) + t^2 f'(x) + \frac{t^3}{2}f''(x) + \dots$$

The integral of $tf(x)$ over the interval $[-\epsilon, \epsilon]$ is zero because $tf(x)$ is odd (with resepect to t!). Similarly, all odd-powered parts vanish. Thus,

$$\lim_{\epsilon\to 0} \frac{3}{\epsilon^3}\int_{-\epsilon}^{\epsilon} tf(x+t)dt = \lim_{\epsilon\to 0} \frac{3}{\epsilon^3}\int_{-\epsilon}^{\epsilon} t^2 f'(x) + \text{h.o.t.}\, dt$$

The expression is

$$\lim_{\epsilon\to 0} \frac{3}{\epsilon^3}(\frac{2\epsilon^3}{3}f'(x) + \text{h.o.t. in } \epsilon) = 2f'(x)$$

□

Exercise 10.66. Which set does not have the same cardinality as the others?

(A) $\{f : \mathbb{N} \to \mathbb{N}\}$

(B) $\mathbb{R}$

(C) $\{f : \{0, 1, \dots, 100\} \to \mathbb{R}\}$

(D) {solutions to polynomial equations with coefficients in $\mathbb{Z}$}

(E) $\{f : \mathbb{Z} \to \{0, 1\}\}$

Solution: (D). (D) are algebraic numbers; there are only countably many polynomials with integer coefficients, and each polynomial has finitely many roots. Hence, this set is countable. (A) is $\mathbb{N}^{\mathbb{N}}$, which has cardinality $2^{\mathbb{N}}$. (B) is $\mathbb{R}$, which has cardinality $2^{\mathbb{N}}$. (C) is just $\mathbb{R}^{101}$, which has same cardinality as $\mathbb{R}$. (E) is $2^{\mathbb{Z}}$, having same cardinality as $2^{\mathbb{N}}$ or $\mathbb{R}$. □

Exercise 10.67. Let $S = \{1, 2, 3, 4\}$ and

$$X = \{f : S \to S \mid x < y \Rightarrow f(x) \leq f(y)\}.$$

Then $|X| =$

(A) 18

(B) 20

(C) 24

(D) 35

(E) 40

Solution: (D). Each function corresponds to selecting a multiset of 4 elements from S. The way we do this is via stars and bars. This is just $\binom{7}{4}$. □

Remark 10.68. For the above, suppose we want to select a multiset of n element from a set S, with k elements. This is done by solving for nonnegative solutions to $n_1+\cdots+n_k = n$. The n_i represent the number of elements from each element in S. Of course, this is set to be n because we want n elements in total.

Exercise 10.69. Suppose y_n and x_n satisfy

$$\begin{pmatrix} x_n \\ y_n \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x_{n-1} \\ y_{n-1} \end{pmatrix}$$

with $x_0 = y_0 = 1$. What is

$$\lim_{n\to\infty} \frac{x_n}{y_n}$$

assuming this limit exists?

(A) ∞

(B) $1 - \sqrt{2}$

(C) $1 + \sqrt{2}$

(D) $\dfrac{1+\sqrt{5}}{2}$

(E) $\dfrac{1-\sqrt{5}}{2}$

Solution: (D). Using the recurrence relation, we have

$$\frac{x_n}{y_n} = \frac{2x_{n-1} + y_{n-1}}{x_{n-1} + y_{n-1}}.$$

Assuming the limit exists, we find that

$$L = \frac{2L+1}{L+1},$$

so $L = \frac{1\pm\sqrt{5}}{2}$. Since $x_0, y_0 = 1$, all terms are positive, so $L = \frac{1+\sqrt{5}}{2}$. □

Exercise 10.70. Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

and define

$$\|A\|_{\text{op}} := \sup_{\|v\|_{\mathbb{R}^2}=1} \{\|Av\|_{\mathbb{R}^2}\}.$$

There is a vector v such that

$$\|A\|_{\text{op}} = \|Av\|_{\mathbb{R}^2}.$$

Also, note that once we find one v that maximizes the norm $\|\cdot\|_{\text{op}}$, $-v$ also does. Find the angle v makes with the positive x-axis assuming v is to the right of the y-axis:

(A) $\arctan 2$

(B) $\dfrac{\pi}{2} - \dfrac{\arctan 2}{2}$

(C) $\dfrac{3\pi}{4}$

(D) $\dfrac{\pi}{4}$

(E) $-\dfrac{\pi}{4}$

Solution: (B). Our vector $v = \begin{pmatrix}\cos\theta & \sin\theta\end{pmatrix}^T$. Our condition is asking when is the $f(\theta) = \cos^2\theta + 2\sin\theta\cos\theta + 2\sin^2\theta$ maximized? We take the derivative:

$$f'(\theta) = 2\cos 2\theta + \sin 2\theta,$$

so this is zero when $\tan(2\theta) = -2$. 2θ is in the second quadrant then, so we have that $2\theta = \pi - \arctan(2)$. □

Exercise 10.71. Which of the following is the largest:

(A) e^π

(B) π^e

(C) 3^π

(D) Not enough information.

Solution: (C). (C) is bigger than (A), clearly. We note however that (A) is bigger than (B). This can be seen by using logs and the function $\frac{\ln(x)}{x}$. Taking the derivative, its maximum is at e and decreases for $x > e$. Therefore, $\frac{\ln(\pi)}{\pi} < \frac{\ln(e)}{e}$. □

Exercise 10.72. Let

$$F(x) = \int_{\sin x}^{\cos x} e^{t^2+xt}\,dt.$$

Then $F'(0) =$

(A) $e\cos 1 - 2$

(B) $-2 + e$

(C) $\dfrac{e}{2} - \dfrac{3}{2}$

(D) $\dfrac{e}{2} + \dfrac{1}{2}$

(E) $\dfrac{e}{2} - \dfrac{1}{2}$

Solution: (C). We use Leibniz; $F'(x)$ is:

$$F'(x) = e^{\cos^2 x + \cos xt} \cdot (-\sin x) - e^{\sin^2 x + \sin xt} \cdot (\cos x) + \int_{\sin x}^{\cos x} te^{t^2+xt} dt$$

Setting $x = 0$, we have that

$$F'(0) = 0 - 1 + \int_0^1 te^{t^2} dt = \frac{1}{2}(e-1) - 1.$$

□

Exercise 10.73. Which functions are entire:

1. $f(z) = 2^{z^2}$
2. $g(z) = z^{2z} \sin z$
3. $h(z) = \sum_{i=0}^{\infty} z^i$

(A) 1, 2 only

(B) 1 only

(C) 1 and 3 only

(D) 2 only

(E) 2 and 3 only

Solution: (B). For 1, $2^{z^2} = e^{z^2 \cdot \ln 2}$. This is a composition of entire functions, so is entire. Next, $z^{2z} = e^{2z \ln z}$ is not entire because $\ln z$ is not entire (it requires a branch cut. This is conventionally taken as $(-\infty, 0]$). Finally, 3 is a geometric series for $|z| < 1$, so is holomorphic only there. □

Exercise 10.74. Which of the following holomorphic functions must be constant functions:

1. An entire function with positive real part.
2. An entire function whose image does not intersect a line $L \subset \mathbb{C}$.
3. An entire function such that $f(0) = 0$ and

$$\frac{f(z)}{z} \to 0 \qquad \text{as } |z| \to \infty.$$

   (Hint: $f(z)/z$ has a removable singularity at 0, so can be extended to an entire function.)

4. A holomorphic function whose range misses two points.

(A) 1, 2, 3, 4

(B) 2, 3, 4 only

(C) 1, 2 only

(D) 3 only

(E) 1, 2, 3 only

Solution: (E). 1 is classic; take $g(z) = e^{f(z)}$, which is bounded and entire. By Liouville, it's constant. 2 is true because we can apply an affine transformation to make this into the statemenet of 1. $g(z) := \frac{f(z)}{z}$ has a removable singularity, so can be extended to an entire function. Now g is bounded outside a large disk, and is also bounded inside it by maximum modulus principle (or EVT more simply). Thus, Liouville says that g is constant, but then this constant must be zero. Thus, $f = 0$. 4 is not true because we don't assume entire. If it was entire, we can use Picard's theorem. □

Exercise 10.75. Let R be a ring. Which of the following are always true?

1. The set of idempotents
$$\{x \in R \mid x^2 = x\}$$
is a subring.
2. If $x^2 = x$ for all $x \in R$, then R is commutative.
3. The only idempotents of an integral domain R are 0 and 1.

a) 1 only

b) 2 only

c) 3 only

d) 2 and 3 only

e) None are true.

Solution: (D). 1 is not true if we take $\mathbb{Z}$. 3 is clearly true. 2 is also true:

$$2a = (a+a)^2 = 4a^2 = 4a \implies 2a = 0.$$

Next,
$$a + b = (a+b)^2 = a^2 + ab + ba + b^2 = a + ab + ba + b \implies ab = ba.$$

In the second line, we really get $ab = -ba$, but $b = -b$, so we get commutativity. □

Exercise 10.76. What is the characteristic polynomial of an $n \times n$ matrix all of whose entries are 1?

(A) $x(x-1)(x-2)\cdots(x-n)$

(B) $x^{n-1}(x-n)$

(C) $x^{n-1}(x-1)$

(D) $x(x+1)(x+2)\cdots(x+n)$

(E) $x^{n-2}(x-1)(x+1)$

Solution: (B). The eigenspaces are given by one with eigenvalue n and all entries one. The other has eigenvalue 0, but $x_1 + \cdots + x_n = 0$, which has dimension $n - 1$. We've found them all, so the characteristic polynomial is (B). □

Exercise 10.77. Evaluate

$$\lim_{n\to\infty} \left(\frac{1}{n+1} + \cdots + \frac{1}{2n} \right).$$

(A) 1

(B) $\frac{3}{4}$

(C) $\frac{1}{2}$

(D) $\ln 2$

(E) $\ln 3$

Solution: (D). We can write the sum as

$$\frac{1}{n} \sum_{k=n+1}^{2n} \frac{1}{\frac{k}{n}},$$

which we can now see as a Riemann sum. This is then

$$\int_1^2 \frac{1}{x} dx = \ln(2) - \ln(1) = \ln 2.$$

□

Exercise 10.78. How many times do the hands of an analog clock (i.e. one with a hour and minute hand) form perpendicular angles in a 24 hour time cycle?

(A) 48

(B) 47

(C) 46

(D) 45

(E) 44

Solution: (E). In 12 hours, the minute hand rotates (fully) 12 times, while the hour hand does once. Thus, the relative difference in (full) revolutions is 11, so in 12 hours, there are 22 times they make a perpendicular angle. □

Exercise 10.79. Determine which of the following statements are true:

1. There is a function $f : \mathbb{R}^3 \to \mathbb{R}$ such that

$$\operatorname{grad} f = \langle y, x + z\cos y, \sin y \rangle.$$

2. There is a vector field $\vec{F} : \mathbb{R}^3 \to \mathbb{R}^3$ such that

$$\operatorname{curl} F = \langle z, y, x \rangle.$$

3. If $u, v : \mathbb{R} \to \mathbb{R}^3$ satisfy
$$u' = v - u, \qquad v' = v + u,$$
then $u \times v$ is constant.

(A) 1, 2, and 3

(B) 2 and 3 only

(C) 1 and 3 only

(D) 2 and 3 only

(E) 1 only

Solution: (A). 1 is true because $\operatorname{curl}\operatorname{grad} f = 0$. 2 is true because $\operatorname{div}\operatorname{curl} F = 0$. FOr 3, we use Leibniz:
$$\frac{d}{dt}(u \times v) = u \times v' + u' \times v = u \times (v + u) + (v - u) \times v = 0.$$
Thus, $u \times v$ is constant. □

Exercise 10.80. Suppose that A and B are 3×3 invertible matrices. Which of the following are always true?

(I) The eigenvalues of AB are the same as BA.

(II) The eigenvectors of AB are the same as BA.

(III) If A and B both only have 3 as an eigenvalue, then they are similar.

(A) 1 only

(B) 2 only

(C) 3 only

(D) 1 and 3 only

(E) None of them are true

Solution: (A). $A^{-1}ABA = BA$, so true. 2 is just not true; if $ABv = \lambda v$, then there is no reason why v should be an eigenvecotr for BA. 3 is not true; consider Jordan decomposition. □

Exercise 10.81. 100 people have bought tickets for the sold-out showing of *Casablanca* playing at the NuArt. The theater has precisely the right amount of seats for this size audience, and so to prevent any confusion, has assigned each person a seat. The people arrive one by one and respectfully sit in the seat they have been assigned. One complication: the first person to arrive is a drunkard who has lost his ticket and does not know his assigned seat. He randomly chooses one. As the 99 other people arrive, they may face the uncomfortable fact that their assigned seat is already filled. If it is, they avoid confrontation and randomly select an open seat. Finally the last person arrives. What is the probability he sits in his assigned seat?

(A) $\frac{1}{100}$

(B) $\frac{2}{77}$

(C) $\frac{1}{2}$

(D) $\frac{99}{100}$

(E) $\frac{3}{4}$

Solution: (C). Note that this chain of incorrectly chosen seats terminates at the 100th guy or before. Therefore, when the 100th guy walks in, he is either sitting in his seat, or the drunkard's seat. □

Exercise 10.82. Let f be the function defined by

$$f(x) = \begin{cases} \sin \frac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Let

$$I = \{(x, f(x)) \in \mathbb{R}^2 : -1 \leq x \leq 1\}.$$

Which of the following are TRUE?

1. The set I is connected.
2. The set I is path connected.
3. The set I is compact.

(A) 1 only

(B) 3 only

(C) 1 and 2 only

(D) 1 and 3 only

(E) 1, 2, and 3

Solution: (A). This is basically the topologist's sine curve, so we have 1 is true and 2 is false. 3 is not true because $(0, 1)$ is a limit point, for instance, but is not contained. Thus, not closed, and by Heine Borel. □

Exercise 10.83. Let

$$f(x) = \int_0^{x^2} \sqrt{t} \sin \frac{1}{t} \, dt,$$

and let $I = [-1, 0) \cup (0, 1]$. Which of the following are TRUE?

1. f is bounded on the set I.
2. f' is bounded on the set I.
3. f'' is bounded on the set I.

(A) 1 only

(B) 2 only

(C) 1 and 2 only

(D) 1 and 3 only

(E) 1, 2, and 3

Solution: (C). The integral is bounded by the same bounds with integrand $\sqrt{t}$, so is bounded. The FTC says that the derivative is $2x|x|\sin\frac{1}{x^2}$, which is bounded. Taking the derivative gives a term with x at the denominator, so is unbounded. □

Exercise 10.84. Suppose

$$\text{for all } \varepsilon > 0 \text{ there exists a } \delta > 0 \text{ such that for all } x \text{ and } y \text{ in } D,$$

$$|x - y| < \delta \quad \Longrightarrow \quad |f(x) - f(y)| < \varepsilon.$$

Consider the following statements:

1. For all $\varepsilon > 0$ there is a $\delta > 0$.
2. For all x and y in D.
3. $|x - y| < \delta$.
4. $|f(x) - f(y)| \geq \varepsilon$.

Using the numbers listed above, the proposition originally stated is which of the following? Denote "not" by $\neg$.

(A) $A(B(C \text{ or } D))$

(B) $A(B(\neg C \text{ and } D))$

(C) $\neg A(B(\neg C \text{ or } D))$

(D) $A(\neg B(\neg C \text{ or } D))$

(E) $A(B(\neg C \text{ or } \neg D))$

Solution: (E). The statement is $A(B(C \implies \neg D))$. We note that $P \implies Q$ is equivalent to $\neg P$ or Q, so we have (E). □

Exercise 10.85. Suppose V is a real vector space of finite dimension n. Let $\mathcal{M}(V)$ denote the set of matrices from V into itself. Let $T \in \mathcal{M}(V)$. Consider the two subspaces

$$\mathcal{U} = \{X \in \mathcal{M}(V) : TX = XT\}$$

and

$$\mathcal{W} = \{TX - XT : X \in \mathcal{M}(V)\}.$$

Which of the following must be TRUE?

1. If V has a basis containing only eigenvectors of T, then $\mathcal{U} = \mathcal{M}(V)$.
2. $\dim(\mathcal{U}) + \dim(\mathcal{W}) = n^2$.
3. $\dim(\mathcal{U}) < n$.

(A) 1 only

(B) 2 only

(C) 3 only

(D) 1 and 2 only

(E) 1, 2, and 3

Solution: (B). 2 is true by rank nullity under the map $L(X) = TX - XT$. For 3, let $T = I$. For 1, there is really no reason why this should be true. □

Exercise 10.86. Suppose today is Wednesday. What day of the week will it be 10^{10} days from now?

(A) Sunday

(B) Monday

(C) Tuesday

(D) Wednesday

(E) Thursday

Solution: (A). $10 \equiv 3 \pmod 7$. Also, $10^{10} \equiv 4 \pmod 6$, so we just need to calculate $3^4 \pmod 7$, which is 4. We move four days forward. □

Exercise 10.87. In the complex plane, let C be the circle

$$|z + 2| = 3$$

with positive (counterclockwise) orientation. Then

$$\oint_C \frac{dz}{z^3(z-2)} =$$

(A) $-\dfrac{\pi i}{4}$

(B) 0

(C) $\dfrac{3\pi i}{8}$

(D) $\dfrac{7\pi i}{8}$

(E) $2\pi i$

Solution: (A). Only the pole $z = 0$ is worth considering. By Cauchy's theorem,

$$\frac{d}{dz}\left(\frac{1}{z-2}\right) \cdot \frac{2\pi i}{2!} = -\frac{\pi i}{4}$$

is what we want. □

Exercise 10.88. In the complex plane, let C be the circle

$$|z| = 3$$

with positive (counterclockwise) orientation. Calculate

$$\oint_C \frac{dz}{z^2(z-1)^2}.$$

(A) $-\pi i$

(B) 0

(C) πi

(D) $2\pi i$

(E) $4\pi i$

Solution: (B). The residue at the pole 0 is 2 and the residue at the pole 1 is -2. □

Exercise 10.89. Suppose the real-valued function f has a continuous derivative for all values of x in $\mathbb{R}$. Which of the following must be FALSE?

1. For some closed interval $[a, b]$ and every natural number N there exists an x in the interval $[a, b]$ such that $|f(x)| > N$.
2. For each real number c there are exactly two solutions of $f(x) = c$.
3. The limit
$$\lim_{x\to\infty} \frac{f(x)}{x} = \infty$$
if and only if
$$\lim_{x\to\infty} f'(x) = \infty.$$

(A) 1 only

(B) 2 only

(C) 3 only

(D) 1 and 2 only

(E) 1, 2, and 3

Solution: (D). f is continuous, so bounded on the compact set $[a, b]$. 2 is clearly false, by MVT (or Rolle's theorem). 3 can be true. Take $f(x) = x^2$. □

Exercise 10.90. Suppose A and B are $n \times n$ matrices with real entries. Which of the following are TRUE?

1. The trace of A^2 is nonnegative.
2. If $A^2 = A$, then the trace of A is nonnegative.
3. The trace of AB is the product of the traces of A and B.

(A) 2 only

(B) 3 only

(C) 1 and 2 only

(D) 2 and 3 only

(E) 1, 2, and 3

Solution: (A). 1 and 3 are clearly wrong. For 2, this is true. Just write out the entries of A. Alternatively, look at eigenvalues of A. We have $\lambda^2 = \lambda$, so $\lambda = 0$ or 1. □

Exercise 10.91. Consider the independent random variables X_i such that either $X_i = 0$ or $X_i = 1$ and each event is equally as likely. Let

$$X = X_1 + X_2 + \cdots + X_{100}.$$

Which of the following values is largest?

(A) $\mathrm{Var}(X)$

(B) $100P(|X - 50| > 25)$

(C) $\displaystyle\sum_{k=0}^{100} k\binom{100}{k}\left(\frac{1}{2}\right)^k$

(D) $100P(X \geq 60)$

(E) 30

Solution: (C). This is a binomial distribution. Thus, $\mathrm{Var}(X) = np(1-p)$, which is 25. (C) is clearly bigger than B and D because these are probabilities (so include a 2^{-100} term). Thus, it's really between (C) and (E). However, the $k = 50$ terms is already humongous. If we wanted to be rigorous, we could eliminate (B) and (D) via Chebyshev. □

Exercise 10.92. Evaluate the following limit:

$$\lim_{n\to\infty}\left(\frac{1}{n} + \frac{1}{n+2} + \frac{1}{n+4} + \cdots + \frac{1}{3n}\right).$$

(A) $\dfrac{1}{2}\log 2$

(B) $\dfrac{3}{4}$

(C) $\log \sqrt{3}$

(D) 1

(E) 2

Solution: (C). Write as a sum and divide top/ bottom by n. Then, we see this is

$$\int_0^1 \frac{1}{1+2x} dx.$$

□

Exercise 10.93. Suppose A and A_k are subsets of $\mathbb{R}$ where k is any positive integer. Which of the following must be TRUE?

1. If A is closed, then A is compact.
2. If for each sequence $\{a_k\}$ with terms in A there is a strictly increasing function $\alpha : \mathbb{Z} \to \mathbb{Z}$ such that
$$\lim_{k\to\infty} a_{\alpha(k)}$$
is in A, then A is compact.
3. If
$$B = \bigcup_{k=1}^{\infty} A_k,$$
then
$$\overline{B} = \bigcup_{k=1}^{\infty} \overline{A_k}.$$

(A) 1 only

(B) 2 only

(C) 3 only

(D) 2 and 3 only

(E) 1, 2, and 3

Solution: (B). 1 is false and 2 is true (clear). 3 is false, and we only usually have

$$\bigcup_{k=1}^{\infty} \overline{A_k} \subseteq \overline{B}.$$

For instance, take $A_k = \left\{\frac{1}{k}\right\}$. □

Remark 10.94. Note that for a finite union, we have $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Exercise 10.95. For each positive integer n, let f_n be the function defined on the interval $[0, 1]$ by

$$f_n(x) = \frac{nx}{1+nx^2}.$$

Which of the following statements are TRUE?

1. The sequence $\{f_n\}$ converges point-wise on $[0,1]$ to a limit function f.
2. The sequence $\{f_n\}$ converges uniformly on $[0,1]$ to a limit function f.
3. $$\left| \int_0^1 f_n(x)dx - \int_{1/n}^1 \lim_{k\to\infty} f_k(x)\,dx \right| \to 0 \qquad \text{as } n \to \infty.$$

(A) 1 only

(B) 3 only

(C) 1 and 2 only

(D) 1 and 3 only

(E) 1, 2, and 3

Solution: (A). The pointwise limit is the function 0 at $x = 0$ and $\frac{1}{x}$ for $x > 0$. This is not continuous, so definitely not uniformly convergent. 3 is not true by calculation. □

Exercise 10.96. Suppose A, B, and C are statements such that C is true if exactly one of A and B is true. If C is false, which of the following statements must be true?

(A) If A is true, then B is false.

(B) If A is false, then B is false.

(C) If A is false, then B is true.

(D) Both A and B are true.

(E) Both A and B are false.

Solution: (B). By contrapositive, this means that A and B are both false or both true. Thus, (B) is certainly true. (D) and (E) are not right because we don't know if they're both true or false. □

Exercise 10.97. What is the value of

$$\int_{-\pi/4}^{\pi/4} \left(\cos t + \sqrt{1+t^2}\sin^3 t \cos^3 t\right) dt?$$

(A) 0

(B) $\sqrt{2}$

(C) $\sqrt{2} - 1$

(D) $\dfrac{\sqrt{2}}{2}$

(E) $\dfrac{\sqrt{2}-1}{2}$

Solution: (B). The sophisticated part is zero because it is an odd function. Thus, we just calculate the cosine part. Remember, odd × even is odd. □

Exercise 10.98. Let V be a finite-dimensional real vector space and let P be a linear transformation of V such that $P^2 = P$. Which of the following must be true?

(A) P is invertible.

(B) P is diagonalizable.

(C) P is either the identity transformation or the zero transformation.

(A) None

(B) I only

(C) II only

(D) III only

(E) II and III

Solution: (C). 1 and 3 are clearly false. For 2, possible eigenvalues are 0 or 1. The minimal polynomial of P divides $x(x-1)$, so is a product of linear factors with no repetition. Thus, P is diagonalizable. □

Exercise 10.99. Let ℓ be the line that is the intersection of the planes

$$x + y + z = 3$$

and

$$x - y + z = 5$$

in $\mathbb{R}^3$. An equation of the plane that contains $(0, 0, 0)$ and is perpendicular to ℓ is

(A) $x - z = 0$

(B) $x + y + z = 0$

(C) $x - y - z = 0$

(D) $x + z = 0$

(E) $x + y - z = 0$

Solution: (A). All we really need is the cross product of $(1, 1, 1)$ and $(1, -1, 1)$. This gives $(2, 0, -2)$, so (A) is really the only one that works. □

Exercise 10.100. A curve in the xy-plane is given parametrically by

$$x = t^2 + 2t, \qquad y = 3t^4 + 4t^3,$$

for all $t > 0$. The value of $\dfrac{d^2y}{dx^2}$ at the point $(8, 80)$ is

(A) 4

(B) 24

(C) 32

(D) 96

(E) 192

Solution: (A). We note that

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = 6t^2.$$

Next, we note that

$$\frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right) \cdot \frac{dt}{dx} = \frac{d}{dt}\left(\frac{dy}{dx}\right) \cdot \frac{1}{\frac{dx}{dt}}.$$

Thus, $\frac{d^2y}{dx^2} = \frac{12t}{2t+2}$. At $t = 2$, this yields 4. □

Exercise 10.101. Let f and g be functions of a real variable such that

$$g(x) = \int_0^x f(y)(y - x)\, dy$$

for all x. If g is three times continuously differentiable, what is the greatest integer n for which f must be n times continuously differentiable?

(A) 1

(B) 2

(C) 3

(D) 4

(E) 5

Solution: (A). By Lebniz rule, we have

$$g'(x) = f(x)(x - x) \cdot 1 - f(0)(0 - x) \cdot 0 + \int_0^x -f(y)dx.$$

Thus,

$$g''(x) = -f(x).$$

g is three time differentiable, so we need f' to exist at the minimum. □

Exercise 10.102. If a and b are positive numbers, what is the value of

$$\int_0^\infty \frac{e^{ax} - e^{bx}}{(1 + e^{ax})(1 + e^{bx})}\, dx?$$

(A) 0

(B) 1

(C) $a - b$

(D) $(a-b)\log 2$

(E) $\dfrac{a-b}{ab}\log 2$

Solution: (E). Note that the integrand is a difference of $\frac{1}{1+e^{bx}}$ and same but b replaced with a. Now,

$$\int_0^\infty \frac{1}{1+e^{bx}}dx = \int_0^\infty \frac{e^{-bx}}{1+e^{-bx}}dx = -\frac{1}{b}\ln|1+e^{-bx}|\Big|_0^\infty = \frac{1}{b}\ln(2).$$

Thus, it is (E). □

Exercise 10.103. For each positive integer n, let x_n be a real number in the open interval $\left(0, \frac{1}{n}\right)$. Which of the following statements must be true?

1. $\lim_{n\to\infty} x_n = 0$.
2. If f is a continuous real-valued function defined on $(0,1)$, then $\{f(x_n)\}_{n=1}^\infty$ is a Cauchy sequence.
3. If g is a uniformly continuous real-valued function defined on $(0,1)$, then $\lim_{n\to\infty} g(x_n)$ exists.

(A) 1 only

(B) 1 and 2 only

(C) 1 and 3 only

(D) 2 and 3 only

(E) 1, 2, and 3

Solution: (C). 1 is true by squeeze theorem, since $\frac{1}{n} \to 0$. 2 is false. Consider $f(x) = \frac{1}{x}$. Then, $f(x_n) > n$, so the given set can't be Cauchy. 3 is true. Uniform continuity and 1 tell us that $|g(x_n) - g(x_m)|$ tend to zero, thus Cauchy. By completeness, Cauchy implies convergence. □

Exercise 10.104. A real-valued function f defined on $\mathbb{R}$ has the following property: For every positive number ε, there exists a positive number δ such that

$$|f(x) - f(1)| \geq \varepsilon \qquad \text{whenever} \qquad |x-1| \geq \delta.$$

This property is equivalent to which of the following statements about f?

(A) f is continuous at $x = 1$.

(B) f is discontinuous at $x = 1$.

(C) f is unbounded.

(D) $\lim\limits_{|x|\to\infty} |f(x)| = \infty$.

(E) $\int_0^\infty |f(x)|\,dx = \infty$.

Solution: (D). This is precisely the definition of (D). Vaguely, we get this because this says that if the input is really far from 1, then the output is really from $f(1)$. □

Exercise 10.105. A tank initially contains a salt solution of 3 grams of salt dissolved in 100 liters of water. A salt solution containing 0.02 grams of salt per liter of water is sprayed into the tank at a rate of 4 liters per minute. The sprayed solution is continually mixed with the salt solution in the tank, and the mixture flows out of the tank at a rate of 4 liters per minute. If the mixing is instantaneous, how many grams of salt are in the tank after 100 minutes have elapsed?

(A) 2

(B) $2 - e^{-2}$

(C) $2 + e^{-2}$

(D) $2 - e^{-4}$

(E) $2 + e^{-4}$

Solution: (E). If $S(t)$ denotes the salt in the tank at time t, then we have

$$S' = 4(0.02) - 4\left(\frac{S}{100}\right).$$

This gives $S(t) = 2 + e^{-\frac{t}{25}}$, using the initial conditions. □

Exercise 10.106. What is the value of the flux of the vector field $\mathbf{F}$, defined on $\mathbb{R}^3$ by

$$\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k},$$

through the surface

$$z = \sqrt{1 - x^2 - y^2}$$

oriented with an upward-pointing normal vector field?

(A) 0

(B) $\frac{2\pi}{3}$

(C) π

(D) $\frac{4\pi}{3}$

(E) 2π

Solution: (E). Use the divergence theorem. The integral is equal to

$$\int_V \nabla \cdot \mathbf{F} dV = \int_V 3dV.$$

Now, the volume is half a sphere of radius 1, so it is $\frac{2}{3}\pi$. Thus, (E). □

Exercise 10.107. How many surjective functions $f : \{1, \ldots, 7\} \to \{1, \ldots, 4\}$ satisfy $f(1) = f(2)$?

(A) 1320

(B) 1440

(C) 1560

(D) 1680

(E) 1800

Solution: (C). The best way to do this is noting that we really only need to count the number of surjective functions from a set of size 6 to a set of size 4. This is:

$$4^6 - \binom{4}{1}3^6 + \binom{4}{2}2^6 - \binom{4}{3}1^6 = 1560.$$

□

Remark 10.108. The number of surjective functions between finite sets N, M of size n, m are as follows:

$$\sum_{k=0}^{m}(-1)^k\binom{m}{k}(m-k)^n.$$

Exercise 10.109. How many surjective functions $f : \{1, \ldots, 8\} \to \{1, \ldots, 3\}$ satisfy both of the following conditions?

- Exactly two elements are mapped to 3.
- $f(1) \neq f(2)$.

(A) 960

(B) 1056

(C) 1176

(D) 1224

(E) 1344

Solution: (D). This requires case work. If 1 or 2 maps to 3, or not. First suppose neither go to 3. Then, they split to 1 and 2, so this gives 2 cases. The remaining 6 can go to wherever, as long as exactly two go to 3, so this gives $\binom{6}{2} \cdot 2^4$. This gives $2 \cdot \binom{6}{2} \cdot 2^4 = 480$. Next, either 1 or 2 maps to 3. Now, 1 or 2 maps to 3, and the reamining goes to either 1 or 2, which is $2 \cdot 2$ cases. Now, of the reamining 6 numbers, exactly one goes to 3, and the reamining can go to 1 or 2, but one of these were already hit, which gives $2^5 - 1$ ways. The number of ways here is $2 \cdot 2 \cdot \binom{6}{1}(2^5 - 1) = 744$. Summing these two cases, we get 1224. □

Exercise 10.110. Let $f : [8] \to [4]$ be surjective. How many such functions have the property that exactly two elements of [4] have preimages consisting of a single element?

(A) 8400

(B) 10080

(C) 13440

(D) 16800

(E) 20160

Solution: (D). We choose the two elements of the domaind and target which are chosen. This is $2 \cdot \binom{8}{2} \cdot \binom{4}{2}$. For the remaining 6 values in the domain, we have 2 elements for the target. We need the 6 to surject onto the 2, but we can't have the preimages to be just one element. This is $2^6 - 2 - 2 \cdot \binom{6}{1}$. Multiplying these, we have

$$2 \cdot \binom{8}{2} \cdot \binom{4}{2} \cdot \left(2^6 - 2 - 2 \cdot \binom{6}{1}\right) = 16800.$$

□

Exercise 10.111. Let

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

be a 7×7 matrix, where A is an invertible 3×3 matrix and D is 4×4. Suppose $\dim(\ker M) = 2$. Which of the following must be true?

(A) $\det(D - CA^{-1}B) = 1$

(B) $\operatorname{rank}(D - CA^{-1}B) = 2$

(C) $D - CA^{-1}B$ is invertible.

(D) D is invertible.

(E) $BC = 0$

Solution: (B). Multiplying on the left by $P = \left(\begin{smallmatrix} I_3 & 0 \\ -CA^{-1} & I_4 \end{smallmatrix}\right)$, we have that

$$PM = \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}$$

Since P is invertible, $\operatorname{Rank}(M) = \operatorname{Rank}(PM)$. Also, since A is inveritble, it has full rank, so $D - CA^{-1}B$ has rank 2 by rank-nullity. □

Remark 10.112. Given a matrix $M := \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ as above, the matrix $S := D - CA^{-1}B$ is called the *Schur compelement* of A, when A is invertible. It satisfies the following properties:

- $\operatorname{Rank}(M) = \operatorname{Rank}(A) + \operatorname{Rank}(S)$
- $\det(M) = \det(A)\det(S)$.
- M is inveritble if and only if S is invertible.

Exercise 10.113. Let

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

be invertible, where A is invertible. Write

$$M^{-1} = \begin{pmatrix} P & Q \\ R & S \end{pmatrix}.$$

Which of the following expressions is necessarily equal to S?

(A) D^{-1}

(B) $(D - CA^{-1}B)^{-1}$

(C) $D - CA^{-1}B$

(D) $A^{-1}B(D - CA^{-1}B)^{-1}$

(E) $(D - BA^{-1}C)^{-1}$

Solution: (B). This comes from $MM^{-1} = I$, and using the rules above/ solving for S using the Schur compelement of A. □

Exercise 10.114. Let

$$\mathbf{F}(x, y, z) = \frac{(x, y, z)}{(x^2 + y^2 + z^2)^{3/2}}.$$

Let S be the sphere

$$\left(x - \frac{1}{2}\right)^2 + y^2 + z^2 = 1$$

with outward orientation. What is

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS?$$

(A) 0

(B) 2π

(C) 4π

(D) 8π

(E) Undefined

Proof. (C). Let $\mathbf{r} = (x, y, z)$, so $|r|^3$ is the denominator. Then, Gauss's flux theorem applies. □

Exercise 10.115. Let $\phi, \psi : \mathbb{R}^3 \to \mathbb{R}$ be twice continuously differentiable scalar functions. Which expression must be identically zero?

(A) $\nabla \cdot (\phi \nabla \psi)$

(B) $\nabla \cdot (\nabla \phi \times \nabla \psi)$

(C) $\nabla \times (\phi \nabla \psi)$

(D) $\nabla \cdot (\nabla \phi)$

(E) $\nabla \cdot (\phi \nabla \phi)$

Solution: (B). Note that $\nabla \cdot (A \times B) = B \cdot (\nabla \times A) - A \cdot (\nabla \times B)$. A and B are now the divergence of some scalar functions, so grad div $= 0$ implies the inside of the brackets are zero. □

Exercise 10.116. Let B denote the unit ball in $\mathbb{R}^3$, and let $\mathbf{G}$ be any twice continuously differentiable vector field on a neighborhood of B. Define

$$\mathbf{H} = \nabla \times \mathbf{G} + (x^2 z, y^2 z, z^3).$$

What is the outward flux of $\mathbf{H}$ through ∂B?

(A) 0

(B) $\dfrac{\pi}{5}$

(C) $\dfrac{4\pi}{5}$

(D) $\dfrac{4\pi}{3}$

(E) $\dfrac{12\pi}{5}$

Solution: (C). The divergence theorem gives flux on the surface side, so the volume side gives

$$\int_V 2xz + 2yz + 3z^2 dV.$$

Now, integrating over the ball, the first two terms vanish because they are odd in z. The final term remains. However, note that

$$\int_V z^2 dV = \int_V x^2 dV = \int_V y^2 dV.$$

Thus, $\int_V 3z^2 dV = \int_V x^2 + y^2 + z^2 dV$. Relying on polar coordiantes, we have the integrand is r^2, while the volume element is $dV = r^2 \sin\theta d\phi d\theta dr$.

$$\int_0^1 \int_0^\pi \int_0^{2\pi} r^4 \sin\theta d\phi d\theta dr = \frac{4}{5}\pi.$$

□

Exercise 10.117. Suppose $f : \mathbb{R} \to \mathbb{R}$ is continuously differentiable, and let $a \in \mathbb{R}$. Evaluate

$$\lim_{h \to 0} \frac{1}{h^2} \left(\int_a^{a+h} f(t)\, dt - h f(a) \right).$$

(A) $-f'(a)$

(B) 0

(C) $\frac{1}{2}f'(a)$

(D) $f'(a)$

(E) $2f'(a)$

Solution: (C). The top and bottom tend to zero, so by L'hôpital, we differentiate the top and bottom. We find that this limit is equal to $\frac{f(a+h)-f(a)}{2h}$. □

Exercise 10.118. Evaluate

$$\lim_{x\to 0} \frac{1}{x^2} \int_{\sin x}^{x^2} \left(\int_0^t e^{u^2}\, du \right) dt.x$$

(A) -1

(B) $-\frac{1}{2}$

(C) 0

(D) $\frac{1}{2}$

(E) 1

Solution: (B). We use L'hôpital again, with FTC. If $F(t)$ is the inner integral, then we have the derivative with respec to x of the above if $2xF(x^2) - \cos(x)F(\sin(x))$. Then, the limit we need to evaluate is $\frac{2xF(x^2)-\cos(x)F(\sin(x))}{2x}$ as $x \to 0$. The first part tends to zero, since $F(0) = 0$. As for the second part, we rewrite it as

$$-\frac{1}{2} \underbrace{\frac{F(\sin(x))}{\sin(x)}}_{\to F'(0)} \cdot \underbrace{\frac{\sin(x)}{x}}_{\to 1} \cdot \underbrace{\cos(x)}_{\to 1}.$$

To finish off, $F'(t) = e^{t^2}$, so $F'(0) = 1$. □

Exercise 10.119. Evaluate

$$\lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{\sqrt{n^2+k}}.$$

(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) $\ln(1+\sqrt{2})$

(E) The sequence diverges to $+\infty$.

Solution: (C). This is the squeeze theorem. Note that

$$\frac{1}{\sqrt{n^2+n}} \leq \frac{1}{\sqrt{n^2+k}} \leq \frac{1}{\sqrt{n^2+1}}.$$

Since there are n terms of the finite sum, so

$$\frac{1}{\sqrt{1+\frac{1}{n}}} = \frac{n}{\sqrt{n^2+n}} \leq \sum_{k=1}^{n} \frac{1}{\sqrt{n^2+k}} \leq \frac{n}{\sqrt{n^2+1}} = \frac{1}{\sqrt{1+\frac{1}{n^2}}}$$

Taking $n \to \infty$, the sum is bounded above and below by 1. □

Exercise 10.120. Evaluate

$$\lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{\sqrt{n^2+nk}}.$$

(A) 1

(B) $2(\sqrt{2}-1)$

(C) $\ln(1+\sqrt{2})$

(D) $\dfrac{\pi}{4}$

(E) $\dfrac{\sqrt{2}}{2}$

Solution: (B). The summand can be reformulated to $\frac{1}{n} \cdot \frac{1}{\sqrt{1+\frac{k}{n}}}$. This has the form of a Riemann sum, and it is equal to the integral

$$\int_0^1 \frac{1}{\sqrt{1+x}} dx.$$

The integral is $\left(2\sqrt{x+1}\right)\Big|_0^1$. □

Exercise 10.121. Evaluate

$$\lim_{n\to\infty} \sum_{k=1}^{n} \frac{k}{n^2+k^2}.$$

(A) $\ln 2$

(B) $\dfrac{1}{2}\ln 2$

(C) $\dfrac{\pi}{4}$

(D) $\dfrac{1}{4}$

(E) The sequence diverges to $+\infty$.

Solution: (B). We divide the summand out. Then, we can write it as:

$$\frac{1}{n} \cdot \frac{\frac{k}{n}}{1 + \left(\frac{k}{n}\right)^2}.$$

This is the form of a Riemann sum, so the integral is

$$\int_0^1 \frac{x}{1+x^2} dx = \frac{1}{2} \ln |1 + x^2| \Big|_0^1 = \frac{1}{2} \ln(2)$$

□

Exercise 10.122. Let

$$a_n = \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n}.$$

Which of the following is true?

(A) $a_n \to 0$

(B) $a_n \to \frac{1}{2}$

(C) $a_n \to \ln 2$

(D) $a_n \to 1$

(E) $a_n \to +\infty$

Solution: (C). This is $\sum_{k=1}^{n} \frac{1}{k+n}$, and the summand is equal to $\frac{1}{n} \cdot \frac{1}{1+\frac{k}{n}}$. Thus, we're taking the integral $\int_0^1 \frac{1}{1+x} dx$. □

Exercise 10.123. For a fixed real number $p > 0$, define

$$a_n(p) = \sum_{k=1}^{n} \frac{1}{\sqrt{n^2 + k^p}}.$$

For which values of p is

$$\lim_{n \to \infty} a_n(p) = 1?$$

(A) $0 < p < 2$

(B) $0 < p \leq 2$

(C) $p = 2$ only

(D) $p \geq 2$

(E) Every $p > 0$

Solution: (A). We focus on the integrad, which can be rewritten as

$$\frac{1}{n} \cdot \frac{1}{\sqrt{1 + n^{p-2} \left(\frac{k}{n}\right)^p}}.$$

We focus on the non-$\frac{1}{n}$ part. Then,

$$1 \geq \frac{1}{\sqrt{1 + n^{p-2}\left(\frac{k}{n}\right)^p}} \geq \frac{1}{\sqrt{1+n^{p-2}}},$$

by noting that $0 \leq \left(\frac{k}{n}\right)^p \leq 1$. This implies that

$$1 \geq a_n(p) \geq \frac{1}{\sqrt{1+n^{p-2}}}.$$

Now, if $0 < p < 2$, then the RHS tends to 1. If $p = 2$, then we can't quite use the same methods, but we can actually use a method of Riemann sums. The integral is

$$\int_0^1 \frac{1}{\sqrt{1+x^2}}dx = \left(\ln|x + \sqrt{1+x^2}|\right)\Big|_0^1 = \ln(1+\sqrt{2}).$$

If $p > 2$, we note that $0 < a_n(p) \leq a_n(2)$, but the limit for $a_n(2) < 1$. □

Exercise 10.124. Suppose (a_n) is a sequence of real numbers such that

$$\sum_{n=1}^{\infty} a_n$$

converges. Which of the following must also converge?

(A) $\displaystyle\sum_{n=1}^{\infty} |a_n|$

(B) $\displaystyle\sum_{n=1}^{\infty} (-1)^n a_n$

(C) $\displaystyle\sum_{n=1}^{\infty} a_n^2$

(D) $\displaystyle\sum_{n=1}^{\infty} \frac{a_n}{n}$

(E) $\displaystyle\sum_{n=1}^{\infty} n a_n$

Solution: (D). First note that convergence of the series implies $a_n \to 0$. Now, $a_n = (-1)^n \frac{1}{n}$ kills options (A), (B), and (E). Also, $a_n = (-1)^n \frac{1}{\sqrt{n}}$ kills (C). (D) is the only one which remains. To rigoorusly prove (D) works, we use Abel's theorem, which states that if $\sum a_n$ is converges and (b_n) is a monotonoic and bounded, then $\sum a_n b_n$ converges. □

Exercise 10.125. Let (a_n) be a sequence of nonnegative real numbers such that

$$\sum_{n=1}^{\infty} a_n^2 < \infty.$$

Which of the following must be true?

(A) $\sum_{n=1}^{\infty} a_n$ converges.

(B) $\sum_{n=1}^{\infty} (-1)^n a_n$ converges.

(C) $\sum_{n=1}^{\infty} \frac{a_n}{n}$ converges.

(D) $\lim_{n\to\infty} n a_n = 0$

(E) The sequence (a_n) is eventually decreasing.

Solution: (C). This uses Cauchy Schwarz:

$$\left(\sum_{n=1}^{N} \frac{a_n}{n}\right)^2 \leq \left(\sum_{n=1}^{N} a_n^2\right)\left(\sum_{n=1}^{N} \frac{1}{n^2}\right)$$

Taking N to infinity, the RHS converges, so the square is bounded. The the sum in (C) is bounded. Since the a_n are positive, the sum as $N \to \infty$ is bounded, positive, and monotonically icnreasing, so converges. □

Remark 10.126. Note that convergence tells us $a_n^2 \to 0$, so $|a_n| \to 0$, because $|a_n| = \sqrt{a_n^2}$. Then, $a_n \to 0$ trivially, since $a_n \to 0$ iff $|a_n| \to 0$.

Exercise 10.127. Let (a_n) be a real sequence such that both series

$$\sum_{n=1}^{\infty} a_n \qquad \text{and} \qquad \sum_{n=1}^{\infty} (-1)^n a_n$$

converge. Which of the following must converge?

(A) $\sum_{n=1}^{\infty} |a_n|$

(B) $\sum_{n=1}^{\infty} a_n^2$

(C) $\sum_{n=1}^{\infty} a_{2n}$

(D) $\sum_{n=1}^{\infty} (-1)^n |a_n|$

(E) $\sum_{n=1}^{\infty} \sqrt{|a_n|}$

Solution: (C). Just add the two convergent sequences together. □

Exercise 10.128. For each real number $p > -1$, define

$$L(p) = \lim_{n\to\infty} \frac{1}{n^{p+1}} \sum_{k=1}^{n} k^p.$$

What is the set of all possible values of $L(p)$?

(A) $(0, 1]$

(B) $[0, 1]$

(C) $(0, \infty)$

(D) $(0, \infty]$

(E) $[0, \infty]$

Solution: (C). We see that

$$\frac{1}{n^{p+1}} \sum_{k=1}^{n} k^p = \frac{1}{n} \sum_{k=1}^{n} \left(\frac{k}{n}\right)^p.$$

Thus, $L(p)$ is actually a Riemann sum. The integral is

$$\int_0^1 x^p dx = \frac{1}{p+1}.$$

Thus, (C). □

Exercise 10.129. Let (a_n) be any sequence of nonnegative real numbers satisfying

$$\sum_{n=1}^{\infty} a_n = 1.$$

What is the set of all possible values of

$$\sum_{n=1}^{\infty} a_n^2?$$

(A) $[0, 1]$

(B) $(0, 1]$

(C) $(0, 1)$

(D) $[0, \infty)$

(E) $\{1\}$

Solution: (B). Cauchy Schwarz gives the lower bound. For the upper bound, we note that all terms are nonnegative, so each $a_n \in [0, 1]$. Therefore, $a_n^2 \leq a_n$, so we can bound the sum of squares above by the linear sum, which is 1. To test whether this is attainable, just take $a_1 = 0$, rest 0. If we want to be precise, then we should show every value is attainable. For this, we just use a geometric series. □

Exercise 10.130. For $p \in \mathbb{R}$, define

$$a_n(p) = \frac{1}{n^{p+1}} \sum_{k=1}^{n} k^p.$$

What is the set of all possible limits of $a_n(p)$ as $n \to \infty$, allowing $+\infty$?

(A) $[0, \infty)$

(B) $(0, \infty)$

(C) $(0, \infty]$

(D) $[0, \infty]$

(E) $\{0, +\infty\}$

Solution: (C). If $p \leq -1$, we have divergence. If $p > -1$, then we have the sum converges to $\frac{1}{p+1}$, so attains all positive reals. □

Exercise 10.131. For each $p > 0$, define

$$a_n(p) = \sum_{k=1}^{n} \frac{1}{n + k^p}.$$

What is the set of all possible values of

$$\lim_{n\to\infty} a_n(p)?$$

(A) $\{0, 1\}$

(B) $[0, 1]$

(C) $\{0, \ln 2, 1\}$

(D) $(0, 1]$

(E) $\{0, \ln 2\}$

Solution: (C). First, if $p = 1$, then this is the integral $\int_0^1 \frac{1}{1+x} dx = \ln(2)$. If $p \in (0, 1)$, then

$$\frac{1}{n} \cdot \frac{1}{1 + n^{p-1}} \leq \frac{1}{n + n^p} \leq \frac{1}{n + k^p} \leq \frac{1}{n + 1} \leq \frac{1}{n}.$$

Therefore, $a_n(p) = 1$ by squeeze. If $p > 1$, then we note that $\frac{1}{n+k^p} \leq \frac{1}{n}$ and $\frac{1}{n+k^p} \leq \frac{1}{k^p}$. This leads to the following:

$$a_n(p) \leq \frac{K}{n} + \sum_{k=K+1}^{\infty} \frac{1}{k^p},$$

for some arbitrary integer K. Since the second part converges by p-series, we can take K to be arbitrarily large and send it to zero. The first term clearly tends to 0, so we have as desired. □

Exercise 10.132. Let $f : [0, 1] \to [0, \infty)$ be continuous and satisfy $f(0) = 1$. Define

$$L(f) = \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} f\left(\frac{k}{n}\right).$$

As f varies over all such functions, what is the set of possible values of $L(f)$?

(A) $[0, 1]$

(B) $(0, 1]$

(C) $(0, \infty)$

(D) $[0, \infty)$

(E) $(0, \infty]$

Solution: (C). This is the integral of $f(x)$ over the interal $[0, 1]$. Since f is positive and $f(0) = 1$, its integral is strictly positive. Also, it is finite because it is a continuous function on a compact set, so is bounded. Thus, (C). For instance, take $f(x) = (1 - x)^t$, for $t \geq 0$. □

Exercise 10.133. Let A be a 3×5 real matrix and B a 5×3 real matrix such that

$$AB = I_3.$$

What is the rank of $I_5 - BA$?

(A) 0

(B) 1

(C) 2

(D) 3

(E) 5

Solution: (C). $AB = I_3$ implies that B is injective. Thus, $\text{Rank}(B) = 3$. Now, we note that $\ker(I_5 - BA) = \text{Im}\,(B)$, since if $x = By$, then

$$(I_5 - BA)By = By - By = 0.$$

Also, if $(I_5 - BA)y = 0$, then $I_5 y = BAy$, so $y = B(Ay)$, i.e. $y \in \text{Im}\,(B)$. Therefore,

$$\text{Rank}(I_5 - BA) = 5 - \dim \ker(I_5 - BA) = 5 - \text{Rank}(B) = 2.$$

□

Exercise 10.134. Let

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

be a 5×5 matrix, where A is an invertible 2×2 matrix. Suppose

$$\det(A) = 3, \qquad \det(M) = -12.$$

What is $\det(D - CA^{-1}B)$?

(A) -12

(B) -4

(C) 0

(D) 4

(E) 36

Solution: (B). This is an application of Schur decomposition: $\det(M) = \det(A)\det(D - CA^{-1}B)$. □

Remark 10.135. Recall that Schur decomposition is the following:

$$\begin{pmatrix} A & B \\ 0 & \underbrace{D - CA^{-1}B}_{=:S} \end{pmatrix} = \underbrace{\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix}}_{=:L} \underbrace{\begin{pmatrix} A & B \\ C & D \end{pmatrix}}_{=:M}$$

As $\det(L) = 1$, we get that $\det(A)\det(D - CA^{-1}B) = \det(M)$. We also get:

- $\mathrm{Rank}M = \mathrm{Rank}A + \mathrm{Rank}S$
- M is invertible if and only if S is invertible.

A nice consequence of Schur decomposition is calculating things like $\det(A + uv^T) = \det(A)\det(1 + v^T A^{-1}u)$

Exercise 10.136. Let $u, v \in \mathbb{R}^5$ be column vectors such that

$$v^T u = -1.$$

What is the rank of the matrix

$$I_5 + uv^T?$$

(A) 1

(B) 2

(C) 3

(D) 4

(E) 5

Solution: (D). We claim $\dim\ker(I_5 + uv^T) = 1$. First, notice that

$$(I_5 + uv^T)u = u - u = 0.$$

Conersely, if some $x \in \ker(I_5 + uv^T)$, then $x = -uv^T x$, but $v^T x$ is a scalar, so our claim is true. □

Exercise 10.137. Let $M(t)$ be the 5×5 matrix whose diagonal entries are t and whose off-diagonal entries are all 1. Which expression equals $\det M(t)$?

(A) $(t+1)^4(t-4)$

(B) $(t-1)^4(t+4)$

(C) $(t-1)^5$

(D) $(t+4)^4(t-1)$

(E) $t^4(t+4)$

Solution: (B). Use Schur decomposition. Let $u = (1,1,1,1,1)$. Then, $M(t) = (t-1)I_5 + u^T u$. Therefore, we have that

$$\det(M(t)) = \det((t-1)I_5)\det(I_5 - u^T\left((t-1)I_5\right)^{-1}u)$$
$$= (t-1)^5 \cdot (1 + \frac{5}{t-1}) = (t-1)^4(t+4)$$

Note that in the second equality, we used Sylvesters determinant theorem:

$$\det(I_5 - u^T\left((t-1)I_5\right)^{-1}u) = \det(1 + \left((t-1)I_5\right)^{-1}uu^T) = \det(1 + \frac{5}{t-1}) = 1 + \frac{5}{t-1}.$$

□

Exercise 10.138. Let A be a real 4×4 matrix satisfying

$$A^2 - 3A + 2I = 0$$

and

$$\mathrm{tr}(A) = 7.$$

What is $\det(A)$?

(A) 2

(B) 4

(C) 6

(D) 8

(E) 16

Solution: (D). Note that every eigenvalue of A must satisfy the polynomial. The possible eigenvalues of A are 1 and 2. The only possible combination is $1, 2, 2, 2$, which multiplies to 8. □

Exercise 10.139. Let A be a 5×5 matrix satisfying

$$A^3 = 0, \qquad \mathrm{rank}(A) = 3.$$

What is $\mathrm{rank}(A^2)$?

(A) 0

(B) 1

(C) 2

(D) 3

(E) Cannot be determined from the information given.

Solution: (B). Use Jordan normal form. $A^3 = 0$ tells us that every Jordan block has size at most 3. Now, note that the number of Jordan blocks of a nilpotent matrix equals its nullity, so there are two Jordan blocks. This means that $A \sim J_3(0) \oplus J_2(0)$. Now, A^2 then has rank 1. □

Exercise 10.140. Let A and B be real 5×5 matrices satisfying

$$AB = -BA.$$

Suppose A is invertible. Which of the following must be true?

(A) A is singular.

(B) B is singular.

(C) B is invertible.

(D) $B = 0$.

(E) $AB = BA$.

Solution: (B). A is invertible, so $ABA^{-1} = -B$ and $\det(B) = -\det(B)$. □

Exercise 10.141. Let A be an arbitrary real 3×3 matrix. Which of the following matrices must be invertible?

(A) $I - A$

(B) $I + A$

(C) $I + A^2$

(D) $I + A^T A$

(E) $A^T A$

Exercise 10.142. Let A be a real 4×4 matrix satisfying

$$A^2 = I, \qquad \operatorname{tr}(A) = 0.$$

What is $\det(A)$?

(A) -2

(B) -1

(C) 0

(D) 1

(E) 2

Solution: (D). Note that all eigenvalues of A satisfy $x^2 - 1$, so candidates are ± 1. These must sum to 0, so the possiblility is $+1, +1, -1, -1$. □

Exercise 10.143. Let $P_4(\mathbb{R})$ denote the vector space of real polynomials of degree at most 4. Define the linear transformation

$$T : P_4(\mathbb{R}) \to P_4(\mathbb{R})$$

by

$$T(p)(x) = p(x+1) - p(x).$$

What is the rank of T?

(A) 0

(B) 1

(C) 3

(D) 4

(E) 5

Solution: (D) We use rank nullity. The only maximal degree-4 polynomials which are invariant under the linear transformation $x+1$ are those which are purely constant. Thus, $\dim \ker T = 1$. □

Exercise 10.144. Let

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

and define

$$T : M_3(\mathbb{R}) \to M_3(\mathbb{R})$$

by

$$T(X) = DX - XD.$$

What is $\dim(\ker T)$?

(A) 1

(B) 3

(C) 4

(D) 5

(E) 6

Solution: (D). We have a block matrix, so we can really just find the kernels of the blocks. The kernel is $M_2(\mathbb{R}) \oplus M_1(\mathbb{R})$, which dimensions is $2^2 + 1^2 = 5$. □

Exercise 10.145. Let Q be a real 3×3 orthogonal matrix satisfying

$$\det(Q) = -1.$$

Which of the following must be true?

(A) 1 is an eigenvalue of Q.

(B) -1 is an eigenvalue of Q.

(C) $\mathrm{tr}(Q) = -1$.

(D) $Q^2 = I$.

(E) Q is symmetric.

Solution: (B). Recall that eigenvalues of orthogonal matrices have absolute value 1. Since the matrix is real, if we have a complex eigenvalue λ, it comes with a complex conjugate $\overline{\lambda}$, where $\lambda\overline{\lambda} = 1$. However, we require -1 to be an eigenvalue due to the determinant. □

Exercise 10.146. Let A be a real 4×4 matrix with

$$\mathrm{rank}(A) = 3.$$

What is the rank of the adjugate matrix $\mathrm{adj}(A)$?

(A) 0

(B) 1

(C) 2

(D) 3

(E) 4

Solution: (B). Since $\mathrm{Rank}(A) = n-1$, $\mathrm{Rank}(\mathrm{adj}(A)) = 1$. Just remember the adjugate criterion. □

Exercise 10.147. Let A be a 2×3 matrix and B a 3×2 matrix. Suppose the eigenvalues of AB are 2 and 5. What is $\det(I_3 + BA)$?

(A) 0

(B) 6

(C) 12

(D) 15

(E) 18

Solution: (E). First note that $\det(I_3 + BA) = \det(I_2 + AB)$. Since 2 and 5 are eigenvalues of AB, we find that 3 and 6 are eigenvalues of $I_2 + AB$. Thus, the determinant is 18. □

Remark 10.148. We prove a super brief lemma: if λ is an eigenvalue of $A \in M_n$, then $cI_n + A$ has $c + \lambda$ as an eigenvalue. The proof is simple: If v is an eigenvector associated to λ, then $(cI + A)v = cv + Av = (c + \lambda)v$.

Exercise 10.149. Let A be a real 4×4 matrix. Which of the following conditions guarantees that A is diagonalizable over $\mathbb{R}$?

(A) $A^2 = 0$

(B) $(A - I)^2 = 0$

(C) $A^3 = A$

(D) The characteristic polynomial of A is $(t-1)^4$.

(E) $\mathrm{tr}(A) = 0$

Solution: (C). Let $m_A(t)$ be the minimal polynomial of A. Then, if $p(t)$ is some polynomial such that $p(A) = 0$, then $m_A(t) \mid p(t)$. Since (C) is a product of linear factors, then $m_A(t)$ is a product of linear parts, so done. □

Exercise 10.150. Let A be a 3×3 matrix satisfying

$$\det(A) = 2.$$

What is $\det(\mathrm{adj}(A))$?

(A) $\frac{1}{2}$

(B) 1

(C) 2

(D) 4

(E) 8

Solution: (D). Use that $\det(\mathrm{adj}(A)) = (\det(A))^{3-1}$. □

Exercise 10.151. Let A be a real symmetric 3×3 matrix such that

$$\mathrm{tr}(A) = 0, \qquad \mathrm{tr}(A^2) = 6, \qquad \det(A) = 0.$$

What is $\mathrm{tr}(A^4)$?

(A) 9

(B) 12

(C) 16

(D) 18

(E) 36

Solution: (D). $\det(A)$ tells us that ate least one of its eigenvalues are 0. For the remaining two, if λ is one, then $\mathrm{Tr}(A) = 0$ implies that the other is $-\lambda$. Then, we must have that $\lambda^2 = 3$ via the $\mathrm{Tr}(A^2) = 6$ condition. Thus, $\lambda = \sqrt{3}$. It follows that A^4 is similar to a matrix with diagonal entries $9, 9, 0$, so we are done. □

Exercise 10.152. Suppose A is a real 4×4 matrix satisfying

$$\operatorname{rank}(A) = 2.$$

Which of the following is the set of all possible values of $\operatorname{rank}(A^2)$?

(A) $\{0, 1, 2\}$

(B) $\{1, 2\}$

(C) $\{0, 2\}$

(D) $\{2\}$

(E) $\{0, 1, 2, 3\}$

Solution: (A); use Sylvester. □

Exercise 10.153. Let A be a real 5×5 matrix with characteristic polynomial

$$p_A(t) = (t-2)^3(t+1)^2.$$

What is

$$\operatorname{tr}(\operatorname{adj}(A))?$$

(A) -8

(B) -4

(C) -2

(D) 2

(E) 4

Solution: (B). We note that $A \operatorname{adj}(A) = \det(A)I$, so

$$\operatorname{Tr}(\operatorname{adj}(A)) = \det(A)\operatorname{Tr}(A^{-1}).$$

The trace of A^{-1} is the sum of reciprocal of eigenvalues, which is $-\frac{1}{2}$. The determinant is 8, so -4. □

Exercise 10.154. Let A and B be commuting nilpotent 4×4 matrices, so that $AB = BA$. Which of the following must be true?

(A) $A + B$ is invertible.

(B) $A - B$ is invertible.

(C) $A + B$ is nilpotent.

(D) $A + B$ is symmetric.

(E) AB is invertible.

Solution: (C).This is basic algebra. □

Exercise 10.155. Consider the linear transformation

$$T : M_3(\mathbb{R}) \to M_3(\mathbb{R})$$

defined by

$$T(X) = \frac{1}{2}(X + X^T).$$

What is the rank of T?

(A) 3

(B) 4

(C) 5

(D) 6

(E) 9

Solution: (D). Assign arbitrary inputs and see what lines up. Alternatively, look at the kernel; it is matrices of 0 diagonal entries and alternating diagonals. □

Exercise 10.156. Let A be a real 2×2 matrix with

$$\mathrm{tr}(A) = 3, \qquad \det(A) = 2.$$

Which expression is equal to A^5?

(A) $16A - 15I$

(B) $31A - 30I$

(C) $32A - 31I$

(D) $25A - 24I$

(E) $A + 30I$

Solution: (B). Note that characteristic polynomial of A is

$$\chi_A(t) = t^2 - \mathrm{Tr}(A)t + \det(A).$$

Thus, $\chi_A(t) = t^2 - 3t + 2$. We iteratively substitute then. □

Exercise 10.157. Evaluate

$$\lim_{x \to 0} \frac{\tan(3x) - 3\tan x}{x^3}.$$

(A) 0

(B) 2

(C) 6

(D) 8

(E) $+\infty$

Solution: (D). Use the Taylor expansion of $\tan(x)$:

$$\tan(x) = x + \frac{x^3}{3} + O(x^5) + \dots$$

Then, the order 3 terms only matter. □

Exercise 10.158. Evaluate

$$\lim_{x \to 2} \frac{\ln x}{\sin(\pi x)}.$$

(A) 0

(B) $\ln 2$

(C) $+\infty$

(D) $-\infty$

(E) The limit does not exist.

Solution: (E). Evaluate from the LHS and the RHS. □

Exercise 10.159. What is the exact range of

$$f(x) = \frac{\ln(1+x)}{x}, \qquad 0 < x < 1?$$

(A) $(0, 1)$

(B) $(\ln 2, 1)$

(C) $[\ln 2, 1]$

(D) $(1, \infty)$

(E) $(0, \ln 2)$

Solution: (B) f is continuous on this interval, so we just evaluate the limits to the end points. as $x \to 1$, $f \to \ln(2)$. As $x \to 0$, the limit is 1. □

Exercise 10.160. Let $(G, \cdot)$ be a group with identity e, and suppose $\star$ is another binary operation on G with the same two-sided identity e, satisfying

$$(a \cdot b) \star (c \cdot d) = (a \star c) \cdot (b \star d)$$

for all $a, b, c, d \in G$. Which of the following must hold?

(A) G is cyclic.

(B) $\star = \cdot$, but G need not be abelian.

(C) G is abelian, but the two operations need not coincide.

(D) $\star = \cdot$ and G is abelian.

(E) G is trivial.

Solution: (D). Consider identities on $\cdot$ and $\star$, and see that they're equal. Next, we can swap around some variables to get commutativity. □

Exercise 10.161. Let

$$a_n = (-1)^n + \frac{1}{n}.$$

Determine

$$\left(\limsup_{n\to\infty} a_n,\ \liminf_{n\to\infty} a_n\right).$$

(A) $(0, 0)$

(B) $(1, -1)$

(C) $(-1, 1)$

(D) $(1, 0)$

(E) Neither limit exists.

Solution: (B). For lim sup, take even n, and for lim inf, take odd n. □

Exercise 10.162. Suppose

$$f : [1, \infty) \to [0, \infty)$$

satisfies

$$\int_1^\infty f(x)\,dx < \infty.$$

Which additional hypothesis guarantees that

$$\lim_{x\to\infty} f(x) = 0?$$

(A) f is continuous.

(B) f is C^∞.

(C) f is bounded.

(D) f is uniformly continuous.

(E) None of the above.

Solution: (D) This is a classical analysis result. □

Exercise 10.163. Consider the subset of $\mathbb{R}^2$ defined by

$$S = \left\{\left(x, \sin\frac{1}{x}\right) : 0 < x \le 1\right\} \cup (\{0\} \times [-1, 1]).$$

Which of the following statements hold?

I. S is connected.

II. S is path-connected.

III. S is closed.

(A) I only

(B) II only

(C) I and III only

(D) II and III only

(E) I, II, and III

Solution: (C). S is the closure of the graph part of $\sin\left(\frac{1}{x}\right)$, so is closed. The graph is connected and closures of connected spaces are connected. Clearly not path connected. □

Exercise 10.164. Consider the hyperbola

$$\frac{x^2}{4} - \frac{y^2}{12} = 1.$$

A circle centered at the origin passes through the foci of the hyperbola. What is the area of the circular sector bounded by the positive x-axis, the first-quadrant asymptote of the hyperbola, and the circle?

(A) 2π

(B) $\frac{4\pi}{3}$

(C) $\frac{8\pi}{3}$

(D) 4π

(E) $\frac{16\pi}{3}$

Solution: (C). The asymptotes are $y = \pm\sqrt{3}x$, so the angle created is $\frac{\pi}{3}$. As for the appropriate radius, we calculate the focus, which is $\sqrt{4+12} = 4$. Thus, the area is $\frac{1}{2} \cdot 4^2 \cdot \frac{\pi}{3}$. □

Exercise 10.165. Let G be a finite group of order $|G| = 56$. Let n_7 denote the number of Sylow 7-subgroups of G. Which of the following gives all possible values of n_7 allowed by Sylow's congruence and divisibility conditions?

(A) $\{1\}$

(B) $\{7\}$

(C) $\{1, 7\}$

(D) $\{1, 8\}$

(E) $\{1, 2, 4, 8\}$

Solution: (D). Use Sylow III. □

Exercise 10.166. Let A be an invertible real 3×3 matrix satisfying

$$\det A = -6.$$

Define

$$E = \{x \in \mathbb{R}^3 : \|Ax\| \leq 1\},$$

where $\|\cdot\|$ denotes the Euclidean norm. What is the volume of E?

(A) $\dfrac{2\pi}{9}$

(B) $\dfrac{4\pi}{3}$

(C) $\dfrac{2\pi}{3}$

(D) 8π

(E) $\dfrac{8\pi}{9}$

Solution: (A). We note that for an inertible matrix A, $\mathrm{vol}(A(S)) = |\det A| \,\mathrm{vol}(S)$, where S is some subset of $\mathbb{R}^3$. Since the set in concern here is really $A^{-1}(B)$, where B is the unit ball, we get as desired. □

Exercise 10.167. Let (a_n) and (b_n) be bounded real sequences. Consider the following statements.

I.
$$\limsup_{n\to\infty}(a_n + b_n) \leq \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.$$

II.
$$\limsup_{n\to\infty}(a_n + b_n) = \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.$$

III.
$$\limsup_{n\to\infty}(-a_n) = -\liminf_{n\to\infty} a_n.$$

Which statements must be true?

(A) I only

(B) II only

(C) III only

(D) I and III only

(E) I, II, and III

Solution: (D). II is incorrect, because if a_n is the sequence of all 1s except for one index which is -1, and b_n is the sequence of all -1 except for one index which is 1, then the LHS is 0 whilst the RHS is 2. III is clearly true. We prove I. Fix N. Then, $\sup_{n\geq N}(a_n + b_n) \leq \sup_{n\geq N} a_n + \sup_{n\geq N}$, so taking the limit $N \to \infty$, we get precisely as desired. □

Exercise 10.168. Let X and Y be independent random variables, each uniformly distributed on $[0, 1]$. Find

$$\mathbb{P}(X + Y \leq 1,\ X \leq Y).$$

(A) $\frac{1}{8}$

(B) $\frac{1}{6}$

(C) $\frac{1}{4}$

(D) $\frac{1}{3}$

(E) $\frac{1}{2}$

Solution: (C). Just find the area. □

Exercise 10.169. Let A be a real 4×4 matrix with characteristic polynomial

$$p_A(t) = (t-2)^3(t+1)$$

and

$$\dim \ker(A - 2I) = 2.$$

What is the minimal polynomial of A?

(A) $(t-2)(t+1)$

(B) $(t-2)^2(t+1)$

(C) $(t-2)^3(t+1)$

(D) $(t-2)^2$

(E) $(t-2)^3$

Solution: (B). $p_A(t)$ is the characteristic polynomial. The algebraic multiplicity of 2 is 3 and the geometric multiplicity of 2 is 2. Therefore, we have that $J_1(2) \oplus J_2(2)$ as a component. Finally, we have -1 as an eigenvalue, so the missing piece is $J_1(-1)$. Since $A \sim J_1(2) \oplus J_2(2) \oplus J_1(-1)$, it is clear that (B) is the one. □

Exercise 10.170. Evaluate

$$\lim_{h \to 0} \frac{1}{h^3} \int_0^h \sin(t^2)\, dt.$$

(A) 0

(B) $\frac{1}{6}$

(C) $\frac{1}{3}$

(D) $\frac{1}{2}$

(E) 1

Solution: (C). L'höpital says that limit is equal to

$$\lim_{h\to 0} \frac{\frac{d}{dh}\int \cdots}{3h^2}.$$

To differentiate this integral with respect to h, we use Leibniz:

$$\frac{d}{dh}\left(\int_0^h \sin(t^2)dt\right) = \sin(h^2)\cdot 1 - 0 + \int_0^h 2t0dt = \sin(h^2)$$

Taking the limit, only the degree-2 term in the Taylor expansion remains from $\sin(h^2)$. In particular, the numereator should be 2 after two derivatives, and the bottom is 6. Thus, (C). □

Exercise 10.171. Five distinguishable balls are placed independently and uniformly at random into three distinguishable boxes. What is the probability that no box is empty?

(A) $\frac{10}{27}$

(B) $\frac{40}{81}$

(C) $\frac{50}{81}$

(D) $\frac{5}{9}$

(E) $\frac{2}{3}$

Solution: (C) Total number of ways is 3^5. As for the condition, we use PIE:

$$3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5 = 150.$$

The above is subtracting when at least 1 box is empty, then adding back in when there were two empty boxes. □

Exercise 10.172. Let R be the region in $\mathbb{R}^2$ defined by

$$x \geq 0, \qquad y \geq 0,$$

$$2x + y \leq 6, \qquad x + 2y \leq 6.$$

What is the area of R?

(A) 5

(B) 6

(C) 7

(D) 8

(E) 9

Solution: (B). Just some coordinate geo. □

Exercise 10.173. Suppose (a_n) is a sequence of nonnegative real numbers satisfying

$$\sum_{n=1}^{\infty} a_n < \infty.$$

Consider the following statements.

I. $\lim_{n\to\infty} a_n = 0$.

II. $\sum_{n=1}^{\infty} a_n^2 < \infty$.

III. $\sum_{n=1}^{\infty} \sqrt{a_n} < \infty$.

Which statements must be true?

(A) I only

(B) II only

(C) I and II only

(D) II and III only

(E) I, II, and III

Solution: (C). I is standard. II also follows because after some fixed N, $a_n^2 < a_n$, so the squared series tends to 0 by comparison. III is not true; just consider $a_n = \frac{1}{n^2}$. □

Exercise 10.174. Let G be a group of order 21. Consider the following statements.

I. G has a normal subgroup of order 7.

II. G has a normal subgroup of order 3.

III. G is cyclic.

Which statements must be true?

(A) I only

(B) II only

(C) I and II only

(D) I and III only

(E) I, II, and III

Solution: (A). III is clearly false. For I and II, we want to show that the number of Sylow-p subgroups is exactly one (or not), since all Sylow subgroups are conjugate to each other. By Sylow III, there exists a Sylow-7 subgroup. The number of such subgroups are $n_7 \mid 21$ and $n_7 \equiv 1 \pmod 7$. n_7 cannot be divisible by 7, so $n_7 \mid 3$, forcing just one Sylow-7 subgroup. For $p = 3$, $n_3 \mid 7$ and congruent to 1 (mod 3). Actually, $n_3 \in \{1, 7\}$ then so n_3 can be not equal to 1. □

Exercise 10.175. Consider the plane $x + y + z = 3$. The plane intersects the three positive coordinate axes at points P, Q, R. What is the area of triangle PQR?

(A) $\frac{9}{2}$

(B) $\frac{9\sqrt{2}}{2}$

(C) $\frac{9\sqrt{3}}{2}$

(D) $9\sqrt{3}$

(E) $\frac{27}{2}$

Solution: (C) We get an equilateral triangle of side $3\sqrt{2}$. □

Exercise 10.176. An urn contains 4 red balls, 3 blue balls, and 2 green balls. Two balls are drawn uniformly at random without replacement. Given that at least one of the two balls is red, what is the probability that both balls are red?

(A) $\frac{1}{6}$

(B) $\frac{2}{9}$

(C) $\frac{3}{13}$

(D) $\frac{1}{4}$

(E) $\frac{4}{13}$

Solution: (E). We use conditional probability. Probability of 2 red is $\frac{4 \cdot 3}{9 \cdot 8}$. Probability of at least one red is $1 - \frac{5 \cdot 4}{9 \cdot 8}$. Now, the intersection of having 2 reds and at least 1 red is just having exactly two reds. Thus, the probability of intersection is equal to probability of 2 red. Thus, the answer is

$$\frac{\frac{1}{6}}{\frac{52}{72}} = \frac{3}{13}.$$

□

Exercise 10.177. Let A be a real symmetric 3×3 matrix with eigenvalues

$$1, \quad 1, \quad 4.$$

What is the set of all possible values of the Rayleigh quotient

$$R(x) = \frac{x^T A x}{x^T x}, \qquad x \in \mathbb{R}^3 \setminus \{0\}?$$

(A) $\{1, 4\}$

(B) $(1, 4)$

(C) $[1, 4]$

(D) $[0, 4]$

(E) $[1, \infty)$

Solution: (C). We diagonalize A, so that $x^T A x = x^2 + y^2 + 4z^2$. Thus,

$$R(x) = \frac{x^2 + y^2 + 4z^2}{x^2 + y^2 + z^2} = 1 + \frac{3z^2}{x^2 + y^2 + z^2}.$$

Setting $x = y = 0$ and $z = 1$, we attain 4. Setting $z = 0$ and $x, y \neq 0$, we attain 1. Also, every intermediate value occurs. □

Exercise 10.178. Consider the function

$$f(x, y) = x^4 + y^4 - 2x^2 - 2y^2.$$

How many distinct local minima does f have in $\mathbb{R}^2$?

(A) 0

(B) 1

(C) 2

(D) 3

(E) 4

Solution: (E). $f_x = 4x^3 - 4x$ and $f_y = 4y^3 - 4y$. Then, $f_{xx} > 0$ at the critical points $(1, \pm 1), (-1, \pm 1)$ and the determinant of the Hessian is positive there. □

Exercise 10.179. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation represented by a matrix A satisfying $\det A = -3$. Let Δ be the triangle with vertices $(0, 0), (2, 1), (1, 3)$. What is the area of $T(\Delta)$?

(A) $\frac{5}{2}$

(B) 5

(C) $\frac{15}{2}$

(D) 10

(E) 15

Solution: (C). The original triangle has area $\frac{5}{2}$. Then, the area of $T(\Delta)$ is $|\det T| \cdot \text{Area}(\Delta)$. □

Remark 10.180. I guess more generally, if you have an invertible linear transformation T acting on $S \subseteq \mathbb{R}^n$, then $\text{vol}(T(S)) = \det(T)\,\text{vol}(S)$, where the volume here is the Euclidean volume.

Exercise 10.181. Let X and Y be independent real-valued random variables satisfying

$$\text{Var}(X) = 2, \qquad \text{Var}(Y) = 3.$$

What is

$$\text{Var}(2X - Y)?$$

(A) 5

(B) 7

(C) 11

(D) 13

(E) 20

Solution: (C). Follows from $\text{Var}(aX + bY) = a^2\,\text{Var}(X) + b^2\,\text{Var}(Y)$. □

Exercise 10.182. Consider the following statements.

I. Every uniformly continuous function $f : (0, 1) \to \mathbb{R}$ has a finite limit as $x \to 0^+$.

II. If $f : (0, 1) \to \mathbb{R}$ is differentiable and

$$|f'(x)| \leq 10 \quad \text{for every } x \in (0, 1),$$

then f is uniformly continuous.

III. If $f : (0, 1) \to \mathbb{R}$ is continuous and the sequence $(f(1/n))_{n\geq 2}$ converges, then $\lim_{x\to 0^+} f(x)$ exists.

Which statements must be true?

(A) I only

(B) II only

(C) I and II only

(D) I and III only

(E) I, II, and III

Solution: (C). For I, take a Cauchy seqences converging to 0. Then, uniform continuity of f implies that f applied to the sequence is Cauchy, therefore converges by completeness of $\mathbb{R}$. For II, note that f is Lipschitz continuous, so is uniformly continuous. That is, by MVT, for all $x, y \in (0,1)$, exists $c \in (x,y)$ such that $f(x) - f(y) = f'(c)(x-y)$. Taking absolute values,

$$|f(x) - f(y)| = |f'(c)(x-y)| \leq 10|x-y|.$$

III is not true. Just consider $f(x) = \sin\left(\frac{2\pi}{x}\right)$. □

Exercise 10.183. Let G be a group of order $|G| = 45$. Consider the following statements.

I. Both the Sylow 3-subgroup and the Sylow 5-subgroup are normal.

II. G is abelian.

III. G is cyclic.

Which statements must be true?

(A) I only

(B) II only

(C) I and II only

(D) I and III only

(E) I, II, and III

Solution: (C). Sylow III says I is true. As for II, we note that the Sylow-5 group P_5 has order 5, so is cyclic (hence abelian). As for the Sylow-3 group P_3, it has order $9 = 3^2$, so it is abelian. Now, we aim to use the inner direct product theorem on G. Since the Sylow subgroups are of coprime order, they have trivial intersection. Also,

$$|P_3P_5| = \frac{|P_3||P_5}{|P_3 \cap P_5|} = |P_3||P_5| = 5 \cdot 9 = |G|,$$

implying that $G = P_3P_5$ Applying the theorem, $G \cong P_3 \times P_5$, and a direct product of abelian groups is clearly abelian. III is not true because P_3 could be acylic. □

Exercise 10.184. Let $F(x,y,z) = (x^3, y^3, z^3)$ and S be the unit sphere oriented outward. Evaluate

$$\iint_S F \cdot n dS.$$

(A) $\frac{4\pi}{5}$

(B) $\frac{8\pi}{5}$

(C) $\frac{12\pi}{5}$

(D) 4π

(E) 12π

Solution: (D). Use divergence theorem. The flux integral is now

$$\int_V 3(x^2+y^2+z^2)dV.$$

We use spherical coordinates now:

$$\int_{\theta=0}^{2\pi}\int_{\phi=0}^{\pi}\int_{r=0}^{1} r^2\cdot r^2\sin\phi dr d\phi d\theta = \frac{2\pi}{5}\int_{\phi=0}^{\pi}\sin\phi d\phi = \frac{4\pi}{5}.$$

□

Exercise 10.185. A permutation of

$$\{1,2,3,4,5\}$$

is chosen uniformly at random. What is the probability that the permutation has no fixed points?

(A) $\frac{1}{3}$

(B) $\frac{11}{30}$

(C) $\frac{3}{8}$

(D) $\frac{2}{5}$

(E) $\frac{1}{2}$

Solution: (B). There are 5! permutations. Those which have nonfixed points are those where all numbers are present. This consists of elements in the conjugacy classes (12345) and (12)(345). The number of these are 4! and $\binom{5}{2}\cdot 2!$. Thus, the answer is $\frac{24+20}{120} = \frac{11}{30}$. □

Exercise 10.186. Let A be a diagonalizable real 5×5 matrix with three distinct real eigenvalues whose eigenspaces have dimensions 2, 2, and 1. Let

$$V = \{X\in M_5(\mathbb{R}) : AX = XA\}.$$

What is $\dim V$?

(A) 5

(B) 7

(C) 9

(D) 11

(E) 13

Solution: (C). This $V = \ker(T)$, where $T(X) = AX - XA$. Those which lie in the kernel of this are those which vanish on each block. In each block, these are essentially scalar multiples of identities, so commute with anything in that block. This gives the sum of $2^2 + 2^2 + 1$. □

Exercise 10.187. Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable and suppose

$$f'(x) \geq 0 \qquad \text{for every } x \in \mathbb{R}.$$

Consider the following statements.

I. f is nondecreasing.

II. f is injective.

III. f has at most one zero.

Which statements must be true?

(A) I only

(B) II only

(C) III only

(D) I and II only

(E) I, II, and III

Solution: (A) I is clearly true by MVT. As for II and III, consider constant $f = 0$. □

Exercise 10.188. Let A be an invertible real 3×3 matrix with

$$\det A = 5.$$

Consider the region

$$R = \left\{ x \in \mathbb{R}^3 : \begin{array}{l} (Ax)_i \geq 0, \quad i = 1, 2, 3, \\ (Ax)_1 + (Ax)_2 + (Ax)_3 \leq 1 \end{array} \right\}.$$

What is the volume of R?

(A) $\dfrac{1}{6}$

(B) $\dfrac{1}{10}$

(C) $\dfrac{1}{15}$

(D) $\dfrac{1}{30}$

(E) $\dfrac{5}{6}$

Solution: (D). The region without A is just the region of positive x, y, z bounded by the origin and $(1,0,0)(0,1,0),(0,0,1)$. The volume of this is $\frac{1}{6}$, so the transformed space is $\frac{1}{5} \cdot \frac{1}{6}$. □

Exercise 10.189. Let σ be a uniformly random permutation of

$$\{1, 2, \ldots, 7\}.$$

Let X be the number of fixed points:

$$X = \#\{i : \sigma(i) = i\}.$$

What is $\mathbb{E}[X]$?

(A) 0

(B) $\frac{1}{7}$

(C) 1

(D) $\frac{7}{2}$

(E) 7

Solution: (C). Define

$$I_i = \begin{cases} 1 & \text{if } \sigma(i) = i, \\ 0 & \text{otherwise} \end{cases}$$

Then, $X = I_1 + \cdots + I_7$, and we can apply linearity of expectation. Now, $\mathbb{E}[I_i] = \frac{1}{7}$, because if one index is fixed, then there are 6! other choices. Thus, the answer is (C). □

Exercise 10.190. What is the minimum value of $f(x, y) = x^2 + y^2$ subject to $x + 2y = 5$?

(A) 1

(B) 4

(C) 5

(D) 10

(E) 25

Solution: (C). Use Cauchy Schwarz. □

Exercise 10.191. Let A be a real 6×6 matrix with characteristic polynomial

$$p_A(t) = (t-3)^4(t+1)^2.$$

Suppose

$$\dim \ker(A - 3I) = 2.$$

The generalized eigenspace corresponding to the eigenvalue 3 is

$$G_3 = \ker((A - 3I)^6).$$

What is $\dim G_3$?

(A) 2

(B) 3

(C) 4

(D) 5

(E) 6

Solution: (C). The possible Jordan decomposition associated to the eigenvalue 3 is $J_1(3) \oplus J_3(3)$ or $J_2(3) \oplus J_2(3)$. In these, we find that the dimension of G_3 is 4, since after powers of 6, all 'shifted' Jordan blocks vanish. □

Exercise 10.192. Let (f_n) be a sequence of continuous real-valued functions on $[0, 1]$ converging uniformly to f. Consider the following statements.

I. f is continuous.

II. If every f_n is differentiable, then f is differentiable.

III.
$$\lim_{n\to\infty} \int_0^1 f_n(x)\,dx = \int_0^1 f(x)\,dx.$$

Which statements must be true?

(A) I only

(B) II only

(C) I and II only

(D) I and III only

(E) I, II, and III

Solution: (D). I is true by a basic result: If f_n are continuous and converges uniformly to some f, then f is continuous. For III, uniform convergence implies point-wise convergence. Also, f is continuous on the compact interval $[0, 1]$, so it's bounded, so we have some dominating function. Thus, DCT says this is true. As for II, this is not necessarily true; we cannot conclude anything about differentiability at this point. (note that uniform convergence of the derivative is sufficient, but not necessary). □

Bibliography

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